# Supplementary material for "New rules, old habits: across seven team sports, each league keeps its own way of using the rules"

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**What each section backs in the main text**

| Section | What it backs |
|---|---|
| S1 | Methods: the sports, leagues and records, each league's limit, and what each test could detect |
| S2 | Results: football's leagues kept their order through the change |
| S3 | Results: the explanations tested for football's order |
| S4 | Methods and the table of tests: the pre-specified tests and their dates; Results: the other sports |
| S5 | Results: the check that each rule did what it was written to do, and the coaches |
| S6 | Results: a league's place persists |
| S7 | Discussion: why the load-management law is not claimed; the table of tests: the timing tests |
| S8 | Results: how widely the leagues of other sports use their players |

# S1 Leagues, records and rules

## S1.1 The football source

Football's counts of substitutions come from transfermarkt-datasets, a public dataset built from Transfermarkt, a website on football players and matches [1]. We read the version saved as commit af8c44a in its public repository, from a snapshot taken on 3 August 2026. The dataset is released under the CC0-1.0 licence. Its maintainers say it is current to 6 July 2026. The checksums of the snapshot and of each file read are archived with the analysis scripts. A checksum is a number computed from a file, and it changes if the file changes. For each match the records carry the starting line-ups, the benches and the minute of each substitution. These come in three separate records. The team sheet has two lists for each team: its starters and its substitutes on the bench. The appearance records list each player who played. The match events list each substitution, with its minute. The same site gives market values, its estimates of each player's transfer fee.

The source holds thirty-one men's top leagues. Of these, fourteen are covered from 2012–13, and the other seventeen only from 2024. So only these fourteen leagues have seasons under both limits (Table S1). They are the source's leagues with long records. They are not all the leagues that changed the rule. In their seasons from 2015–16 to 2025–26 the source lists 43,817 league games. Of these, 43,765 have a player on record. A league-season is one league's season. Table S1 gives 153 of them: every season of these leagues except Ukraine's 2021–22. That season stopped in February 2022, and the source misses most of its starters. The 153 league-seasons hold 43,159 matches on record. This is the count the main text gives.

A team-match is one team's side of one match. A team-match enters the counts only if every starter on its team sheet also has an appearance on record. We call this rule the record check. It keeps 84,229 team-matches in the 153 league-seasons of Table S1.

The leagues differ from one analysis to another. A league's increment is its mean under five minus its own mean under three (Section S2.7). The increment also reads Brazil, the MLS and South Korea. These were collected separately from the same site. So the increment reads 17 leagues: Belgium, Brazil, Denmark, England, France, Germany, Greece, Italy, the MLS, the Netherlands, Portugal, Russia, Scotland, South Korea, Spain, Turkey and Ukraine. The timing measures add leagues outside Europe and seven women's leagues (Section S7). One timing measure, the timing product, is the timing departure counted per starter who began the match. It reads 17 men's leagues: Argentina, Australia, Brazil, Denmark, England, France, Germany, Greece, Italy, Japan, the MLS, Mexico, the Netherlands, Portugal, South Korea, Spain and Turkey. The two sets are the same size, but they are not the same leagues. Table S4 gives every analysis with the leagues it reads. Across all analyses, football's records hold 49,067 matches in 28 leagues, men's and women's. The leagues that only the explanation models read are not in this count. The six other sports' records hold 207,899 games and innings. Each league-season is counted once. Where analyses read different counts for it, the largest is used.

### S1.2 Each league-season's limit

Each league-season's limit, three or five, is the one its competition's regulations set (Table S1). For each limit, the table gives the document it comes from, with its publisher and date. It also says how directly the document names the league and season. For 26 league-seasons the document names the league and season, or the change within the season. For 62 it is the Laws of the Game, read in the season's own edition or carried from the nearest edition read. For 42 it is the document of the season before, because the rule did not change. IFAB, football's rule-maker, let competitions choose a limit of five. For the other 23 league-seasons the document is IFAB's option or a general news report, because the league's own decision was not found. In each of them, the substitutions made show a limit of five.

A season whose limit changed within it is split at the date of the change. In 2019–20, eight leagues played under three and then, from their restart, under five. After 2019–20, two seasons changed limit. Belgium's 2020–21 changed on 1 January 2021, and Scotland's 2021–22 on 17 January 2022. But Belgium's teams began before the date their league announced. They made four or five substitutions in twelve team-matches from 26 December 2020.

An earlier reading took each season's limit from the substitutions its teams made. It left out the whole of any season whose mean showed no single limit. That reading is kept as a check. It disagrees with the regulations in five of the 153 league-seasons: Belgium's 2020–21 and 2021–22, Russia's 2021–22 and Scotland's 2020–21 and 2021–22. It had left four of them out, and had counted Scotland's 2021–22 under three.

Table S1 also gives the number of named substitutes: the substitutes listed on each team sheet. The regulations we read give that number for only 7 league-seasons. So the table gives the number that the team sheets name most often.

Table S1: Every league-season of the fourteen European football leagues, except Ukraine's 2021–22, in which almost no team-match was kept. The limit is the one the competition's regulations set. A season that changed limit is split at the date given. Named is the most common number of substitutes named on the team sheets. Matches are those on record. Kept is the share of team-matches that the record check keeps: those in which every starter on the team sheet has an appearance on record. The last column gives the document behind each limit: its publisher, its date, and how directly it names the league and season. The full reference of each is in the archived repository.

| Season | Limit | Five from | Named | Matches | Kept | Document behind the limit |
|---|---|---|---|---|---|---|
| *Belgium* | | | | | | |
| 2015–16 | three | | 7 | 240 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2016–17 | three | | 7 | 240 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2017–18 | three | | 7 | 240 | 100% | The IFAB, 2017-06-01; Laws of the season |
| 2018–19 | three | | 7 | 240 | 100% | The IFAB, 2018-06-01; Laws of the season |
| 2019–20 | three | | 7 | 232 | 93% | The IFAB, 2018-06-01; Laws 2018/19 and 2019/20 |
| 2020–21 | three, then five | 2021-01-01 | 7 | 306 | 100% | La DH, 2020-12-15; names the change |
| 2021–22 | five | | 7 | 305 | 71% | Sporza, 2022-06-01; names the league |
| 2022–23 | five | | 7 | 306 | 100% | Sporza, 2022-06-01; names the league |
| 2023–24 | five | | 7 | 240 | 100% | Sporza, 2022-06-01; as the season before |
| 2024–25 | five | | 9 | 240 | 100% | Sporza, 2022-06-01; as the season before |
| 2025–26 | five | | 9 | 240 | 100% | Sporza, 2022-06-01; as the season before |
| *Denmark* | | | | | | |
| 2015–16 | three | | 7 | 198 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2016–17 | three | | 7 | 182 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2017–18 | three | | 7 | 182 | 100% | The IFAB, 2017-06-01; Laws of the season |
| 2018–19 | three | | 7 | 182 | 100% | The IFAB, 2018-06-01; Laws of the season |
| 2019–20 | three, then five | 2020-05-28 | 7 | 182 | 100% | The Copenhagen Post, 2020-05-25; names the change |
| 2020–21 | five | | 7 | 132 | 100% | The Jakarta Post, 2020-11-17; general report |
| 2021–22 | five | | 7 | 132 | 56% | The IFAB, 2022-06-13; IFAB's temporary option |
| 2022–23 | five | | 9 | 132 | 91% | The IFAB, 2022-06-13; IFAB's permanent option |
| 2023–24 | five | | 9 | 132 | 100% | The IFAB, 2022-06-13; as the season before |
| 2024–25 | five | | 9 | 132 | 100% | The IFAB, 2022-06-13; as the season before |
| 2025–26 | five | | 9 | 132 | 98% | The IFAB, 2022-06-13; as the season before |
| *England* | | | | | | |
| 2015–16 | three | | 7 | 380 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2016–17 | three | | 7 | 380 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2017–18 | three | | 7 | 380 | 100% | The IFAB, 2017-06-01; Laws of the season |
| 2018–19 | three | | 7 | 380 | 99% | The IFAB, 2018-06-01; Laws of the season |
| 2019–20 | three, then five | 2020-06-17 | 7 | 380 | 97% | Chelsea FC, 2020-06-04; names the change |
| 2020–21 | three | | 7 | 380 | 96% | Sky Sports, 2020-09-04; names the league |
| 2021–22 | three | | 9 | 380 | 92% | Sports Illustrated, 2022-03-31; names the league |
| 2022–23 | five | | 9 | 380 | 100% | Sports Illustrated, 2022-03-31; names the league |

| Season | Limit | Five from | Named | Matches | Kept | Document behind the limit |
|---|---|---|---|---|---|---|
| 2023–24 | five | | 9 | 380 | 100% | Sports Illustrated, 2022-03-31; as the season before |
| 2024–25 | five | | 9 | 380 | 100% | Sports Illustrated, 2022-03-31; as the season before |
| 2025–26 | five | | 9 | 380 | 100% | Sports Illustrated, 2022-03-31; as the season before |
| *France* | | | | | | |
| 2015–16 | three | | 7 | 380 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2016–17 | three | | 7 | 380 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2017–18 | three | | 7 | 380 | 100% | The IFAB, 2017-06-01; Laws of the season |
| 2018–19 | three | | 7 | 380 | 99% | The IFAB, 2018-06-01; Laws of the season |
| 2019–20 | three | | 7 | 279 | 95% | The IFAB, 2018-06-01; Laws 2018/19 and 2019/20 |
| 2020–21 | five | | 9 | 380 | 95% | Maxifoot, 2020-07-31; names the league |
| 2021–22 | five | | 9 | 380 | 86% | The IFAB, 2022-06-13; IFAB's temporary option |
| 2022–23 | five | | 9 | 380 | 98% | The IFAB, 2022-06-13; IFAB's permanent option |
| 2023–24 | five | | 9 | 306 | 100% | The IFAB, 2022-06-13; as the season before |
| 2024–25 | five | | 9 | 306 | 100% | The IFAB, 2022-06-13; as the season before |
| 2025–26 | five | | 9 | 305 | 100% | The IFAB, 2022-06-13; as the season before |
| *Germany* | | | | | | |
| 2015–16 | three | | 7 | 306 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2016–17 | three | | 7 | 306 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2017–18 | three | | 7 | 306 | 100% | The IFAB, 2017-06-01; Laws of the season |
| 2018–19 | three | | 7 | 306 | 100% | The IFAB, 2018-06-01; Laws of the season |
| 2019–20 | three, then five | 2020-05-16 | 9 | 306 | 100% | Sport1, 2020-05-14; names the change |
| 2020–21 | five | | 9 | 306 | 99% | Sport1, 2020-09-03; names the league |
| 2021–22 | five | | 9 | 306 | 92% | Malay Mail, 2021-07-15; names the league |
| 2022–23 | five | | 9 | 306 | 94% | The IFAB, 2022-06-13; IFAB's permanent option |
| 2023–24 | five | | 9 | 306 | 100% | The IFAB, 2022-06-13; as the season before |
| 2024–25 | five | | 9 | 306 | 100% | The IFAB, 2022-06-13; as the season before |
| 2025–26 | five | | 9 | 306 | 100% | The IFAB, 2022-06-13; as the season before |
| *Greece* | | | | | | |
| 2015–16 | three | | 7 | 239 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2016–17 | three | | 7 | 240 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2017–18 | three | | 7 | 239 | 100% | The IFAB, 2017-06-01; Laws of the season |
| 2018–19 | three | | 7 | 240 | 100% | The IFAB, 2018-06-01; Laws of the season |
| 2019–20 | three | | 7 | 182 | 100% | The IFAB, 2018-06-01; Laws 2018/19 and 2019/20 |
| 2020–21 | five | | 9 | 182 | 87% | The Jakarta Post, 2020-11-17; general report |
| 2021–22 | five | | 9 | 182 | 76% | The IFAB, 2022-06-13; IFAB's temporary option |
| 2022–23 | five | | 9 | 182 | 100% | The IFAB, 2022-06-13; IFAB's permanent option |

| Season | Limit | Five from | Named | Matches | Kept | Document behind the limit |
|---|---|---|---|---|---|---|
| 2023–24 | five | | 9 | 182 | 100% | The IFAB, 2022-06-13; as the season before |
| 2024–25 | five | | 9 | 182 | 100% | The IFAB, 2022-06-13; as the season before |
| 2025–26 | five | | 12 | 182 | 100% | The IFAB, 2022-06-13; as the season before |
| *Italy* | | | | | | |
| 2015–16 | three | | 12 | 380 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2016–17 | three | | 12 | 380 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2017–18 | three | | 12 | 380 | 100% | The IFAB, 2017-06-01; Laws of the season |
| 2018–19 | three | | 12 | 380 | 100% | The IFAB, 2018-06-01; Laws of the season |
| 2019–20 | three, then five | 2020-06-20 | 12 | 380 | 100% | Fantacalcio, 2020-06-06; names the change |
| 2020–21 | five | | 12 | 380 | 97% | Sky Sport, 2020-09-09; names the league |
| 2021–22 | five | | 12 | 380 | 90% | Sky Sport, 2021-07-30; names the league |
| 2022–23 | five | | 12 | 380 | 100% | The IFAB, 2022-06-13; IFAB's permanent option |
| 2023–24 | five | | 12 | 380 | 100% | The IFAB, 2022-06-13; as the season before |
| 2024–25 | five | | 12 | 380 | 100% | The IFAB, 2022-06-13; as the season before |
| 2025–26 | five | | 12 | 380 | 100% | The IFAB, 2022-06-13; as the season before |
| *Netherlands* | | | | | | |
| 2015–16 | three | | 7 | 306 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2016–17 | three | | 9 | 306 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2017–18 | three | | 9 | 306 | 100% | The IFAB, 2017-06-01; Laws of the season |
| 2018–19 | three | | 9 | 306 | 100% | The IFAB, 2018-06-01; Laws of the season |
| 2019–20 | three | | 10 | 232 | 100% | The IFAB, 2018-06-01; Laws 2018/19 and 2019/20 |
| 2020–21 | five | | 10 | 306 | 96% | Eredivisie CV, 2020-07-30; names the league |
| 2021–22 | five | | 12 | 306 | 80% | KNVB, 2021-07-06; names the league |
| 2022–23 | five | | 12 | 306 | 100% | The IFAB, 2022-06-13; IFAB's permanent option |
| 2023–24 | five | | 12 | 306 | 100% | The IFAB, 2022-06-13; as the season before |
| 2024–25 | five | | 12 | 306 | 100% | The IFAB, 2022-06-13; as the season before |
| 2025–26 | five | | 12 | 306 | 100% | The IFAB, 2022-06-13; as the season before |
| *Portugal* | | | | | | |
| 2015–16 | three | | 7 | 306 | 70% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2016–17 | three | | 7 | 306 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2017–18 | three | | 7 | 306 | 100% | The IFAB, 2017-06-01; Laws of the season |
| 2018–19 | three | | 7 | 306 | 95% | The IFAB, 2018-06-01; Laws of the season |
| 2019–20 | three, then five | 2020-06-09 | 7 | 306 | 98% | Rádio Renascença, 2020-06-08; names the change |
| 2020–21 | five | | 9 | 306 | 100% | The Jakarta Post, 2020-11-17; general report |
| 2021–22 | five | | 9 | 306 | 67% | The IFAB, 2022-06-13; IFAB's temporary option |
| 2022–23 | five | | 9 | 306 | 100% | The IFAB, 2022-06-13; IFAB's permanent option |

| Season | Limit | Five from | Named | Matches | Kept | Document behind the limit |
|---|---|---|---|---|---|---|
| 2023–24 | five | | 9 | 306 | 96% | The IFAB, 2022-06-13; as the season before |
| 2024–25 | five | | 9 | 306 | 100% | The IFAB, 2022-06-13; as the season before |
| 2025–26 | five | | 9 | 306 | 94% | The IFAB, 2022-06-13; as the season before |
| *Russia* | | | | | | |
| 2015–16 | three | | 9 | 240 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2016–17 | three | | 12 | 240 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2017–18 | three | | 10 | 240 | 100% | The IFAB, 2017-06-01; Laws of the season |
| 2018–19 | three | | 12 | 240 | 97% | The IFAB, 2018-06-01; Laws of the season |
| 2019–20 | three, then five | 2020-06-19 | 10 | 236 | 100% | Sport-Express, 2020-05-15; names the change |
| 2020–21 | five | | 12 | 238 | 100% | The Jakarta Post, 2020-11-17; general report |
| 2021–22 | five | | 12 | 240 | 54% | Sport-Express, 2021-12-04; names the league |
| 2022–23 | five | | 12 | 240 | 100% | The IFAB, 2022-06-13; IFAB's permanent option |
| 2023–24 | five | | 12 | 240 | 100% | The IFAB, 2022-06-13; as the season before |
| 2024–25 | five | | 12 | 240 | 100% | The IFAB, 2022-06-13; as the season before |
| 2025–26 | five | | 12 | 240 | 100% | The IFAB, 2022-06-13; as the season before |
| *Scotland* | | | | | | |
| 2015–16 | three | | 7 | 228 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2016–17 | three | | 7 | 228 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2017–18 | three | | 7 | 198 | 100% | The IFAB, 2017-06-01; Laws of the season |
| 2018–19 | three | | 7 | 198 | 98% | The IFAB, 2018-06-01; Laws of the season |
| 2019–20 | three | | 7 | 179 | 100% | The IFAB, 2018-06-01; Laws 2018/19 and 2019/20 |
| 2020–21 | five | | 9 | 198 | 88% | SPFL, 2020-07-30; names the league |
| 2021–22 | three, then five | 2022-01-17 | 7 | 198 | 38% | SPFL, 2022-01-13; names the change |
| 2022–23 | five | | 9 | 198 | 92% | The IFAB, 2022-06-13; IFAB's permanent option |
| 2023–24 | five | | 9 | 198 | 100% | The IFAB, 2022-06-13; as the season before |
| 2024–25 | five | | 9 | 198 | 100% | The IFAB, 2022-06-13; as the season before |
| 2025–26 | five | | 9 | 198 | 100% | The IFAB, 2022-06-13; as the season before |
| *Spain* | | | | | | |
| 2015–16 | three | | 7 | 380 | 97% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2016–17 | three | | 7 | 380 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2017–18 | three | | 7 | 380 | 100% | The IFAB, 2017-06-01; Laws of the season |
| 2018–19 | three | | 7 | 380 | 98% | The IFAB, 2018-06-01; Laws of the season |
| 2019–20 | three, then five | 2020-06-11 | 7 | 380 | 96% | Goal, 2020-05-08; names the change |
| 2020–21 | five | | 12 | 380 | 78% | Football España, 2020-09-08; names the league |
| 2021–22 | five | | 12 | 380 | 90% | Goal, 2021-10-02; names the league |
| 2022–23 | five | | 12 | 380 | 100% | The IFAB, 2022-06-13; IFAB's permanent option |

| Season | Limit | Five from | Named | Matches | Kept | Document behind the limit |
|---|---|---|---|---|---|---|
| 2023–24 | five | | 12 | 380 | 100% | The IFAB, 2022-06-13; as the season before |
| 2024–25 | five | | 12 | 380 | 100% | The IFAB, 2022-06-13; as the season before |
| 2025–26 | five | | 12 | 380 | 100% | The IFAB, 2022-06-13; as the season before |
| *Turkey* | | | | | | |
| 2015–16 | three | | 7 | 306 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2016–17 | three | | 7 | 306 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2017–18 | three | | 10 | 306 | 100% | The IFAB, 2017-06-01; Laws of the season |
| 2018–19 | three | | 10 | 306 | 100% | The IFAB, 2018-06-01; Laws of the season |
| 2019–20 | three, then five | 2020-06-12 | 10 | 306 | 100% | A Spor, 2020-05-29; names the change |
| 2020–21 | five | | 10 | 420 | 100% | The Jakarta Post, 2020-11-17; general report |
| 2021–22 | five | | 10 | 380 | 45% | The IFAB, 2022-06-13; IFAB's temporary option |
| 2022–23 | five | | 10 | 313 | 100% | The IFAB, 2022-06-13; IFAB's permanent option |
| 2023–24 | five | | 10 | 380 | 100% | The IFAB, 2022-06-13; as the season before |
| 2024–25 | five | | 10 | 342 | 100% | The IFAB, 2022-06-13; as the season before |
| 2025–26 | five | | 10 | 306 | 100% | The IFAB, 2022-06-13; as the season before |
| *Ukraine* | | | | | | |
| 2015–16 | three | | 7 | 172 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2016–17 | three | | 7 | 192 | 100% | The IFAB, 2017-06-01; Laws 2017/18, carried back |
| 2017–18 | three | | 7 | 131 | 100% | The IFAB, 2017-06-01; Laws of the season |
| 2018–19 | three | | 7 | 132 | 100% | The IFAB, 2018-06-01; Laws of the season |
| 2019–20 | three | | 7 | 132 | 100% | The IFAB, 2018-06-01; Laws 2018/19 and 2019/20 |
| 2020–21 | five | | 9 | 181 | 100% | The Jakarta Post, 2020-11-17; general report |
| 2022–23 | five | | 12 | 240 | 89% | The IFAB, 2022-06-13; IFAB's permanent option |
| 2023–24 | five | | 12 | 240 | 100% | The IFAB, 2022-06-13; as the season before |
| 2024–25 | five | | 12 | 240 | 100% | The IFAB, 2022-06-13; as the season before |
| 2025–26 | five | | 12 | 239 | 100% | The IFAB, 2022-06-13; as the season before |

## S1.3 Completeness

The record check leaves out a team-match whose sheet lists a starter with no appearance on record. Table S1 gives the share of team-matches kept in each league-season. In fifteen of the 153 league-seasons, the check leaves out more than a tenth of the team-matches. Of these league-seasons, ten are in 2021–22. The lowest share kept was 38 per cent, in Scotland's 2021–22. The team-matches left out made about as many substitutions as those kept. For each league's 2021–22, two means show this. One takes the match events of every team-match, kept or not. The other takes the corrected counts of the team-matches kept (Section S1.4). The two lie within 0.08 substitutions a match of each other.

Table S2: The leagues of the split record (Section S4), and cricket's leagues. The leagues are listed by sport with the seasons and games on record. Cricket's T20 leagues, which play twenty overs a side, are not in the split record. They are listed with the full-length innings of the seasons used to set their order. Three cricket competitions, marked described, have other formats and enter no test.

| League | Seasons | Games |
|---|---|---|
| *Baseball* | | |
| Taiwan, CPBL | 2018 to 2025 | 2,340 |
| South Korea, KBO | 2019 to 2025 | 5,583 |
| Mexico, LMB | 2017 to 2025 | 6,649 |
| MLB | 2008 to 2025 | 42,201 |
| Japan, NPB | 2018 to 2025 | 5,578 |
| *Basketball* | | |
| ABA League | 2006–07 to 2018–19 | 2,343 |
| Spain, Liga ACB | 2006–07 to 2022–23 | 5,438 |
| Turkey, BSL | 2006–07 to 2018–19 | 3,374 |
| Greece, Basket League | 2006–07 to 2018–19 | 1,633 |
| South Korea, KBL | 2006–07 to 2018–19 | 3,781 |
| Italy, LBA | 2006–07 to 2018–19 | 3,649 |
| Lithuania, LKL | 2006–07 to 2018–19 | 1,684 |
| Brazil, NBB | 2013–14 to 2019–20 | 1,453 |
| Israel, Winner League | 2006–07 to 2018–19 | 2,389 |
| *Handball* | | |
| Spain, ASOBAL | 2014–15 to 2019–20 | 1,345 |
| Denmark | 2014–15 to 2019–20 | 1,256 |
| Hungary | 2010–11 to 2019–20 | 1,773 |
| France, LNH | 2011–12 to 2019–20 | 1,538 |
| Poland | 2009–10 to 2014–15 | 872 |
| *Ice hockey* | | |
| Germany, DEL | 2011–12 to 2018–19 | 2,912 |
| Austria, ICE Hockey League | 2011–12 to 2018–19 | 2,528 |
| Finland, Liiga | 2011–12 to 2018–19 | 3,477 |
| Switzerland, National League | 2011–12 to 2018–19 | 2,876 |
| Sweden, SHL | 2011–12 to 2018–19 | 2,776 |
| *Volleyball* | | |
| France, men | 2012–13 to 2025–26 | 2,615 |
| Germany, men | 2009–10 to 2025–26 | 2,355 |
| Italy, men | 2008–09 to 2025–26 | 3,411 |
| Italy, women | 2008–09 to 2025–26 | 3,046 |
| Japan, men | 2006–07 to 2025–26 | 2,798 |
| Japan, women | 2006–07 to 2025–26 | 3,073 |
| Turkey, men | 2013–14 to 2023–24 | 1,991 |
| Turkey, women | 2013–14 to 2023–24 | 2,008 |
| *Cricket, full-length innings of the seasons opening 2019 to 2022* | | |
| BBL | 4 seasons | 290 |
| BPL | 3 seasons | 149 |
| CPL | 4 seasons | 152 |
| CSA T20 | 3 seasons | 89 |
| Hundred (men) (described) | 2 seasons | 81 |
| Hundred (women) (described) | 2 seasons | 74 |
| ILT20 | 1 season | 38 |
| IPL | 4 seasons | 345 |
| LPL | 3 seasons | 77 |
| PSL | 4 seasons | 160 |
| RHF Trophy (described) | 3 seasons | 62 |
| SA20 | 1 season | 36 |
| Super Smash (men) | 4 seasons | 139 |
| Super Smash (women) | 2 seasons | 70 |
| WBBL | 4 seasons | 260 |
| WPL | 1 season | 22 |

### S1.4 Counts corrected from the match events

In 2021–22 the appearance records miss some incoming substitutes in every league. They miss 2.5 to 12.9 per cent of the substitutes that the match events name. So the appearances fall short of the events by 0.07 to 0.50 substitutions a match. The record check looks only for the listed starters, so it cannot catch this.

Each team-match is therefore counted from the match events where the two records disagree, and from the appearances elsewhere. Three findings support the fix. First, outside 2018–19 the events fell short of the appearances in only two team-matches. So the events almost never miss a substitution that the appearances record. Second, 2018–19 had another defect. It ran both ways: the events counted more substitutions than the appearances in 379 team-matches, and fewer in 174. In each of these 174, the appearances list a player who is on neither list of the team sheet. So there the error is in the appearances. Third, the events disagreed with StatsBomb in only 0.8 to 2.9 per cent of team-matches (Section S1.5).

The fix changes 1,922 of the 84,229 team-matches. Of these, 1,045 are in 2021–22, 553 in 2018–19 and the rest in the other seasons. The fix raises each league's 2021–22 mean by 0.07 to 0.50 substitutions a match. For example, England's 2021–22 mean rises from 2.72 to 2.80. The fix moves no other season by more than 0.09. We call the counts after the fix the corrected counts. Every count in this supplement reads the corrected counts, with each season under the limit its regulations set. The exceptions are pre-specified tests reported as they were first run.

### S1.5 The check against StatsBomb

Earlier studies counted 2.86 to 2.9 substitutions a team-match under three, and 4.2 in LaLiga's first full season under five [2, 3, 4].

StatsBomb, a data company, publishes free match data. These open data give an independent record of some of the same matches [5]. They cover five of the fourteen leagues. We paired all 1,815 of their matches in those leagues' seasons with the source's, by league, score, date and team names. Table S3 compares the two records team-match by team-match. Over all team-matches, Spain's counts disagreed most often. That comes from team-matches whose listed starters have no appearance on record. The record check leaves these out. On record-checked team-matches the counts disagreed in 0.8 to 2.9 per cent. The minute agreed to within one minute for 99 per cent of substitutions.

The check reaches both ends of the order: the leagues' ranking by substitutions per team-match. In 2025–26, England is the lowest league and Spain the highest. Italy was among the highest under three. But the sample is not random. It holds full seasons of 2015–16 in four leagues, and otherwise one club's matches in a season. No independent record we hold covers the other leagues.

The source writes a substitution made in second-half added time at the ninetieth minute. Of the substitutions StatsBomb times in that added time, the source wrote 99 per cent at the ninetieth minute. All eleven made in first-half added time were written at the forty-fifth minute. Substitutions at half-time were written at the second half's first minute in 95 to 100 per cent of cases. Of the source's ninetieth-minute substitutions, 59 to 73 per cent were made in added time. Across all league-seasons, 4.9 per cent of substitutions were written at the ninetieth minute under three, and 5.7 per cent under five. The timing measures use the minute as written. So a starter taken off in added time counts as taken off at the ninetieth minute.

### S1.6 Edge cases

The source lists 52 games with no player on record. Of them, 50 carry an awarded score. Of these, 29 are in Turkey's 2022–23, after clubs withdrew following the earthquake. No game without a player on

Table S3: The source's counts and minutes against StatsBomb's open data. A count disagrees when StatsBomb and the source give a team-match different numbers of substitutions. The column All covers every team-match. Checked covers the record-checked team-matches, whose every listed starter has an appearance on record. Corrected takes the match events where the source's appearances and events disagree. The minutes compare substitutions paired by player. Same is the share timed at the same minute, and Within one the share within one minute. Added time is the share of substitutions in second-half added time that the source writes at the ninetieth minute.

| | | | Counts disagree (%) | | | Minutes (%) | | |
|---|---|---|---|---|---|---|---|---|
| League | Matches read | Team-matches | All | Checked | Corrected | Same | Within one | Added time |
| England | full season, 2015–16 | 760 | 0.8 | 0.8 | 0.8 | 90 | 99.8 | 99 |
| Italy | full season, 2015–16 | 760 | 1.7 | 1.7 | 1.4 | 84 | 98.9 | 100 |
| France | full season, 2015–16; one club, 2021–22 and 2022–23 | 849 | 2.2 | 1.8 | 1.3 | 84 | 99.4 | 98 |
| Spain | full season, 2015–16; one club, 2016–17 to 2020–21 | 1,104 | 8.2 | 2.5 | 1.1 | 84 | 98.6 | 100 |
| Germany | one club, 2015–16 and 2023–24 | 136 | 2.9 | 2.9 | 2.9 | 84 | 99.2 | 100 |

record enters any count. The source does not mark a match that was abandoned and then completed or replayed. So such a match cannot be told apart from the others.

The source does not mark a concussion substitute. So a concussion substitute counts as an ordinary substitution, and it can take a team beyond the limit. Under five, 201 team-matches made six substitutions or more. In each, the match events show as many substitutions as the appearances. None of them has a player who is missing from its team sheet. So these are real substitutions. Of them, 159 fall in 2024–25 and 2025–26, after IFAB's option of permanent concussion substitutes took effect in July 2024.

Under three, 179 team-matches recorded four or more substitutions. 148 of them fall in 2018–19. In each of these, the events show three at most, and the appearances list a player on neither list of the team sheet. So these are errors in the appearance records. England's fourteen came in 2020–21 and 2021–22, after its trial of concussion substitutes began. Belgium's twelve came in December 2020, before its announced change. The rest fall in Italy's 2016–17, Portugal's 2017–18 and Ukraine's 2017–18.

### S1.7 The other sports' records

The timing measures also read women's leagues (Section S7). The women's records come from the appearance tables of Soccerdonna, a site on women's football. Their minutes come from FBref, a football statistics site, and are matched to those tables. Each competition says whether it is for men or for women. So every contrast here between men's and women's football is a contrast between competitions. The club-wealth and manager tests are men's only, because the women's records carry no club column.

The records of basketball, volleyball, handball, ice hockey and baseball come from the leagues' and federations' own sites (Table S2). The first Japanese baseball records missed every game of one club, because the program that collected them did not follow that club's identifier. A prediction written down later counts Japan's league again on all its clubs (Section S4.4). Cricket's records come from Cricsheet, a public archive of ball-by-ball records [6]. A cricket innings, one side's turn to bat, counts only when it ran its full length. So no side was bowled out, no chase ended early and no rain shortened it. Cricket's seasons are named by the year they open. Every source is public and free to read. Nothing in the result tables names a player.

Table S4: The leagues each analysis reads. Football's leagues are named by their country, and the other sports' by their country and league.

| Analysis | Sport | Leagues | Which |
|---|---|---|---|
| Figure 1's leagues | football | 14 | Belgium; Denmark; England; France; Germany; Greece; Italy; Netherlands; Portugal; Russia; Scotland; Spain; Turkey; Ukraine |
| order test (C26) | football | 12 | Belgium; France; Germany; Greece; Italy; Netherlands; Portugal; Russia; Scotland; Spain; Turkey; Ukraine |
| increment | football | 17 | Belgium; Brazil; Denmark; England; France; Germany; Greece; Italy; Netherlands; Portugal; Russia; Scotland; South Korea; Spain; Turkey; Ukraine; United States |
| timing measures | football | 17 | Argentina; Australia; Brazil; Denmark; England; France; Germany; Greece; Italy; Japan; Mexico; Netherlands; Portugal; South Korea; Spain; Turkey; United States |
| bench-value test (C29) | football | 14 | Belgium; Denmark; England; France; Germany; Greece; Italy; Netherlands; Portugal; Russia; Scotland; Spain; Turkey; Ukraine |
| club wealth (C27) | football | 12 | Belgium; Denmark; England; France; Germany; Italy; Netherlands; Portugal; Russia; Spain; Turkey; Ukraine |
| rotation (C28) | football | 13 | Belgium; England; France; Germany; Greece; Italy; Netherlands; Portugal; Russia; Scotland; Spain; Turkey; Ukraine |
| absences | football | 13 | Belgium; England; France; Germany; Greece; Italy; Netherlands; Portugal; Russia; Scotland; Spain; Turkey; Ukraine |
| explanation models (C98 to C100) | football | 31 | Argentina; Australia; Austria; Belgium; Brazil; Croatia; Czechia; Denmark; England; France; Germany; Greece; Italy; Japan; Mexico; Netherlands; Norway; Poland; Portugal; Romania; Russia; Saudi Arabia; Scotland; Serbia; South Korea; Spain; Sweden; Switzerland; Turkey; Ukraine; United States |
| European timing tests (C275, C277, C280) | football | 8 | England; France; Germany; Italy; Netherlands; Portugal; Spain; Turkey |
| the withdrawn law's twelve (C276, C278, C279) | football | 12 | Argentina; Brazil; England; France; Germany; Italy; Japan; Mexico; Netherlands; Portugal; Spain; Turkey |
| women's leagues | football, women | 7 | England, FA Women's Super League; Germany, Frauen-Bundesliga; Italy, Serie A Femminile; Norway, Toppserien; Spain, Primera División Femenina; Sweden, Damallsvenskan; Switzerland, Women's Super League |
| R01 R15: basketball, the line's six | basketball | 6 | ABA League; Greece, Basket League; Israel, Winner League; Italy, LBA; Lithuania, LKL; Spain, Liga ACB |
| R02 R14: basketball, pace's seven | basketball | 7 | ABA League; Greece, Basket League; Israel, Winner League; Italy, LBA; Lithuania, LKL; South Korea, KBL; Spain, Liga ACB |
| R03 R12: basketball, minutes' eight | basketball | 8 | ABA League; Greece, Basket League; Israel, Winner League; Italy, LBA; Lithuania, LKL; South Korea, KBL; Spain, Liga ACB; Turkey, BSL |
| R13: basketball, the line's seven | basketball | 7 | ABA League; Greece, Basket League; Israel, Winner League; Italy, LBA; Lithuania, LKL; South Korea, KBL; Spain, Liga ACB |
| R16 R17: basketball, coaches who move | basketball | 3 | Brazil, NBB; Israel, Winner League; Spain, Liga ACB |

| Analysis | Sport | Leagues | Which |
|---|---|---|---|
| R04 R18: handball, the third time-out | handball | 1 | France, LNH |
| R05 R26: volleyball, the sheets that grew | volleyball | 5 | France, men; Japan, men; Japan, women; Turkey, men; Turkey, women |
| R06 R19 R20 R21: volleyball, the eight | volleyball | 8 | France, men; Germany, men; Italy, men; Italy, women; Japan, men; Japan, women; Turkey, men; Turkey, women |
| R07: ice hockey, three-on-three overtime | ice hockey | 5 | Finland, Liiga; Germany, DEL; ICE Hockey League; Sweden, SHL; Switzerland, National League |
| R08 R22 R23: ice hockey, the goalie pull | ice hockey | 4 | Finland, Liiga; ICE Hockey League; Sweden, SHL; Switzerland, National League |
| R09 R10 R11 R24: baseball, the pitch clock | baseball | 4 | MLB; Mexico, LMB; South Korea, KBO; Taiwan, CPBL |
| R25: baseball, MLB's drift | baseball | 1 | MLB |
| G01: basketball, the line in new competitions | basketball | 2 | EuroCup; Germany, BBL |
| G02: ice hockey, overtime in a new league | ice hockey | 1 | Sweden, HockeyAllsvenskan |
| G03 to G06: basketball, standings in new seasons | basketball | 7 | ABA League; Greece, Basket League; Israel, Winner League; Italy, LBA; Lithuania, LKL; South Korea, KBL; Spain, Liga ACB |
| G07: basketball, moves in new seasons | basketball | 3 | Brazil, NBB; Israel, Winner League; Spain, Liga ACB |
| G08: ice hockey, the order over six | ice hockey | 6 | Finland, Liiga; ICE Hockey League; Sweden, HockeyAllsvenskan; Sweden, SHL; Switzerland, National League; Switzerland, Swiss League |
| G09: persistence on new seasons | ice hockey; basketball | 13 | ABA League; Finland, Liiga; Greece, Basket League; ICE Hockey League; Israel, Winner League; Italy, LBA; Lithuania, LKL; South Korea, KBL; Spain, Liga ACB; Sweden, HockeyAllsvenskan; Sweden, SHL; Switzerland, National League; Switzerland, Swiss League |
| G10: basketball, the line's proportion | basketball | 6 | EuroLeague; Poland, PLK; Portugal, LPB; Portugal, Liga Feminina; Spain, LEB Oro; Spain, Liga Femenina |
| G11 G11a: baseball, the order over eight | baseball | 8 | Dominican Republic, LIDOM; Japan, NPB; MLB; Mexico, LMB; Mexico, LMP; South Korea, KBO; Taiwan, CPBL; Venezuela, LVBP |
| G12: baseball, the first pitch clocks | baseball | 6 | Dominican Republic, LIDOM; Double-A; High-A; Single-A (not California); Triple-A; Venezuela, LVBP |
| G13 G14: cricket, thirteen T20 leagues | cricket | 13 | BBL; BPL; CPL; CSA T20; ILT20; IPL; LPL; PSL; SA20; Super Smash (men); Super Smash (women); WBBL; WPL |
| G15: ice hockey, how minutes are shared | ice hockey | 8 | Czechia, Extraliga; Finland, Liiga; Finland, Mestis; Germany, DEL; NHL; Sweden, SHL; Switzerland, National League; Switzerland, Swiss League |
| G16: basketball beyond the fit | basketball | 5 | France, Betclic ELITE; G League; Japan, B.LEAGUE; NBA; WNBA |
| persistence fit (stage 290): basketball, volleyball, ice hockey, baseball | basketball; volleyball; ice hockey; baseball | 25 | ABA League; Finland, Liiga; France, men; Germany, men; Greece, Basket League; ICE Hockey League; Israel, Winner League; Italy, LBA; Italy, men; Italy, women; Japan, men; Japan, women; Lithuania, LKL; MLB; Mexico, LMB; South Korea, KBL; South Korea, KBO; Spain, Liga ACB; Sweden, HockeyAllsvenskan; Sweden, SHL; Switzerland, National League; Switzerland, Swiss League; Taiwan, CPBL; Turkey, men; Turkey, women |

### S1.8 What each test could detect

With few leagues, a test can detect only a strong link. Table S5 shows how strong a link each group of tests could detect. It gives each group's number of leagues or moves, and whether its tests are one-sided or two-sided. Every value assumes a test that passes when its $P$ is below $\alpha = 0.05$. The table gives the rank correlation needed to pass. Up to nine leagues, this pass mark comes from counting every possible order of the leagues. Beyond that, it comes from 125,000 to 400,000 random orders. For example, with seven leagues a rank correlation of 0.71 is needed to pass. Power is a test's chance of passing when the true rank correlation has a given size. The table gives the power at a true rank correlation of 0.7. It also gives the true rank correlation that a test detects with a power of 0.80. The power comes from 40,000 samples drawn by computer at each size. The computations used Python 3.12.10, numpy 2.3.5 and SciPy 1.16.3.

Two formulas are in common use to find the detectable correlation without simulation. Both use Fisher's $z$, a standard formula for correlations. They differ in its variance, a measure of its uncertainty. With the variance $1.06/(n-3)$, Fisher's $z$ agrees with the exact and simulated values to within 0.04. Here $n$ is the number of leagues or moves. The variance of Bonett and Wright grows with the correlation, so it gives a larger detectable correlation. For example, twelve leagues detect 0.69 by simulation and 0.73 by that variance. The table gives both, so that each reading can be checked. Every 95% interval for a rank correlation in the paper uses Fisher's $z$ with the variance $1.06/(n-3)$.

For a difference between two leagues, the smallest effect of interest is 0.2 substitutions per team-match. That is a tenth of the room the change added, or about 7.6 substitute appearances per club in a season of 38 matches. For an order, the smallest correlation of interest is 0.5 (main text, Methods). Only the tests of the coach's carry on 56 or more moves could detect it. The order and explanation tests detect only links well above it. Section S6.3 gives the difference from the other sports that each sport's persistence could show.

## S2 Football's order, spread and increments

### S2.1 The order before the change

A league's pre-change mean is its mean substitutions per team-match over 2015–16 to 2018–19, each season weighted by its matches. Table S6 sets it against the leagues' later levels and increments, in three sets of leagues. The first is the fourteen European leagues of the main text. The second is the 12 leagues of the pre-specified order test: Belgium, France, Germany, Greece, Italy, the Netherlands, Portugal, Russia, Scotland, Spain, Turkey and Ukraine. The third is the twelve leagues that opened 2020–21 under five. For them, the first season under five is 2020–21. For the other leagues it is a later season.

Denmark and England are outside the test. The test set a floor of 150 matches a season in advance, and each Danish season in its records held only 132 matches. The test opened on 2020–21, and England played that season under three.

England's use of its three substitutions had climbed for two decades from 1996. Its teams' share of the limit used rose from 0.74 to 0.89. Under five, one analyst expected England to catch up with the other leagues [7]. By 2025–26 it had not: its teams made the fewest substitutions of the fourteen leagues.

The order under five was already there under three. Under three, one season's order predicted another's at a rank correlation of 0.80 to 0.87, over six pairs of seasons. The pre-change mean ranked the fourteen leagues much as their first season under five did (rank correlation 0.89, 95% interval 0.67 to 0.97). Against each separate season under five it correlated 0.60 to 0.87.

The increments relate to the level before the change more weakly than the levels do. Leagues that used more of the old limit tended to add more under the new one (rank correlation 0.56, 95% interval

Table S5: What each group of tests could detect, at $\alpha = 0.05$. The column $n$ gives the number of leagues or moves, and Sided says whether a test is one-sided or two-sided. To pass is the rank correlation a test needs to pass. Power is the chance of passing when the true rank correlation is 0.7. Detectable is the true rank correlation that a test detects with a power of 0.80, by exact counting and simulation. The last two columns give the same by Fisher's $z$. One uses the variance $1.06/(n-3)$, and the other Bonett and Wright's variance. Section S7 describes the timing tests.

| Tests | $n$ | Sided | To pass | Power | Detectable | Fisher's $z$ | Bonett–Wright |
|---|---|---|---|---|---|---|---|
| *Football* | | | | | | | |
| The order, the pre-specified test | 12 | one | 0.50 | 0.81 | 0.69 | 0.69 | 0.73 |
| Explanations, fourteen leagues | 14 | one | 0.46 | 0.87 | 0.65 | 0.65 | 0.68 |
| Explanations, the increment's leagues | 17 | one | 0.41 | 0.94 | 0.59 | 0.59 | 0.62 |
| Timing, the resting test | 8 | one | 0.64 | 0.60 | 0.82 | 0.82 | 0.86 |
| Timing, the coefficient before the change | 12 | one | 0.50 | 0.81 | 0.69 | 0.69 | 0.73 |
| *Orders in the other sports* | | | | | | | |
| Ice hockey and baseball, the smallest tests | 4 | one | 1.00 | 0.22 | 0.99 | 0.99 | 1.00 |
| Ice hockey | 6 | one | 0.83 | 0.32 | 0.94 | 0.90 | 0.94 |
| Basketball | 7 | one | 0.71 | 0.49 | 0.87 | 0.86 | 0.90 |
| Volleyball | 8 | one | 0.64 | 0.60 | 0.82 | 0.82 | 0.86 |
| Cricket | 13 | one | 0.48 | 0.85 | 0.67 | 0.67 | 0.71 |
| *The coach's carry* | | | | | | | |
| MLB's managers | 24 | two | 0.41 | 0.97 | 0.55 | 0.56 | 0.58 |
| Basketball's moves in new seasons | 56 | one | 0.22 | 1.00 | 0.33 | 0.34 | 0.34 |
| Football's managers who changed league | 99 | two | 0.20 | 1.00 | 0.28 | 0.29 | 0.28 |
| Basketball's moves on half B | 134 | one | 0.14 | 1.00 | 0.21 | 0.22 | 0.22 |
| Football's managers within one league | 192 | two | 0.14 | 1.00 | 0.20 | 0.21 | 0.20 |

0.03 to 0.85). The pre-change mean correlated 0.76 with the first season's increment and 0.34 with the latest season's. The increments kept an order of their own too, from the first season under five to the latest (rank correlation 0.69). The pre-specified test is about levels, and the levels kept their order (rank correlation 0.74 over the test's leagues).

When the order test was saved, the seasons under five of England, France, Germany, Italy and Spain had been read. The other nine leagues' seasons of 2020–21 and 2025–26 had not been downloaded. The test holds eight of these nine. On them, the order pointed the same way (rank correlation 0.57, 95% interval $-0.25$ to 0.91; exact one-sided $P = 0.076$). On the four test leagues that had been read, it was 0.60 ($P = 0.21$). The leagues not read would not have passed alone, so the result rests on all twelve. The ninth league not read is Denmark, which the test's floor of matches left out. With Denmark, the nine leagues not read give a rank correlation of 0.68 (0.01 to 0.93; $P = 0.025$).

## S2.2 The order under each reading of the rules and the records

The pre-specified test gave 0.73 over its 12 leagues (95% interval 0.28 to 0.92, one-sided $P = 0.0033$). The counts were corrected after the test was first run (Section S1.4). On the corrected counts, the same test gives 0.74 (0.27 to 0.93). The test read each league's first season under five, as the earlier reading assigned it, against 2025–26. That first season was 2020–21 for ten leagues and 2022–23 for Belgium and Scotland. The table below holds every league to 2020–21 instead, so none of its cells repeats 0.73. Its measure is the rank correlation between each league's level in 2020–21 and in 2025–26, both under five. Table S7 computes that rank correlation again, as description, under eight readings. The readings take the limit from the substitutions made, as the earlier reading did, or from the regulations. The

Table S6: The order before and after the change. Every value is description, read after the change. Each cell is a rank correlation with its 95% interval by Fisher's $z$. The first season under five is 2020–21 for the leagues that opened under five then, and a later season for the others.

| Correlation | Fourteen leagues | The test's twelve | Opened in 2020–21 |
|---|---|---|---|
| Pre-change mean, first season under five | 0.89 (0.67 to 0.97) | 0.90 (0.65 to 0.97) | 0.87 (0.57 to 0.96) |
| Pre-change mean, 2025–26 | 0.60 (0.08 to 0.86) | 0.59 (0.00 to 0.87) | 0.54 (−0.07 to 0.86) |
| Pre-change mean, mean under five | 0.82 (0.51 to 0.94) | 0.85 (0.52 to 0.96) | 0.78 (0.36 to 0.94) |
| Pre-change mean, increment | 0.56 (0.03 to 0.85) | 0.52 (−0.09 to 0.85) | 0.48 (−0.15 to 0.83) |
| Pre-change mean, first season's increment | 0.76 (0.37 to 0.92) | 0.78 (0.35 to 0.94) | 0.69 (0.16 to 0.91) |
| Pre-change mean, latest season's increment | 0.34 (−0.25 to 0.74) | 0.32 (−0.33 to 0.76) | 0.19 (−0.45 to 0.70) |
| Increments, first season against latest | 0.69 (0.24 to 0.90) | 0.69 (0.18 to 0.91) | 0.71 (0.20 to 0.91) |
| Levels, first season against latest | 0.75 (0.36 to 0.92) | 0.74 (0.27 to 0.93) | 0.77 (0.33 to 0.93) |

earlier reading drops Belgium and Scotland, because their seasons of 2020–21 showed no single limit. The readings take seasons whole or split at the change. They keep or drop the league-seasons that lose more than a tenth of their team-matches. They count from the appearances or from the corrected counts. Over the test's leagues the rank correlation ran from 0.71 to 0.81. Over every league with both seasons under five, Denmark included, it ran from 0.77 to 0.84.

Table S7: The measure of the pre-specified order test under each reading of the limits and the records. The measure is the rank correlation, $\rho$, between each league's level in 2020–21 and in 2025–26. Every value is description: the pre-specified test is not read again. Each cell gives $\rho$, its 95% interval and the number of leagues, $n$.

| | The test's leagues | | Every league with both seasons | |
|---|---|---|---|---|
| Reading | $\rho$ (95% interval) | $n$ | $\rho$ (95% interval) | $n$ |
| Limits from the substitutions made | 0.81 (0.34 to 0.95) | 10 | 0.84 (0.46 to 0.96) | 11 |
| the same, at most a tenth left out | 0.71 (−0.01 to 0.95) | 8 | 0.80 (0.27 to 0.96) | 9 |
| Regulations, whole seasons | 0.73 (0.21 to 0.93) | 11 | 0.78 (0.35 to 0.94) | 12 |
| Regulations, seasons split | 0.74 (0.27 to 0.93) | 12 | 0.79 (0.40 to 0.94) | 13 |
| the same, at most a tenth left out | 0.77 (0.19 to 0.95) | 9 | 0.83 (0.40 to 0.96) | 10 |
| Regulations, whole seasons, corrected counts | 0.73 (0.21 to 0.93) | 11 | 0.77 (0.33 to 0.93) | 12 |
| Regulations, seasons split, corrected counts | 0.74 (0.27 to 0.93) | 12 | 0.78 (0.39 to 0.93) | 13 |
| the same, at most a tenth left out | 0.77 (0.19 to 0.95) | 9 | 0.82 (0.37 to 0.96) | 10 |

## S2.3 The order on every scale

A ranking is the same on every scale that keeps the order of values. A scale that turns every season's order upside down keeps the rank correlation between seasons too. The scales used here are substitutions per team-match, the share of the limit used, log substitutions, the logit of that share and the room left unused. On the log scale, equal ratios are equal steps. The logit of a share is the log of the share used over the share left unused. The room left unused is the limit minus the substitutions made. It turns the order upside down. So the rank correlation between two single seasons is the same on all these scales. From the first season under five to the latest, it is 0.75 on each of these five scales. A league's mean over several seasons is taken after the scale is applied. So on the logit scale the pre-change mean

can rank a pair of leagues differently. Its correlations then move by up to 0.02.

How the increment relates to the level before the change depends on the scale. Over the fourteen leagues, the rank correlation between the two is 0.56 on substitutions and on the room left unused. It is 0.47 on log substitutions, 0.36 on the share of the limit and −0.79 on its logit. The logit stretches shares close to the limit. So on that scale, leagues close to the limit of three show small increments because of how the scale is built. The increments' own order also depends on the scale. From the first season under five to the latest, its rank correlation is 0.69 on substitutions. It is 0.30 on log substitutions, 0.30 on the share and 0.72 on the logit.

### S2.4 The distribution of substitutions per team-match

Under three, 87 per cent of team-matches used all three substitutions and 11 per cent used two (Figure S1). Under five, 55 per cent used all five, 27 per cent four and 13 per cent three. Counts above the limit are kept as recorded. They are 0.2 per cent of team-matches under three and 0.6 per cent under five (Section S1.6). By league, the share at the limit ran from 76 per cent (Scotland) to 95 per cent (Portugal) under three. Under five it ran from 39 per cent (Scotland) to 69 per cent (Italy). A limit binds when most team-matches use all of it. So the old limit bound in every league. The new one binds again at the top. In 2025–26, the highest league made 4.73 substitutions a match.

### S2.5 The spread between the leagues

Whether the leagues spread apart depends on the scale. The spread between the leagues is read on six scales (Table S8, Figure S2). Each era's value is the mean of its seasons' values. Its 95% interval comes from resampling leagues: drawing the leagues again at random, many times. In substitutions per team-match the spread grew by a factor of 2.63. That factor is the ratio of the spread under five to the spread under three. Its interval comes from resampling the same leagues in both eras at once. The coefficient of variation is the spread divided by the mean. It grew by a factor of 1.76. As a share of the limit, the spread grew by a factor of 1.58. But on the logit of the share, the spread fell to 0.60 times its size under three. The coefficient of variation of the room left unused fell to 0.57 times its size. So on these two scales the leagues grew more alike. Season by season, the spread in substitutions ran from 0.07 to 0.10 under three and from 0.15 to 0.30 under five. From 2022–23, all fourteen leagues played under five. Over those seasons, the spread was 2.34 times the spread under three (1.71 to 3.41).

We lead with substitutions per team-match, because playing minutes are shared out on that scale. Each substitution moves minutes from a starter to a substitute, whatever the limit. The order is a ranking, so its rank correlations are the same on every scale (Section S2.3).

Suppose each league had kept its share of the limit. Then its level under five would have been five thirds of its level under three. The spread under five would have been five thirds of the spread under three, about 0.15 substitutions a match. We call this the benchmark. The spread observed over 2020–21 to 2025–26 was 1.59 times the benchmark (95% interval 1.17 to 2.15). A league's mean over the era averages out its movement from season to season. On these means, the ratio was 1.42 (1.02 to 1.94). Season by season, the spread was 1.63 to 1.99 times the benchmark in each of the first four seasons under five. In 2024–25 it was 0.98 times the benchmark (0.68 to 1.49). In 2025–26 it was 1.26 times (0.81 to 1.95). So part of the leagues' spreading apart in the main text's first figure comes from the larger limit itself. The rest is new spread. It had narrowed by 2024–25.

### S2.6 Diffusion

Diffusion is one alternative to a kept order. Under diffusion, the leagues sit at different points on one common climb. Diffusion predicts that the gaps shrink as the leagues near the new limit. Over the leagues present in each season, the coefficient of variation ran from 3.3 to 7.3 per cent. But that set of

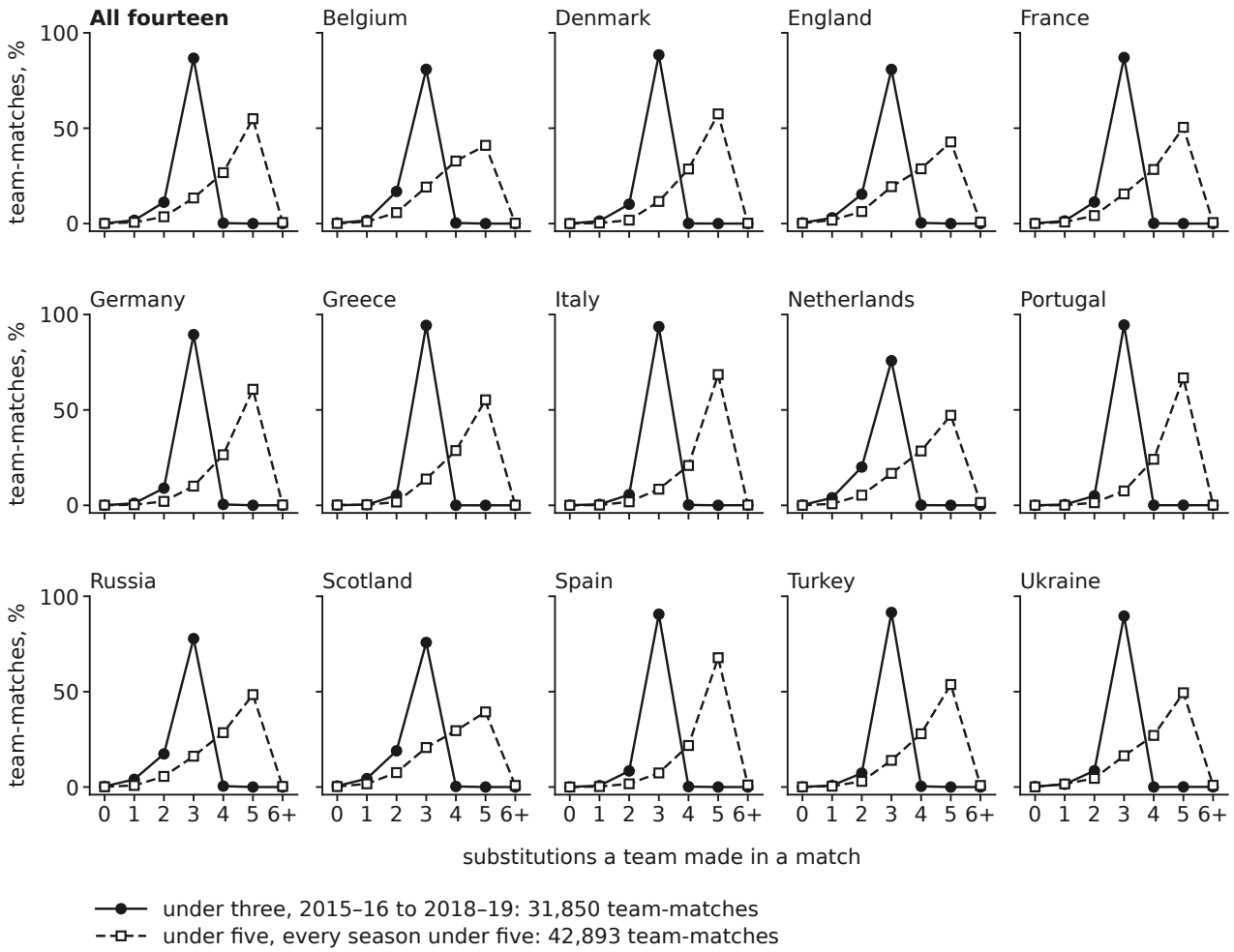



Figure S1: Under three the counts pile up at the limit. Under five they spread over three to five. Each panel gives the share of team-matches with each number of substitutions, for the fourteen leagues together and league by league. Filled circles are the era under three, 2015–16 to 2018–19. Open squares are the era under five, every season played under five.

Table S8: The spread between the fourteen leagues on six scales, with the mean share of the limit used. Under three is 2015–16 to 2018–19, and under five 2020–21 to 2025–26. Each value is the mean of the era's seasons, with a 95% interval from resampling leagues. Five over three is the ratio of the value under five to the value under three.

| Statistic | Under three | Under five | Five over three |
|---|---|---|---|
| Standard deviation, substitutions | 0.09 (0.06 to 0.10) | 0.23 (0.16 to 0.28) | 2.63 (1.97 to 3.43) |
| Coefficient of variation (%) | 3.1 (2.0 to 3.6) | 5.5 (3.8 to 6.6) | 1.76 (1.32 to 2.29) |
| Standard deviation, share of the limit (points) | 3.0 (1.9 to 3.4) | 4.7 (3.2 to 5.5) | 1.58 (1.18 to 2.06) |
| Standard deviation, log substitutions | 0.031 (0.020 to 0.036) | 0.056 (0.038 to 0.067) | 1.78 (1.32 to 2.32) |
| Standard deviation, logit of the share | 0.66 (0.48 to 0.75) | 0.39 (0.29 to 0.45) | 0.60 (0.47 to 0.75) |
| Coefficient of variation, room left unused (%) | 59 (42 to 70) | 34 (24 to 39) | 0.57 (0.42 to 0.74) |
| Mean share of the limit used | 0.950 (0.934 to 0.964) | 0.861 (0.839 to 0.881) | 0.91 (0.89 to 0.92) |

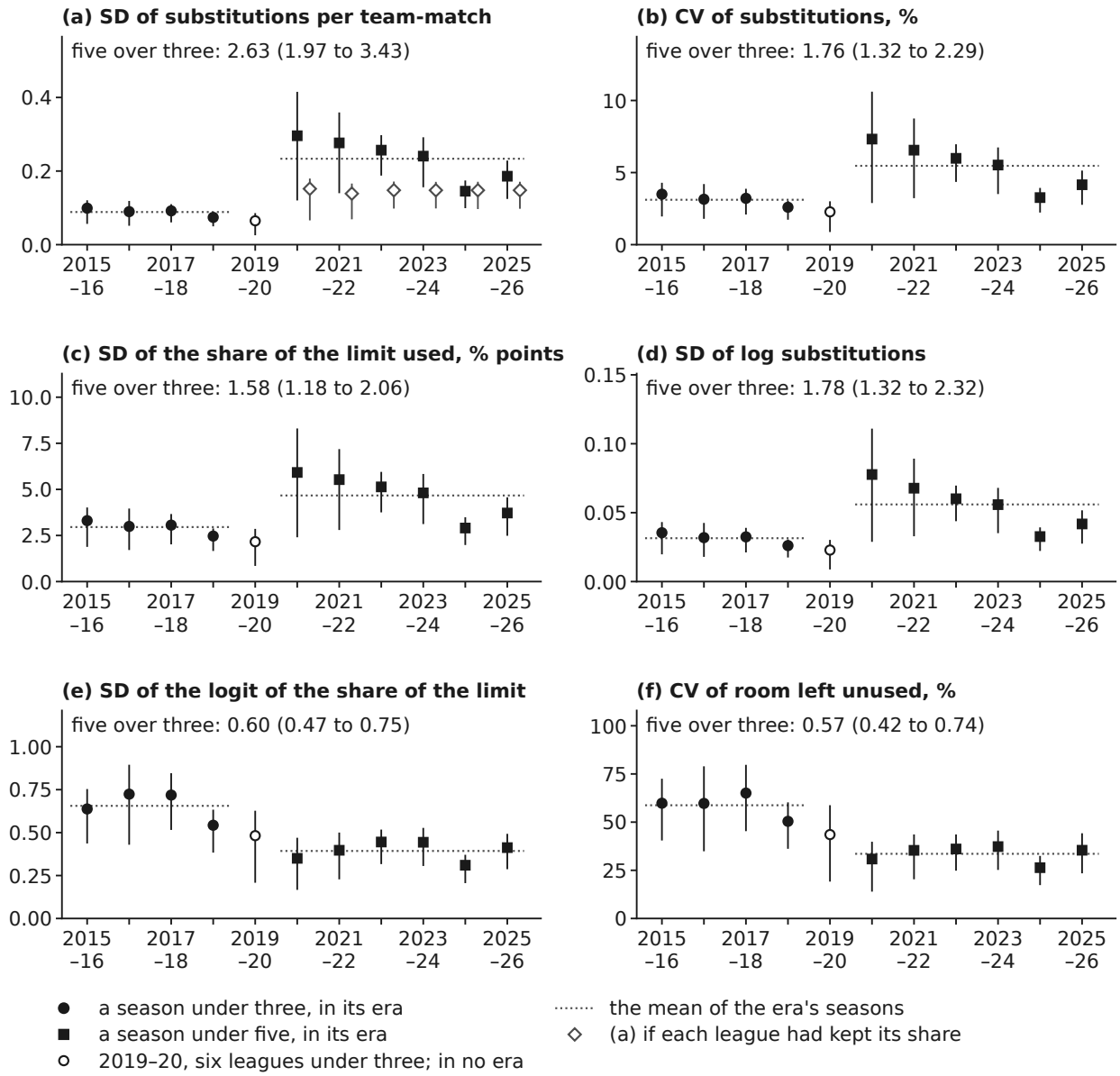



Figure S2: The spread between the leagues, season by season, on six scales. On the scale of substitutions the leagues spread apart under five. On the logit of the share and in the room left unused they drew together. The scales are (a) the standard deviation (SD) of substitutions per team-match, (b) its coefficient of variation (CV), (c) the SD of the share of the limit used, in percentage points, (d) the SD of log substitutions, (e) the SD of the logit of the share and (f) the CV of the room left unused. The CV is the SD divided by the mean. Circles are seasons under three and squares seasons under five, each with its 95% interval from resampling leagues. The open circle is 2019–20, which belongs to neither era. Each season's spread is taken over the leagues that played it under one limit. So 2019–20 to 2021–22 read fewer leagues. The seasons 2020–21 and 2021–22 also rest partly on league-seasons whose records were incomplete. Dotted lines give each era's mean. Each panel prints the ratio of the spread under five to the spread under three, with its 95% interval. Every panel starts at zero. The open diamonds in (a) give the benchmark with its intervals. The benchmark is the spread the leagues would show under five if each had kept its share of the limit. By 2024–25 the spread is near the benchmark (Section S2.5).

leagues changes from season to season, because Belgium, England and Scotland moved to five late. On a fixed set of leagues, the gaps shrank as the leagues neared the new limit. Over the fourteen leagues from 2022–23, the coefficient of variation fell from 6.0 to 4.2 per cent. That is a slope of −0.78 points a season (95% interval −1.21 to −0.30). A second fixed set holds the ten leagues under five in every season from 2020–21. Over it, the slope was −0.37 (−0.62 to 0.02). So the first set supports diffusion. The second points the same way, but its interval reaches zero.

The leagues that opened lowest also climbed most. A league's gain to 2025–26 correlated −0.75 with its level in 2022–23 (−0.92 to −0.35). Over twelve leagues, it correlated −0.49 with the level in 2020–21 (−0.84 to 0.14). But regression to the mean pushes these correlations down even without any narrowing. A league that sits low by chance in one season tends to sit nearer the middle later. The order held all the same (rank correlation 0.75 from the first season under five to the latest). So diffusion and a kept order can both hold.

### S2.7 The increments

A league's increment is its mean under five minus its own mean under three, each season weighted by its matches. The mean under three covers every season the league played under three. For a league that stayed under three through 2019–20, it includes that season. It also includes England's 2020–21 and 2021–22. The mean under five covers every season played under five. Brazil, the MLS and South Korea were collected separately from the same site. Each has one calendar-year season under each limit: 2019 and 2024. Their counts take only substitutes with recorded minutes. So a substitute who came on at the ninetieth minute or later is not counted. Table S9 gives each league's two means, its increment and the increment's standard error. The errors are clustered by team through both eras. Clustering allows for a team's matches being alike. The errors use the usual correction for few clusters: the variance is scaled by the number of teams over that number minus one. The median clustered error was 0.056. The design effect is the clustered variance divided by the variance that treats every team-match as separate. Its median was 6.7. So each team keeps its own way of substituting across seasons.

The plain mean of the 17 increments is 1.47 (95% interval 1.39 to 1.54). Judged by that mean, Scotland and Brazil, the leagues at the two ends, would be misjudged by about 0.22 and 0.24 substitutions per team-match. A random-effects model combines the increments instead. It gives more weight to the more precise increments. Its combined value was 1.47 (1.40 to 1.54). The model also gives $\tau$, the spread of the true increments around the combined value. It was 0.14 (0.09 to 0.22). $\tau$ is estimated by restricted maximum likelihood, with an interval from the Q-profile. Cochran's $Q$ tests whether the increments differ by more than chance. It was 136.2 on 16 degrees of freedom ($P = 4.0 \times 10^{-21}$). $I^2$ is the share of the differences between the increments that lies beyond chance. It was 0.88 (0.78 to 0.95). Over the fourteen European leagues alone $\tau$ was 0.13 (0.09 to 0.22).

The largest increment, Brazil's 1.70, was 1.37 times the smallest, Scotland's 1.25. The ratio's basic bootstrap interval, from resampling teams within each league, runs from 1.13 to 1.41. The difference between the two increments was 0.46 substitutions a match (0.25 to 0.50). Which leagues sit at the two ends changes from resample to resample. Brazil was the largest in 56 per cent of resamples, and Scotland the smallest in 41 per cent. Resampling pushes a ratio of the largest to the smallest upward. So the basic interval and the model's ratio are the better summaries. The model puts 95 per cent of the true increments within a range. The two ends of that range differ by a factor of 1.44 (1.29 to 1.82). Among the European leagues the largest increment was Spain's, 1.67.

Within the era under five, every league made more substitutions in 2025–26 than in its first season under five. But only six of the fourteen rose in every season: Belgium, Denmark, France, Germany, the Netherlands and Spain. The other eight fell at least once. The largest fall was Turkey's, 0.21 substitutions a match in 2023–24.

Table S9: Each league's mean substitutions per team-match under three and under five, its increment and the increment's standard error, clustered by team. Teams is the number of teams, the clusters of the error. Brazil, the MLS and South Korea were collected separately from the same site. The others come from the corrected counts.

| League | Under three | Under five | Increment | Standard error | Teams |
|---|---|---|---|---|---|
| Belgium | 2.80 | 4.06 | 1.25 | 0.043 | 25 |
| Brazil | 2.88 | 4.59 | 1.70 | 0.061 | 35 |
| Denmark | 2.87 | 4.42 | 1.54 | 0.059 | 20 |
| England | 2.76 | 4.06 | 1.30 | 0.072 | 34 |
| France | 2.86 | 4.25 | 1.39 | 0.041 | 33 |
| Germany | 2.89 | 4.46 | 1.56 | 0.040 | 29 |
| Greece | 2.94 | 4.36 | 1.42 | 0.054 | 27 |
| Italy | 2.94 | 4.56 | 1.62 | 0.030 | 34 |
| MLS | 2.63 | 3.97 | 1.34 | 0.071 | 30 |
| Netherlands | 2.72 | 4.20 | 1.47 | 0.052 | 28 |
| Portugal | 2.95 | 4.56 | 1.62 | 0.035 | 32 |
| Russia | 2.74 | 4.18 | 1.44 | 0.056 | 32 |
| Scotland | 2.73 | 3.97 | 1.25 | 0.074 | 17 |
| South Korea | 2.78 | 4.37 | 1.60 | 0.108 | 15 |
| Spain | 2.90 | 4.57 | 1.67 | 0.047 | 31 |
| Turkey | 2.91 | 4.34 | 1.43 | 0.059 | 39 |
| Ukraine | 2.89 | 4.21 | 1.32 | 0.060 | 36 |

# S3 The material explanations

## S3.1 How the candidates were tested

A material explanation of football's order rests on a club's money or its players. Table S10 gives every candidate explanation tested for this order, 51 tests in all. Figure S3 draws sixteen of them, league by league. Of the 51, 29 are pre-specified, one is a look and 21 are stated, not pre-specified. A look only describes. This one read the battery on a league's culture, a set of tests of one idea run together. As pre-specified, the battery read 20 competitions, six of them cups. With the cups taken out, 14 leagues were left. That is below the smallest number of leagues the battery allowed, so its test could not be run on them. The look read the 14 leagues anyway.

Most candidates are set against a league's increment or its level. A league's level is its mean substitutions per team-match under one limit. Each correlation is signed so that a positive value means that the larger the candidate, the larger the quantity it explains. For wealth, a positive value means that richer clubs or leagues use more of the limit. Each pre-specified test said in advance whether it expected a direction. Those that named a direction are one-sided, and the others are two-sided. A link in the other direction counts as no support for the explanation. For example, the test of bench value required a negative correlation with the increment (Section S3.3). Bench value correlated +0.27 instead. Every stated test is two-sided.

With several tests in a family, one may pass by luck. So Holm's correction sets a stricter pass mark for each test in the family. A family is a set of tests written down together, and there are eleven. They hold one test of club wealth, one of rotation and one of bench value. The battery on a league's culture holds three tests, and the battery on the level under five holds five. The two increment batteries hold five and three tests, and the absences nine. The stated tests of league wealth, competition rules and the calendar hold six, eight and four. A second, stricter correction treats all 46 tests of these families as one family. The look and the sensitivity rows of Table S10 belong to no family.

The written text of a pre-specified test is its clause. The two increment batteries and the absences have no clause of their own. Each was written into the description of the program that ran it, and that description was saved with the program's result. So these tests were written down only when they were

run. They are pre-specified only in that weaker sense.

The pre-specified candidates were tested before this paper's increment was built. Most of them explained an earlier form of it, which Table S10 calls the earlier increment. The tests of rotation, bench value, culture and absences used an increment by calendar era. It counted every season from 2020–21 as played under five, and it read every record. Its order of the leagues agrees with that of this paper's increment at a rank correlation of 0.82 (0.50 to 0.94). The two increment batteries used an increment from an earlier count of substitutions. Its order agrees with this paper's increment at 0.67. The battery on the level used the level under five minus a constant, so it tested the level and not the increment. That level's order agrees with the level read here at 0.85.

Within its family, only two candidates survived Holm's correction. Both are stated tests of one share: the share of team-matches in which teams had used all three substitution windows with substitutions left (Section S3.4). Three pre-specified candidates had a $P$ below 0.05 before Holm's correction. They were bookings per match in the battery on culture ($P = 0.048$), goals per match in the battery on the level ($P = 0.038$) and players fielded per team-match ($P = 0.018$). After Holm's correction within their families they had $P = 0.14$, 0.19 and 0.053. Players fielded per team-match are eleven plus the substitutions, so that measure restates the quantity it explains.

Table S10: Every candidate explanation of football's order. Each row gives the candidate, the quantity it explains, the number of leagues ($n$), the rank correlation with its 95% interval, the $P$ of the test as run and that $P$ after Holm's correction within the family. In the row of England's clubs, $n$ counts clubs. A positive correlation means that the larger the candidate, the larger the quantity. The earlier increment is the form each earlier test used (Section S3.1). A dash marks a value that does not apply. The mean over the leagues was held to a pass mark, not tested. The look and the sensitivity rows belong to no family. A sensitivity row repeats a stated test with another form of its candidate.

| Candidate | Explains | $n$ | $\rho$ | 95% interval | $P$ | Holm $P$ |
|---|---|---|---|---|---|---|
| *Club wealth inside a league: pre-specified* | | | | | | |
| England's clubs: market value per player | club's use of the limit | 16 | −0.30 | −0.70 to +0.25 | 0.13 | 0.13 |
| Mean over the leagues (a pass mark, not a test) | club's use of the limit | 12 | +0.02 | −0.16 to +0.20 | – | – |
| *Squad structure: pre-specified* | | | | | | |
| Rotation between matches | earlier increment | 13 | +0.23 | −0.38 to +0.70 | 0.22 | 0.22 |
| Bench value | earlier increment | 14 | +0.27 | −0.32 to +0.71 | 0.82 | 0.82 |
| *A league's culture: pre-specified* | | | | | | |
| Bookings per match | earlier increment | 20 | +0.38 | −0.09 to +0.71 | 0.048 | 0.14 |
| Share of draws | earlier increment | 20 | +0.21 | −0.27 to +0.61 | 0.37 | 0.75 |
| Goals per match | earlier increment | 20 | −0.10 | −0.53 to +0.37 | 0.68 | 0.75 |
| Bookings per match, fourteen domestic leagues (a look) | earlier increment | 14 | +0.41 | −0.18 to +0.78 | 0.075 | – |
| *The level under five: pre-specified* | | | | | | |
| Injury-marked share of withdrawals | level under five | 31 | −0.30 | −0.60 to +0.07 | 0.10 | 0.30 |
| Share of one-sided results | level under five | 31 | +0.07 | −0.30 to +0.42 | 0.71 | 1.00 |
| Goals per match | level under five | 31 | −0.38 | −0.65 to −0.01 | 0.038 | 0.19 |
| Starters' median tenure at the club | level under five | 14 | +0.13 | −0.44 to +0.63 | 0.66 | 1.00 |
| Familiarity-load slope | level under five | 14 | +0.49 | −0.07 to +0.82 | 0.075 | 0.30 |
| *Increment battery one: written with its program* | | | | | | |
| Load-management coefficient | earlier increment | 13 | +0.51 | −0.08 to +0.83 | 0.078 | 0.39 |
| Timing departure left after load management | earlier increment | 13 | +0.46 | −0.14 to +0.81 | 0.11 | 0.45 |
| Workload gradient of minutes | earlier increment | 13 | +0.36 | −0.26 to +0.77 | 0.23 | 0.69 |

| Candidate | Explains | $n$ | $\rho$ | 95% interval | $P$ | Holm $P$ |
|---|---|---|---|---|---|---|
| Retention coefficient | earlier increment | 13 | +0.19 | −0.42 to +0.68 | 0.54 | 1.00 |
| Level under three | earlier increment | 13 | +0.09 | −0.50 to +0.62 | 0.78 | 1.00 |
| *Increment battery two: written with its program* | | | | | | |
| Players fielded per team-match | earlier increment | 14 | +0.62 | +0.12 to +0.87 | 0.018 | 0.053 |
| Median days between league matches | earlier increment | 17 | −0.41 | −0.75 to +0.11 | 0.10 | 0.21 |
| Spread of goal margins | earlier increment | 17 | −0.32 | −0.70 to +0.20 | 0.21 | 0.21 |
| *Absences from matchday squads: written with their program* | | | | | | |
| Absent for a fifth of fixtures or more | rotation between matches | 13 | −0.51 | −0.83 to +0.07 | 0.074 | 0.67 |
| Absent for a fifth of fixtures or more | earlier increment | 13 | −0.45 | −0.81 to +0.16 | 0.13 | 1.00 |
| Absent for a fifth of fixtures or more | bench value | 13 | −0.16 | −0.66 to +0.44 | 0.60 | 1.00 |
| Absent for eight fixtures or more | rotation between matches | 13 | −0.38 | −0.78 to +0.23 | 0.20 | 1.00 |
| Absent for eight fixtures or more | earlier increment | 13 | +0.12 | −0.48 to +0.64 | 0.71 | 1.00 |
| Absent for eight fixtures or more | bench value | 13 | +0.31 | −0.31 to +0.74 | 0.31 | 1.00 |
| Longest absence, share of fixtures | rotation between matches | 13 | −0.05 | −0.60 to +0.52 | 0.86 | 1.00 |
| Longest absence, share of fixtures | earlier increment | 13 | −0.16 | −0.67 to +0.44 | 0.59 | 1.00 |
| Longest absence, share of fixtures | bench value | 13 | −0.16 | −0.66 to +0.44 | 0.60 | 1.00 |
| *League wealth: stated, not pre-specified* | | | | | | |
| Mean squad value, 2022–23 to 2025–26 | level under five | 14 | +0.39 | −0.19 to +0.77 | 0.16 | 0.82 |
| Mean squad value, 2022–23 to 2025–26 | increment | 14 | +0.45 | −0.12 to +0.80 | 0.11 | 0.64 |
| Mean squad value, 2015–16 to 2018–19 | level under three | 14 | +0.16 | −0.42 to +0.65 | 0.58 | 0.94 |
| Mean squad value, 2015–16 to 2018–19 | increment | 14 | +0.38 | −0.21 to +0.76 | 0.19 | 0.82 |
| Top-eighteen value, December 2022 | level under five | 17 | +0.25 | −0.27 to +0.66 | 0.32 | 0.94 |
| Top-eighteen value, December 2022 | increment | 17 | +0.26 | −0.27 to +0.67 | 0.31 | 0.94 |
| Value per player, 2022–23 to 2025–26 (sensitivity) | increment | 14 | +0.44 | −0.14 to +0.79 | 0.12 | – |
| Top-eighteen value, 2022–23 to 2025–26 (sensitivity) | increment | 14 | +0.45 | −0.12 to +0.80 | 0.11 | – |
| *Competition rules: stated, not pre-specified* | | | | | | |
| Named substitutes under five | level under five | 14 | +0.26 | −0.33 to +0.71 | 0.36 | 1.00 |
| Named substitutes under five | increment | 14 | +0.45 | −0.12 to +0.80 | 0.11 | 0.64 |
| Named substitutes added at the change | increment | 14 | +0.12 | −0.46 to +0.62 | 0.69 | 1.00 |
| Named substitutes under three | level under three | 14 | −0.15 | −0.64 to +0.43 | 0.62 | 1.00 |
| Share above five (concussion substitutes) | level under five | 14 | −0.34 | −0.74 to +0.25 | 0.24 | 1.00 |
| Share above five (concussion substitutes) | increment | 14 | −0.20 | −0.67 to +0.39 | 0.50 | 1.00 |

| Candidate | Explains | $n$ | $\rho$ | 95% interval | $P$ | Holm $P$ |
|---|---|---|---|---|---|---|
| Three windows used, substitutions left | level under five | 14 | −0.73 | −0.91 to −0.30 | 0.0032 | 0.022 |
| Three windows used, substitutions left | increment | 14 | −0.82 | −0.94 to −0.51 | $2.9 \times 10^{-4}$ | 0.0023 |
| Windows used, among those below five (sensitivity) | increment | 14 | +0.49 | −0.07 to +0.82 | 0.075 | – |
| *Calendar: stated, not pre-specified* | | | | | | |
| European matches per club and season | level under five | 14 | +0.19 | −0.39 to +0.66 | 0.52 | 1.00 |
| European matches per club and season | increment | 14 | +0.14 | −0.44 to +0.64 | 0.63 | 1.00 |
| Longest break between matchdays | level under five | 14 | −0.37 | −0.76 to +0.21 | 0.19 | 0.76 |
| Longest break between matchdays | increment | 14 | −0.20 | −0.67 to +0.38 | 0.49 | 1.00 |

### S3.2 Money

A league's wealth is measured by its mean squad market value. The records hold no revenue by league, so market value is the only measure of wealth held for every league. For each club and season, the measure adds up the market values of the players who appeared in a league match. Each value is taken as it stood at the end of the season's first calendar year. The measure then averages over the league's clubs and over the seasons common to every league. So the rise of market values over the years cannot reorder the leagues. Under five, a league's mean squad value correlated +0.39 with its level (−0.19 to +0.77) and +0.45 with its increment (−0.12 to +0.80). Over the 17 leagues of the increment, the mean value of each club's eighteen most valuable players correlated +0.26 with the increment (−0.27 to +0.67). None of these tests survived Holm's correction. With fourteen leagues a two-sided test detects only a correlation of 0.71 or more. Holm's correction checks the smallest $P$ first, against its strictest pass mark. Against that mark the test detects only 0.81 or more. So these tests could find only a strong link between money and the order, and not a modest one. England was the richest league by every measure. Under five its mean squad value was 596 million euros a club, against 274 million for Spain, the next.

Inside a league, a pre-specified test set each club's market value per player against its use of the limit. A positive correlation means that richer clubs used more. In England the correlation was −0.30 (−0.70 to +0.25), over 16 clubs. So richer English clubs used less of the limit, as the test predicted. But the link was not beyond chance (one-sided $P = 0.13$). Over 12 leagues the clubs' correlations averaged +0.02 (−0.16 to +0.20). Combined by Fisher's $z$, the usual way to combine correlations, they gave +0.01 (−0.17 to +0.19). The test is inside leagues, so it speaks to clubs and not to the difference between leagues.

Nevill and colleagues read six English seasons, one in each decade from 1973–74 to 2023–24. Successful clubs used fewer substitutes than less successful ones [8]. England's richer clubs here also used less of the limit, but not beyond chance.

### S3.3 Squad structure and availability

Rotation between matches is how often a league changes its starting eleven. It correlated +0.23 with the increment (−0.38 to +0.70; one-sided $P = 0.22$, 13 leagues). Bench value compares a club's eleven most valuable players with its next eleven. It is the mean market value of the first eleven divided by that of the next eleven. A high value means a first eleven worth far more than its bench. The

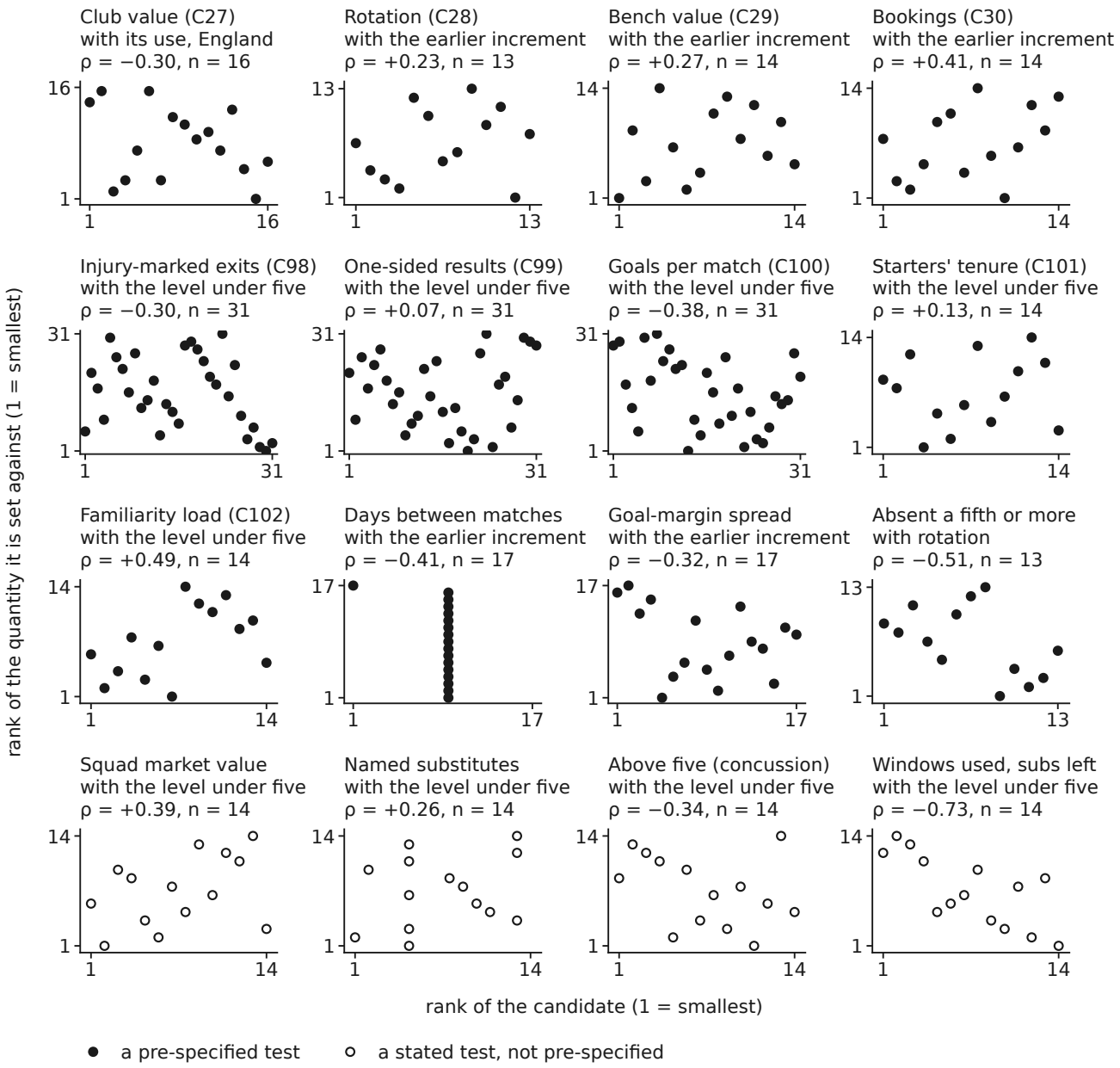



Figure S3: Sixteen candidate explanations of football's order, each drawn as ranks. None of the twelve pre-specified candidates held. Of the stated tests, one held after Holm's correction within its family. It is the share of team-matches in which teams had used all three substitution windows with substitutions left, in the panel titled "Windows used, subs left". That share restates the level itself (Table S10). Each point sets a league's rank on the candidate, from the smallest, against its rank on the quantity the candidate is tested against. In the test inside England each point is one of its sixteen clubs. Tests on thirty-one or seventeen leagues add leagues beyond the fourteen, from the records collected for those tests. Tests on thirteen leave out a league without the measure. Filled points are pre-specified tests, and open points stated tests, not pre-specified. A panel whose test has a clause carries the clause's number. Each panel names its test and prints its rank correlation $\rho$ and $n$. Tied values stand in columns. In the panel on days between matches sixteen of the seventeen leagues share one value, so its $\rho$ is set by the one league that differs. The panel on bookings is drawn on the fourteen leagues. The pre-specified test read six cups as well, and gave $\rho = +0.38$ over twenty competitions.

test expected leagues with such squads to rotate less and to add fewer substitutions. So it required a negative correlation with the increment. Across 14 leagues, bench value correlated +0.27 instead (−0.32 to +0.71).

A league whose players are absent more often has fewer of them to bring on. Absence was read from matchday squads, which every league records alike. An established member of a squad is one named for at least half the club's fixtures that season. An absence is an unbroken run of the club's fixtures in which an established member is not named. Absence is counted in fixtures and never in days. There are three measures of absence. Two are shares of established players: those whose longest absence ran for a fifth of their club's fixtures or more, and those whose longest absence ran for eight fixtures or more. The third is the longest absence itself, as a share of the fixtures. Each measure was set against three quantities: the increment, rotation between matches and bench value. That makes nine pairings.

No pairing survived Holm's correction. The strongest set the first share, of players absent for a fifth of their club's fixtures or more, against rotation between matches. Its rank correlation was −0.51 (95% interval −0.83 to +0.07), with $P = 0.074$ before Holm's correction over the pairings and 0.67 after it. Leagues with more such absences rotated their starting eleven less. That is the opposite of what a shortage of players would force.

The score in a match could also explain a league's substitutions. Coaches substitute by the score, and teams behind their expected result make more attacking changes [9, 10]. So a league with more one-sided matches could make fewer changes. Two tests read this idea: the share of one-sided results and the spread of goal margins. Neither held.

### S3.4 Bench size and the other competition rules

Each league sets its own matchday bench. Under five the named substitutes ran from 7.7 to 12 across the leagues. The bench comes from the team sheets, as the number named most often in each season. The sheets agree with the regulations in all six league-seasons where both can be read. Bench size under five correlated +0.26 with the level under five (−0.33 to +0.71) and +0.45 with the increment (−0.12 to +0.80). The bench added at the change correlated +0.12 with the increment (−0.46 to +0.62). None of these links was clear.

Some leagues enlarged their bench while their limit stood still. Each such change is set against the leagues whose bench and limit did not change over the same seasons. Under five, across five changes, each added place on the bench went with a change of −0.004 substitutions a match in a league's level (−0.051 to +0.042; $P = 0.86$). Under three, across six changes, each added place went with a change of +0.021 (+0.003 to +0.039; $P = 0.026$). So under three, when the limit held teams back, a larger bench went with slightly more substitutions. Under five, when teams had room to spare, it went with no change. England named nine substitutes instead of seven from December 2020. That change fell inside a season under three, with no league to compare it with, so it has no estimate.

Concussion substitutes can explain at most a small part of the differences between leagues. The records do not mark a concussion substitute. But under five a sixth substitution in a league match is possible only through one. Teams made more than five in 0.1 to 1.5 per cent of team-matches, the most in the Netherlands. Before the change, the same records show a team above three in 0.0 to 0.5 per cent of team-matches. That share is the source's own rate of over-counting, and it is of the same size as the share above five. A team may use at most two concussion substitutes. So concussion substitutes can have added at most 0.03 substitutions a team-match in any league. The spread between the leagues' levels is 0.20, so that is at most 0.15 of the spread.

Under five, every league allowed its five substitutions in at most three substitution windows during play. Half-time does not count as a window. The Laws of the Game set this limit, so it is the same in every league. Read minute by minute, 0.3 per cent of team-matches in Belgium and 18.0 per cent in France show more than three windows. That gap comes from how the source writes the minutes.

With neighbouring minutes merged, the share is at most 1.2 per cent. StatsBomb counts substitutions a minute or less apart as one stoppage. On the team-matches it shares with the source, it shows more than three windows in 0.0 to 1.5 per cent. So in most of them, one stoppage was written over two minutes.

Teams had used all three windows with substitutions left in 18 per cent of team-matches in Italy to 31 per cent in Scotland. By league, the limit on windows held back at most 0.22 to 0.42 substitutions a match. That share is the one candidate that survived Holm's correction over all the tests. Against the increment it correlated $-0.82$ ($P = 0.013$ after correction over all the tests). Against the level it correlated $-0.73$. That survives within its family ($P = 0.022$) but not over all the tests ($P = 0.14$). Within its family of eight stated tests, two survived. The share largely restates the level, because it is high wherever teams stop short of five.

Among team-matches that stopped short of five, the share that had used all three windows correlated $+0.49$ with the increment ($-0.07$ to $+0.82$; $P = 0.075$). The link points to choice. In the leagues that used less of the limit, a team that stopped short more often still had a window left. But the interval runs across zero, so the link is not clear. The order also predates the rule: it was there under three, where no limit on windows existed (Section S2.1).

European matches per club and season, and the longest break between matchdays, measure the load of each league's calendar. Neither showed a clear link to the order (Table S10).

Leagues in the other sports also set rules of their own. We did not measure two of them: the size of the game-day roster and the limits on players from abroad. These differ between leagues and change rarely. So they alone could keep a league in its place.

### S3.5 What these tests could detect

A test's power is its chance of passing when a link is real. A test here passes at a $P$ below 0.05. With thirteen or fourteen leagues, a one-sided test has power 0.80 only for a true rank correlation of about 0.65 to 0.67 or more. A two-sided test needs about 0.71 to 0.72. So "did not explain" means that these tests found no strong link (Table S5).

## S4 The split record and its pre-specification

### S4.1 The split record

The split record holds 32 leagues of basketball, volleyball, handball, ice hockey and baseball, with 130,745 games. Every game went to one of two halves, A or B. Its half was fixed by a hash, a number computed from its league, its season and its identifier. Half A was read freely to choose each measure and pass mark. Then 26 pre-specified tests were written down, each in its clause. The clauses were saved before half B was read, and so was the program that runs them. Half B was read once, and only its results count as tests. Table S11 gives every clause word for word as it was saved, with its statistic on half B and its outcome. Of the 26 clauses, 14 held and 12 failed.

The registers name files, numbered scripts and manuscripts by the labels the project used. Table S12 gives what each label refers to in the archive. The copy for anonymous review prints each manuscript's label in neutral words.

The split guards against choosing a measure on the luck of single games. It does not guard against choosing it on the leagues themselves, which both halves share. Each league's mean is nearly the same in the two halves. On every measure, the leagues' values in half A and in half B correlate at 0.92 or more (median 0.98). So an order between leagues seen in half A is likely to appear again in half B. Half B is therefore not an independent test of a claim about leagues. The stronger evidence is the predictions for new seasons, leagues and competitions (Section S4.4).

Table S11 prints half A's statistic beside half B's, with the verdict half A would have given. The verdicts agree in 24 of the 26 clauses. They differ in R08 and R22. Half A would have passed 16 clauses, and half B passed 14.

Every rule moved what it was written to move. In basketball, the three-point line was moved further out in 2010–11. The move cut the share of shots from three in each of the six leagues that made it. In ice hockey, three-on-three overtime raised the odds that a tied game was settled before a shootout. On half B the odds were 2.21 times as high as before, across five leagues ($P = 4.6 \times 10^{-8}$). Section S5.1 explains why this is above the value over both halves. In baseball, the pitch clock shortened the game in each of four leagues.

The clauses that failed are reported as fully as those that held. The pitch clock spread the baseball leagues apart instead of drawing them together. The spread of their game lengths grew from 7.7 to 13.8 minutes. The spread of their pitchers per team and game grew from 0.31 to 0.50. The leagues' clocks differ, so this failure was not a test of one rule (Section S5.2). In ice hockey, a team that trails late in a game may pull its goalie, that is, take the goalie off the ice. Goalie pulls came earlier across the leagues that time them (3.3 seconds a season, $P = 2.7 \times 10^{-7}$). But they did not come earlier in one of them, and the clause asked for each league.

Most clauses on a loosened limit held. For example, France's top men's handball league allowed each team a third team time-out. The share of a team's games in which it used that third time-out rose from 35 per cent to 47 per cent (one-sided $P = 0.0020$). The clause dates the third time-out from the start of 2012. We found no league or federation document that gives the date, but the league's records fit it. A team took three time-outs in one of the 130 team-games before January 2012. From February to June 2012 it took three in 37 per cent of team-games. Larger match sheets in volleyball put more players into use in each of the five leagues whose sheets grew (+0.33 players a team and match, one-sided $P = 1.1 \times 10^{-15}$). The spread between the eight volleyball leagues in substitutions a set widened from 0.13 to 0.18 (one-sided $P = 0.0060$).

The clauses on the leagues' order compare two windows of seasons. Across the shot clock's change of 2014, the basketball leagues kept their order in the share of a team's minutes played by its five busiest players (rank correlation +0.86, exact one-sided $P = 0.0054$). They also kept it in how many players get ten minutes (+0.79, exact one-sided $P = 0.014$). Across the 2010 line they kept it in the share of shots taken from three (+0.79, exact one-sided $P = 0.024$). The volleyball leagues kept theirs in the number of players used (+0.83, exact one-sided $P = 0.0077$).

Other orders were lost. Basketball's pace, a team's possessions per forty minutes, lost its order when the 2014 rules drew the leagues together (+0.39, exact one-sided $P = 0.20$). Volleyball's substitutions a set kept almost none of their order (+0.10). In ice hockey, the two windows' orders in when teams pull the goalie had a rank correlation of +0.60. In baseball, their orders in the number of pitchers a team uses had one of +0.60. Each of those two was read on four leagues, where an order passes only when it is perfect.

No climb was shown to level off as it went. Three climbs were tested. One was the use of the third team time-out in France's top men's handball league. Another was the move to earlier goalie pulls. The third was the number of pitchers used in MLB, the top baseball league of the United States and Canada. That climb quickened after 2014. Two other clauses on a loosened limit failed as well. In volleyball, the second libero was not used more over the seasons. And a volleyball league's use of its bench before its match sheet grew did not predict how much that use rose.

Each verdict is read over every league its clause names. Most leagues in the other sports were not shown to be full-time. That matters for these measures, because a part-time squad may rotate for availability rather than by choice. So each statistic is also computed on the full-time leagues, whose players do not need another job to live. A league is graded full-time when its lowest regular pay is above its country's poverty line for one adult. It is graded likely full-time when the evidence points that way but no pay floor is published. On the leagues graded full-time, every verdict that could be computed

was unchanged. With the leagues graded likely added, the clause that goalie pulls came earlier in each league held. But two clauses that held on every league failed there, one read on three leagues and one on two. Four of the clauses on the leagues' order kept only three leagues there. Three leagues are too few for such a clause to pass at all.

### S4.2 Japan's volleyball sets

One count in the volleyball records was wrong. The Japanese records keep five slots for the durations of a match's sets, and leave the slots of sets not played empty. The program that read them counted the slots. So every Japanese match counted five sets, and Japan's substitutions per set were its substitutions divided by five, whatever was played. Counted as played, Japan's two leagues make the most substitutions a set of the eight in every season read. The volleyball leagues' persistence in substitutions per set is then 0.98 a season, and not 0.82 (Section S6.3). Two clauses of the split record read substitutions per set, one on the leagues' order and one on their spread (Table S11). Both were read on the count of five slots, and both verdicts stand as first read.

Table S11: The 26 clauses of the split record, printed word for word as pre-specified. Each row also gives the statistic on half A, on which the clause was chosen, with its $P$, its $n$ and the verdict it would have had, and the statistic on half B, the test, with its outcome.

| Clause | Its words, as pre-specified | Half A ($P$; $n$) | Half B ($P$; $n$) | Outcome |
|---|---|---|---|---|
| R01 | In half B of stage 275's split, the three-point line moved in 2010-11 lowered the share of shots taken from three in each of the six leagues that moved it that season or are taken to have: Spain, Italy, Lithuania and Greece, where the date is documented or indicated, and the ABA League and Israel, bound by FIBA's Level 1 date. In each league the team-game mean of three-point attempts over field-goal attempts (stage 277) is lower in 2010-11 than in 2009-10, with a one-sided Welch P below 0.05. Korea, whose date is unknown, and Turkey and Brazil, which have no box scores for 2009-10, are left out. | 6 (0.0013; 6); held | 6 (0.0047; 6) | held |
| R02 | In half B of stage 275's split, FIBA's 2014 rules, which reset the shot clock to 14 seconds after an offensive rebound and so shorten an attack, drew the leagues' pace together. For each league with possessions in both windows (Spain, Italy, Greece, the ABA League, Israel, Korea and Lithuania), the change in its mean possessions per 40 minutes (stage 277) runs from the three seasons before the rules (2011-12 to 2013-14; for Spain, which reset the clock in 2013-14, 2010-11 to 2012-13) to the three seasons from the third after them (2016-17 to 2018-19). The least-squares slope of that change on the league's level before, the leagues weighted equally, is negative with a one-sided P below 0.05. | -0.991 (0.0074; 7); held | -0.796 (0.011; 7) | held |
| R03 | In half B of stage 275's split, the leagues keep their order in how far coaches lean on a few players across FIBA's 2014 rules. The Spearman correlation between the league means of the five busiest players' share of team minutes (stage 277) in 2011-12 to 2013-14 and in 2016-17 to 2018-19 is positive with an exact one-sided permutation P below 0.05, over the eight leagues with minutes in both windows: Spain, Italy, Greece, the ABA League, Israel, Korea, Turkey and Lithuania. | 0.857 (0.0054; 8); held | 0.857 (0.0054; 8) | held |

| Clause | Its words, as pre-specified | Half A ($P$; $n$) | Half B ($P$; $n$) | Outcome |
|---|---|---|---|---|
| R04 | In half B of stage 275's split, the use of a loosened limit climbs, as the use of five substitutions did in football. France allowed a third team time-out from 1 January 2012. The share of team-games in which a team took all three (stage 278) is higher in 2016-17 to 2019-20 than in 2012-13, the first full season with three, with a one-sided two-proportion z-test P below 0.05. | 0.121 (0.0028; 818); held | 0.118 (0.0020; 864) | held |
| R05 | In half B of stage 275's split, the larger match sheet widened the use of the bench. Five leagues' sheets grew within the record: Japan's men and women (2006-07 and 2007-08 against 2008-09 and 2009-10), Turkey's men and women (2013-14 against 2014-15 and 2015-16) and France's men (2016-17 to 2018-19 against 2020-21 to 2022-23, the season of change left out). The players used per team and match (stage 276: those who played in at least one set) rise. The league-stratified mean difference, after less before, weighted by the inverse of each league's variance, is positive with a one-sided P below 0.05, and at least four of the five leagues' differences are positive. | 0.289 ($6.3 \times 10^{-12}$; 5); held | 0.332 ($1.1 \times 10^{-15}$; 5) | held |
| R06 | In half B of stage 275's split, the leagues keep their order in how many substitutions a coach makes. The Spearman correlation between the league means of substitutions per team and set, liberos left out (stage 276), in 2013-14 to 2015-16 and in 2021-22 to 2023-24 is positive with an exact one-sided permutation P below 0.05, over the eight leagues: Italy's, Turkey's and Japan's men and women, France's men and Germany's men. | 0.381 (0.18; 8); failed | 0.0952 (0.42; 8) | failed |
| R07 | In half B of stage 275's split, three-on-three overtime settled more tied games in overtime rather than by shootout. Each league's extra-time games in the two seasons before its adoption are set against those in the two seasons after: Sweden 2012-13 and 2013-14 against 2015-16 and 2016-17 (its switch of December 2014 left out); Finland and the ICE league 2013-14 and 2014-15 against 2015-16 and 2016-17; Germany and Switzerland 2014-15 and 2015-16 against 2016-17 and 2017-18. The Mantel-Haenszel odds ratio of an overtime decision against a shootout, stratified by league, exceeds 1 with a one-sided P below 0.05. The decision is read as stage 279 reads it. | 1.33 (0.022; 5); held | 2.21 ($4.6 \times 10^{-8}$; 5) | held |
| R08 | In half B of stage 275's split, coaches pull the goalie earlier season by season in every league that times it. The seconds left in regulation at a trailing team's first pull in the last ten minutes, trailing by one or two goals (stage 279), rise with the season. The pooled least-squares slope on the season, with a separate intercept per league and standard errors clustered by league and season, is positive with a one-sided P below 0.05, and each league's own slope is positive: Sweden 2011-12 to 2018-19, Finland 2014-15 to 2018-19, Switzerland in its timed games of 2011-12 to 2018-19 and the ICE league 2015-16 to 2018-19. | 3.01 (0.0015; 4); held | 3.27 ($2.7 \times 10^{-7}$; 4) | failed |
| R09 | In half B of stage 275's split, the pitch clock shortened games in each league's first enforced season. The mean game length in minutes (stage 280) is lower than in the season before in MLB (2023 against 2022), the Mexican League (2023 against 2022), Taiwan's CPBL (2024, its full clock, against 2023) and Korea's KBO (2025, enforced, against 2024), each with a one-sided Welch P below 0.05. | 4 (0.0021; 4); held | 4 ($3.3 \times 10^{-6}$; 4) | held |

| Clause | Its words, as pre-specified | Half A ($P$; $n$) | Half B ($P$; $n$) | Outcome |
|---|---|---|---|---|
| R10 | In half B of stage 275's split, the pitch clock drew the four leagues' game lengths together. The standard deviation of the four league means of game length (stage 280) is smaller in 2025, when all four enforced the clock, than in 2022, before any did, with a one-sided bootstrap P below 0.05: games resampled within league and season, 2,000 draws, seed 20260927. | 7.07 (1.00; 4); failed | 6.14 (1.00; 4) | failed |
| R11 | In half B of stage 275's split, the pitch clock drew the four leagues' use of pitchers together. The standard deviation of the four league means of pitchers per team and game (stage 280) is smaller in 2025 than in 2022, with a one-sided bootstrap P below 0.05: games resampled within league and season, 2,000 draws, seed 20260927. | 0.245 (1.00; 4); failed | 0.192 (1.00; 4) | failed |
| R12 | In half B of stage 275's split, the leagues keep their order in how many players a coach gives ten minutes or more across FIBA's 2014 rules. The Spearman correlation between the league means of players on ten minutes or more (stage 277) in 2011-12 to 2013-14 and in 2016-17 to 2018-19 is positive with an exact one-sided permutation P below 0.05, over the eight leagues of R03. | 0.905 (0.0023; 8); held | 0.786 (0.014; 8) | held |
| R13 | In half B of stage 275's split, the leagues keep their order in how often they shoot from three across the line move of 2010. The Spearman correlation between the league means of the share of shots from three (stage 277) in 2007-08 to 2009-10 and in 2011-12 to 2013-14 is positive with an exact one-sided permutation P below 0.05, over the seven leagues with box scores in both windows: Spain, Italy, Greece, the ABA League, Israel, Korea and Lithuania. | 0.821 (0.017; 7); held | 0.786 (0.024; 7) | held |
| R14 | In half B of stage 275's split, the leagues keep their order in pace across FIBA's 2014 rules. The Spearman correlation between the league means of possessions per 40 minutes (stage 277) in 2011-12 to 2013-14 and in 2016-17 to 2018-19 is positive with an exact one-sided permutation P below 0.05, over the seven leagues of R02. | 0 (0.52; 7); failed | 0.393 (0.20; 7) | failed |
| R15 | In half B of stage 275's split, after the line moved back in 2010-11 the share of shots from three climbed season by season. Over 2010-11 to 2018-19, the pooled least-squares slope of each team-game's share of shots from three (stage 277) on the season, with a separate intercept per league and standard errors clustered by league and season, is positive with a one-sided P below 0.05, and each league's own slope is positive, in the six leagues of R01. | 0.00742 ($4.7 \times 10^{-17}$; 6); held | 0.00714 ($1.4 \times 10^{-15}$; 6) | held |
| R16 | In half B of stage 275's split, a coach's habit of leaning on a few players travels with the coach who moves. A coach moves when, in consecutive seasons, fewer than 15 per cent of the players of the coach's new team's box scores played for the coach's team the season before; coaches and players are compared by code (src/player_codes.py), never by name. Spain, Israel and Brazil, the three leagues whose box scores name the coach, are pooled. For each move, the coach's team's mean share of minutes to its five busiest players (stage 277), less the mean of the league in that season, is taken in the coach's last season before the move and first season after it. The Spearman correlation of the two across moves is positive with a one-sided P below 0.05, from 10,000 random permutations, seed 20260927. | 0.249 (0.0018; 136); held | 0.268 (0.0015; 134) | held |

| Clause | Its words, as pre-specified | Half A ($P$; $n$) | Half B ($P$; $n$) | Outcome |
|---|---|---|---|---|
| R17 | In half B of stage 275's split, a coach's pace travels with the coach who moves. The moves are those of R16. For each move, the coach's team's mean possessions per 40 minutes (stage 277), less the mean of the league in that season, is taken in the coach's last season before the move and first season after it. The Spearman correlation of the two across moves is positive with a one-sided P below 0.05, from 10,000 random permutations, seed 20260927. | 0.166 (0.029; 136); held | 0.296 ($4.0 \times 10^{-4}$; 134) | held |
| R18 | In half B of stage 275's split, France's climb in the use of the third time-out slows as it goes, as a relaxation towards a ceiling does. The least-squares slope of the team-game indicator of taking all three time-outs (stage 278) on the season is larger in 2012-13 to 2015-16 than in 2016-17 to 2019-20, the difference positive with a one-sided P below 0.05, standard errors clustered by team and season. | 0.0305 (0.20; 1542); failed | 0.0356 (0.18; 1534) | failed |
| R19 | In half B of stage 275's split, the leagues keep their order in how many players a coach uses in a match. The Spearman correlation between the league means of players used per team and match (stage 276) in 2013-14 to 2015-16 and in 2021-22 to 2023-24 is positive with an exact one-sided permutation P below 0.05, over the eight leagues of R06. | 0.667 (0.042; 8); held | 0.833 (0.0077; 8) | held |
| R20 | In half B of stage 275's split, as larger sheets spread through volleyball, the leagues' use of the bench grew apart. The standard deviation of the eight leagues' mean substitutions per team and set, liberos left out (stage 276), is larger in 2021-22 to 2023-24 than in 2013-14 to 2015-16, with a one-sided bootstrap P below 0.05: matches resampled within league and season, 2,000 draws, seed 20260927. | 0.0365 (0.013; 8); held | 0.0413 (0.0060; 8) | held |
| R21 | In half B of stage 275's split, the use of the second libero, allowed from 2009, climbs. The pooled least-squares slope of the team-match indicator of listing two liberos (stage 276) on the season, with a separate intercept per league and standard errors clustered by league and season, is positive with a one-sided P below 0.05, and at least six of the eight leagues' own slopes are positive. Japan enters from 2008-09, its first season with two. | 0.000586 (0.45; 8); failed | 0.00089 (0.43; 8) | failed |
| R22 | In half B of stage 275's split, the leagues keep their order in when a trailing team pulls its goalie. The Spearman correlation between the league medians of seconds left at the first late pull (stage 279) in 2015-16 and 2016-17 and in 2017-18 and 2018-19 is positive with an exact one-sided permutation P below 0.05, over the four leagues that time the goalie; at four leagues this needs the same order in both windows. | 1 (0.042; 4); held | 0.6 (0.21; 4) | failed |
| R23 | In half B of stage 275's split, the climb towards earlier goalie pulls slows as it goes. Each league's timed seasons (those of R08) are split into an earlier and a later half, the middle season of an odd count in the later. The pooled slope of seconds left at the first late pull (stage 279) on the season, with a separate intercept per league, is larger in the earlier halves than in the later, the difference positive with a one-sided P below 0.05, standard errors clustered by league and season. | -4.06 (0.94; 4); failed | -2.27 (0.88; 4) | failed |

| Clause | Its words, as pre-specified | Half A ($P$; $n$) | Half B ($P$; $n$) | Outcome |
|---|---|---|---|---|
| R24 | In half B of stage 275's split, the leagues keep their order in how many pitchers a manager uses, across the pitch clock. The Spearman correlation between the league means of pitchers per team and game (stage 280) in 2021 and 2022 and in 2024 and 2025 is positive with an exact one-sided permutation P below 0.05, over MLB, the Mexican League, the CPBL and the KBO; at four leagues this needs the same order in both windows. | 0.8 (0.17; 4); failed | 0.6 (0.21; 4) | failed |
| R25 | In half B of stage 275's split, MLB's long drift towards more pitchers a game slows before any new rule caps it. The least-squares slope of pitchers per team and game (stage 280) on the season is larger in 2008 to 2013 than in 2014 to 2019, the difference positive with a one-sided P below 0.05, standard errors clustered by team and season. | -0.074 (1.00; 28826); failed | -0.0781 (1.00; 29484) | failed |
| R26 | In half B of stage 275's split, the leagues that used their bench least before their sheets grew widened it most, as the habit before a change predicting the response after it would have them do. Over the five leagues and windows of R05, the least-squares slope of each league's change in players used per team and match (stage 276) on its level before is negative with a one-sided P below 0.05. | -0.0841 (0.31; 5); failed | -0.0389 (0.44; 5) | failed |

Table S12: The key to the internal labels in Tables S11 and S15: each file path, stage number and manuscript name a pre-specified test names, the tests that name it, and what it refers to in the archive. A stage is a numbered script of the archive.

| Label | Tests | What it refers to |
|---|---|---|
| `cricket_bowling_standings.csv` | G13 | a result table in the archive's data/processed/results/ |
| `data/manual/paper_b_ifab_2026_rule.csv` | G17 | a hand-entered input in the archive, with a source for each row |
| `data/manual/paper_b_second_tier_limits.csv` | G19 | a hand-entered input in the archive, with a source for each row |
| `data/manual/pitch_clocks.csv` | G12 | a hand-entered input in the archive, with a source for each row |
| `data/raw/cricsheet/all_json.zip` | G13 | a source record or document in the archive's raw folder (not redistributed) |
| `data/raw/harvest_documents/ice_hockey/sb-2016-2017.pdf` | G02 | a source record or document in the archive's raw folder (not redistributed) |
| `docs/rule_change_harvest/basketball.md` | G01 | a working note in the archive |
| `docs/rule_change_harvest/general_formulas.md` | G10 | a working note in the archive |
| `fetch_feb_shooting.py` | G10 | an archived file |
| `fetch_fpb_shooting.py` | G10 | an archived file |
| `fetch_plk_shooting.py` | G10 | an archived file |
| `g11a_standings.csv` | G11a | a result table in the archive's data/processed/results/ |
| `general_forms_baseball_standings.csv` | G11 | a result table in the archive's data/processed/results/ |
| `general_forms_clock.csv` | G12 | a result table in the archive's data/processed/results/ |
| `general_forms_drift.csv` | G09 | a result table in the archive's data/processed/results/ |
| `general_forms_hockey_standings.csv` | G08 | a result table in the archive's data/processed/results/ |
| `general_forms_line.csv` | G10 | a result table in the archive's data/processed/results/ |
| `general_formula_constants.csv` | G01, G02, G07 | a result table in the archive's data/processed/results/ |

| Label | Tests | What it refers to |
|---|---|---|
| general_formula_standings.csv | G03, G04, G05, G06 | a result table in the archive's data/processed/results/ |
| jsams_cross_sport_eras.csv | G15 | a result table in the archive's data/processed/results/ |
| paper_b_order_new_leagues_coverage.csv | G19 | a result table in the archive's data/processed/results/ |
| src/player_codes.py | R16 | an archived script |
| tools/fetch_eurocup_shooting.py | G10 | an archived script |
| tools/fetch_kbo_mykbostats.py | G11a | an archived script |
| tools/fetch_lf_endesa.py | G10 | an archived script |
| tools/fetch_npb_games.py | G11a | an archived script |
| tools/fetch_statsapi_baseball.py | G11, G12 | an archived script |
| Paper 1 | G10, G15, G16 | the denominator study: its preprint is arXiv:2608.11228v3, and its extension beyond football, which holds the gate of stage 259, is unpublished (in clauses written before the renumbering of 2026-10-05, 'Paper 1' is this study); in the anonymous copy its reference is withheld |
| Paper A | G16 | an unpublished companion manuscript on who starts and who is withdrawn |
| Paper B | G15, G16, G17, G18, G19 | this manuscript |
| Paper C | G15 | an unpublished manuscript on what a rule forces, not pursued |
| the former Paper 2 | G16 | the nine-sport preprint, arXiv:2609.05584v1, since split into three manuscripts; in the anonymous copy its reference is withheld |
| stage 240 | G16 | the archived script src/240_nba_substitution_counts.py: NBA substitution spend, set against the EuroLeague's. |
| stage 259 | G15, G16 | the archived script src/259_cross_sport_contract.py: The cross-sport data contract, enforced before anything is pooled. |
| stage 275 | R01, R02, R03, R04, R05, R06, R07, R08, R09, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26 | the archived script src/275_rule_change_split.py: The rule-change games, split into a fitting half and a testing half. |
| stage 276 | R05, R06, R19, R20, R21, R26 | the archived script src/276_half_a_volleyball.py: Half A, volleyball: who each team listed and used, match by match. |
| stage 277 | R01, R02, R03, R12, R13, R14, R15, R16, R17, G03, G04, G05, G06, G16 | the archived script src/277_half_a_basketball.py: Half A, basketball: how each team spread its minutes, and how it played. |
| stage 278 | R04, R18 | the archived script src/278_half_a_handball.py: Half A, handball: the sheet, the time-outs and the extra attacker. |
| stage 279 | R07, R08, R22, R23, G02, G08 | the archived script src/279_half_a_ice_hockey.py: Half A, ice hockey: how games were decided, and when coaches pulled the goalie. |
| stage 280 | R09, R10, R11, R24, R25, G11, G11a | the archived script src/280_half_a_baseball.py: Half A, baseball: how long games ran, and how managers used their pitchers. |

| Label | Tests | What it refers to |
|---|---|---|
| stage 283 | G01, G02, G03, G04, G05, G06, G07 | the archived script src/283_general_formulas.py: One formula with constants by league and by sport: fitted, then held to new records. |
| stage 290 | G08, G09, G10, G11, G12, G14, G15, G16, G17 | the archived script src/290_general_forms_with_terms.py: The general forms with added terms, fitted on records already read. |
| stage 291 | G11a | the archived script src/291_general_form_terms_tests.py: The forms with added terms, read once on records not read before their registration. |
| stage 292 | G15, G16 | the archived script src/292_allocation_beside_panels.py: How each league shares out its minutes, on the records and seasons of Paper 1's extension. |
| stage 293 | G15 | the archived script src/293_finishing_by_position.py: Who plays the whole match, position by position, under each sport's rule. |
| stage 294 | G16 | the archived script src/294_paper_a_all_sports.py: Paper A beyond football: the cause of the workload gradient in every competition of Paper 1's panel. |
| stage 295 | G13, G14 | the archived script src/295_cricket_bowling.py: How cricket's leagues share an innings among their bowlers, fitted on seasons before 2023. |
| stage 298 | G17, G18 | the archived script src/298_paper_b_fan.py: Paper B's football fan: substitutions per team-match, league by league. |
| stage 300 | G11a | the archived script src/300_baseball_standings_amended.py: G11a: baseball's order over eight leagues, with the NPB and the KBO read again. |
| stage 336 | G19 | the archived script src/336_paper_b_order_and_scale.py: Stage 336: was football's league order there before the rule change, and on what scale did the leagues part? |
| stage 338 | G18 | the archived script src/338_paper_b_explanations.py: Stage 338: the tests of explanations for football's league order, their unit, sign and power. |
| stage 343 | G17 | the archived script src/343_paper_b_ifab_prediction.py: Stage 343: a prediction for IFAB's ten-second substitution rule of 2026-27, drafted before its records are read. |
| stage 344 | G19 | the archived script src/344_paper_b_order_new_leagues.py: Stage 344: does a league's order under three substitutions predict its order under five, in leagues never read? |
| stage 62 | G17, G18, G19 | the archived script src/62_extended_league_family.py: Further leagues admitted to the panel by a rule stated in advance. |

### S4.3 The audit of the pre-specification

Each pre-specified test was saved, with its date, in the project's repository. A commit is a saved, dated version of the project's files. The host is the website that holds the repository, and a push is an upload to it. A commit's date is set on the author's own computer, and others cannot check it. Each test is therefore also dated by the time its commit reached the repository's host, read from the host's push log (Table S13). The working repository is private, so only those given access can read that log. Its history was rewritten twice, in September 2026, and a rewrite gives the commits it touches new names. So every push in the log is matched to today's commits by the host's own record of their commit times.

The split record's 26 clauses were committed 34 to 42 minutes before the commit that records half B's results. All these commits reached the host in one push. So only the commits' order shows that the clauses came first. Each of the eighteen predictions read reached the host before its records were read. Of football's 26 pre-specified tests, thirteen were committed before their outcome was recorded,

and twelve of those reached the host before it. The other thirteen were declared in the commit that ran them. For those, nothing but the author's word dates the writing before the reading. No clause's words changed between its writing and its reading.

Of the 108 pre-specified tests that bear on the paper's claims, 79 are reported in the main text or in this supplement. Table S14 gives the other 29, each with its outcome and why it is not reported. Most read the load-management law, which this version does not claim, or a quantity of a companion manuscript. The others were replaced by an amended test.

Table S13: The audit of every pre-specified test bearing on the paper's claims. Each test is dated by the commit that registered it (author time, UTC), by the time that commit reached the repository's host (the host's push log, UTC) and by the commit that first recorded its outcome. Order: whether the registration came before the recorded outcome, or in the same commit. Words kept: whether the test's words were unchanged when its outcome was recorded. Commits are named by the first seven characters of their hash in the archived history.

| Test | Registered | On the host | Outcome recorded | Order | Words kept |
|---|---|---|---|---|---|
| *the split record* | | | | | |
| R01 | dd69d43 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:35 | 21:59 | 2026-09-27 21:18 | | |
| R02 | dd69d43 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:35 | 21:59 | 2026-09-27 21:18 | | |
| R03 | dd69d43 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:35 | 21:59 | 2026-09-27 21:18 | | |
| R04 | dd69d43 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:35 | 21:59 | 2026-09-27 21:18 | | |
| R05 | dd69d43 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:35 | 21:59 | 2026-09-27 21:18 | | |
| R06 | dd69d43 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:35 | 21:59 | 2026-09-27 21:18 | | |
| R07 | dd69d43 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:35 | 21:59 | 2026-09-27 21:18 | | |
| R08 | dd69d43 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:35 | 21:59 | 2026-09-27 21:18 | | |
| R09 | dd69d43 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:35 | 21:59 | 2026-09-27 21:18 | | |
| R10 | dd69d43 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:35 | 21:59 | 2026-09-27 21:18 | | |
| R11 | dd69d43 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:35 | 21:59 | 2026-09-27 21:18 | | |
| R12 | 7c66b67 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:44 | 21:59 | 2026-09-27 21:18 | | |
| R13 | 7c66b67 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:44 | 21:59 | 2026-09-27 21:18 | | |
| R14 | 7c66b67 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:44 | 21:59 | 2026-09-27 21:18 | | |
| R15 | 7c66b67 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:44 | 21:59 | 2026-09-27 21:18 | | |
| R16 | 7c66b67 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:44 | 21:59 | 2026-09-27 21:18 | | |
| R17 | 7c66b67 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:44 | 21:59 | 2026-09-27 21:18 | | |
| R18 | 7c66b67 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:44 | 21:59 | 2026-09-27 21:18 | | |
| R19 | 7c66b67 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:44 | 21:59 | 2026-09-27 21:18 | | |
| R20 | 7c66b67 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:44 | 21:59 | 2026-09-27 21:18 | | |

| Test | Registered | On the host | Outcome recorded | Order | Words kept |
|---|---|---|---|---|---|
| R21 | 7c66b67 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:44 | 21:59 | 2026-09-27 21:18 | | |
| R22 | 7c66b67 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:44 | 21:59 | 2026-09-27 21:18 | | |
| R23 | 7c66b67 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:44 | 21:59 | 2026-09-27 21:18 | | |
| R24 | 7c66b67 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:44 | 21:59 | 2026-09-27 21:18 | | |
| R25 | 7c66b67 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:44 | 21:59 | 2026-09-27 21:18 | | |
| R26 | 7c66b67 | 2026-09-27 | 00abd1a | before | yes |
| | 2026-09-27 20:44 | 21:59 | 2026-09-27 21:18 | | |
| *predictions on new records* | | | | | |
| G01 | 40c610a | 2026-09-27 | 5a0cf33 | before | yes |
| | 2026-09-27 22:16 | 22:16 | 2026-09-28 11:38 | | |
| G02 | 40c610a | 2026-09-27 | 5a0cf33 | before | yes |
| | 2026-09-27 22:16 | 22:16 | 2026-09-28 11:38 | | |
| G03 | 40c610a | 2026-09-27 | 5a0cf33 | before | yes |
| | 2026-09-27 22:16 | 22:16 | 2026-09-28 11:38 | | |
| G04 | 40c610a | 2026-09-27 | 5a0cf33 | before | yes |
| | 2026-09-27 22:16 | 22:16 | 2026-09-28 11:38 | | |
| G05 | 40c610a | 2026-09-27 | 5a0cf33 | before | yes |
| | 2026-09-27 22:16 | 22:16 | 2026-09-28 11:38 | | |
| G06 | 40c610a | 2026-09-27 | 5a0cf33 | before | yes |
| | 2026-09-27 22:16 | 22:16 | 2026-09-28 11:38 | | |
| G07 | 40c610a | 2026-09-27 | 5a0cf33 | before | yes |
| | 2026-09-27 22:16 | 22:16 | 2026-09-28 11:38 | | |
| G08 | 5acd063 | 2026-09-28 | 73e2b20 | before | yes |
| | 2026-09-28 13:25 | 14:15 | 2026-09-28 16:44 | | |
| G09 | 5acd063 | 2026-09-28 | 73e2b20 | before | yes |
| | 2026-09-28 13:25 | 14:15 | 2026-09-28 16:44 | | |
| G10 | b7ea758 | 2026-09-28 | 31372cd | before | yes |
| | 2026-09-28 13:50 | 14:15 | 2026-09-28 14:27 | | |
| G11 | 24fcc19 | 2026-09-28 | 7aefb6c | before | yes |
| | 2026-09-28 16:14 | 16:14 | 2026-10-17 20:29 | | |
| G11a | c652ac3 | 2026-09-29 | 7aefb6c | before | yes |
| | 2026-09-29 11:50 | 16:14 | 2026-10-17 20:29 | | |
| G12 | 24fcc19 | 2026-09-28 | e8f328c | before | yes |
| | 2026-09-28 16:14 | 16:14 | 2026-09-28 16:19 | | |
| G13 | 9dfd054 | 2026-09-28 | 77b62b7 | before | yes |
| | 2026-09-28 20:00 | 20:00 | 2026-09-28 20:02 | | |
| G14 | 9dfd054 | 2026-09-28 | 77b62b7 | before | yes |
| | 2026-09-28 20:00 | 20:00 | 2026-09-28 20:02 | | |
| G15 | 53e05d2 | 2026-09-29 | 5c69d86 | before | yes |
| | 2026-09-29 11:59 | 16:14 | 2026-09-29 16:31 | | |
| G16 | 53e05d2 | 2026-09-29 | 5c69d86 | before | yes |
| | 2026-09-29 11:59 | 16:14 | 2026-09-29 16:31 | | |
| G17 | 18ce13c | 2026-10-15 | – | not read | – |
| | 2026-10-15 20:45 | 20:46 | | | |
| G18 | 18ce13c | 2026-10-15 | – | not read | – |
| | 2026-10-15 20:45 | 20:46 | | | |
| G19 | 18ce13c | 2026-10-15 | 4708ec8 | before | yes |
| | 2026-10-15 20:45 | 20:46 | 2026-10-16 19:40 | | |
| *the other pre-specified tests behind the claims* | | | | | |
| C10 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |
| C11 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |
| C12 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |

| Test | Registered | On the host | Outcome recorded | Order | Words kept |
|---|---|---|---|---|---|
| C13 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |
| C14 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |
| C15 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |
| C16 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |
| C17 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |
| C18 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |
| C19 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |
| C20 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |
| C25 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |
| C38 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |
| C39 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |
| C47 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |
| C48 | d26b35c | 2026-09-02 | d26b35c | same | yes |
| | 2026-09-02 23:33 | 23:33 | 2026-09-02 23:33 | commit | |
| C64 | fe2897d | 2026-09-06 | fe2897d | same | yes |
| | 2026-09-06 20:49 | 20:49 | 2026-09-06 20:49 | commit | |
| C137 | 3ac30d9 | 2026-09-07 | 3ac30d9 | same | yes |
| | 2026-09-07 02:40 | 02:40 | 2026-09-07 02:40 | commit | |
| C138 | 3ac30d9 | 2026-09-07 | 3ac30d9 | same | yes |
| | 2026-09-07 02:40 | 02:40 | 2026-09-07 02:40 | commit | |
| C176 | 28093dc | 2026-09-07 | 3ebb7e1 | before | yes |
| | 2026-09-07 14:42 | 14:49 | 2026-09-07 14:49 | | |
| C179 | 28093dc | 2026-09-07 | 3ebb7e1 | before | yes |
| | 2026-09-07 14:42 | 14:49 | 2026-09-07 14:49 | | |
| C250 | f88e9be | 2026-09-15 | f88e9be | same | yes |
| | 2026-09-15 19:09 | 19:25 | 2026-09-15 19:09 | commit | |
| C251 | f88e9be | 2026-09-15 | f88e9be | same | yes |
| | 2026-09-15 19:09 | 19:25 | 2026-09-15 19:09 | commit | |
| C252 | f88e9be | 2026-09-15 | f88e9be | same | yes |
| | 2026-09-15 19:09 | 19:25 | 2026-09-15 19:09 | commit | |
| C253 | f88e9be | 2026-09-15 | f88e9be | same | yes |
| | 2026-09-15 19:09 | 19:25 | 2026-09-15 19:09 | commit | |
| C254 | f88e9be | 2026-09-15 | f88e9be | same | yes |
| | 2026-09-15 19:09 | 19:25 | 2026-09-15 19:09 | commit | |
| C62 | 8dead17 | 2026-09-06 | 8dead17 | same | yes |
| | 2026-09-06 20:24 | 20:25 | 2026-09-06 20:24 | commit | |
| C63 | fe2897d | 2026-09-06 | fe2897d | same | yes |
| | 2026-09-06 20:49 | 20:49 | 2026-09-06 20:49 | commit | |
| C135 | 3ac30d9 | 2026-09-07 | 3ac30d9 | same | yes |
| | 2026-09-07 02:40 | 02:40 | 2026-09-07 02:40 | commit | |
| C136 | 3ac30d9 | 2026-09-07 | 3ac30d9 | same | yes |
| | 2026-09-07 02:40 | 02:40 | 2026-09-07 02:40 | commit | |
| C139 | 3ac30d9 | 2026-09-07 | 3ac30d9 | same | yes |
| | 2026-09-07 02:40 | 02:40 | 2026-09-07 02:40 | commit | |
| C69 | fe2897d | 2026-09-06 | fe2897d | same | yes |
| | 2026-09-06 20:49 | 20:49 | 2026-09-06 20:49 | commit | |
| C71 | a21cddd | 2026-09-06 | a21cddd | same | yes |
| | 2026-09-06 22:05 | 22:05 | 2026-09-06 22:05 | commit | |

| Test | Registered | On the host | Outcome recorded | Order | Words kept |
|---|---|---|---|---|---|
| C73 | a21cddd | 2026-09-06 | a21cddd | same | yes |
| | 2026-09-06 22:05 | 22:05 | 2026-09-06 22:05 | commit | |
| C103 | 0181d5a | 2026-09-07 | 0181d5a | same | yes |
| | 2026-09-07 01:15 | 01:15 | 2026-09-07 01:15 | commit | |
| C128 | 273bd5c | 2026-09-07 | 28192f4 | before | yes |
| | 2026-09-07 02:21 | 02:24 | 2026-09-07 02:24 | | |
| *football's own pre-specified tests* | | | | | |
| C26 | 2f094ce | 2026-09-02 | 6fd4f6a | before | yes |
| | 2026-09-02 02:20 | 02:21 | 2026-09-02 05:18 | | |
| C27 | 05c16c2 | 2026-09-02 | ac0bc5b | before | yes |
| | 2026-09-02 16:05 | 16:06 | 2026-09-02 16:17 | | |
| C28 | d9f758e | 2026-09-01 | 0193eb6 | before | yes |
| | 2026-09-01 20:20 | 20:20 | 2026-09-01 21:01 | | |
| C29 | 30dfb20 | 2026-09-01 | 0193eb6 | before | yes |
| | 2026-09-01 20:38 | 20:38 | 2026-09-01 21:01 | | |
| C30 | 2c04ef8 | 2026-08-31 | f369b67 | before | yes |
| | 2026-08-31 17:56 | 18:02 | 2026-08-31 18:02 | | |
| C93 | a03e8ce | 2026-09-07 | a03e8ce | same | yes |
| | 2026-09-07 00:05 | 00:05 | 2026-09-07 00:05 | commit | |
| C94 | a03e8ce | 2026-09-07 | a03e8ce | same | yes |
| | 2026-09-07 00:05 | 00:05 | 2026-09-07 00:05 | commit | |
| C95 | a03e8ce | 2026-09-07 | a03e8ce | same | yes |
| | 2026-09-07 00:05 | 00:05 | 2026-09-07 00:05 | commit | |
| C96 | a03e8ce | 2026-09-07 | a03e8ce | same | yes |
| | 2026-09-07 00:05 | 00:05 | 2026-09-07 00:05 | commit | |
| C98 | 0181d5a | 2026-09-07 | 0181d5a | same | yes |
| | 2026-09-07 01:15 | 01:15 | 2026-09-07 01:15 | commit | |
| C99 | 0181d5a | 2026-09-07 | 0181d5a | same | yes |
| | 2026-09-07 01:15 | 01:15 | 2026-09-07 01:15 | commit | |
| C100 | 0181d5a | 2026-09-07 | 0181d5a | same | yes |
| | 2026-09-07 01:15 | 01:15 | 2026-09-07 01:15 | commit | |
| C101 | 0181d5a | 2026-09-07 | 0181d5a | same | yes |
| | 2026-09-07 01:15 | 01:15 | 2026-09-07 01:15 | commit | |
| C102 | 0181d5a | 2026-09-07 | 0181d5a | same | yes |
| | 2026-09-07 01:15 | 01:15 | 2026-09-07 01:15 | commit | |
| C132 | 3ac30d9 | 2026-09-07 | 3ac30d9 | same | yes |
| | 2026-09-07 02:40 | 02:40 | 2026-09-07 02:40 | commit | |
| C165 | 67c8d7e | 2026-09-07 | 82eabfd | before | yes |
| | 2026-09-07 11:42 | 12:06 | 2026-09-07 13:22 | | |
| C166 | 67c8d7e | 2026-09-07 | 82eabfd | before | yes |
| | 2026-09-07 11:42 | 12:06 | 2026-09-07 13:22 | | |
| C275 | aba863e | 2026-09-25 | 2e7a0f7 | before | yes |
| | 2026-09-25 01:05 | 01:10 | 2026-09-25 02:17 | | |
| C276 | aba863e | 2026-09-25 | 2e7a0f7 | before | yes |
| | 2026-09-25 01:05 | 01:10 | 2026-09-25 02:17 | | |
| C277 | aba863e | 2026-09-25 | 2e7a0f7 | before | yes |
| | 2026-09-25 01:05 | 01:10 | 2026-09-25 02:17 | | |
| C278 | aba863e | 2026-09-25 | 2e7a0f7 | before | yes |
| | 2026-09-25 01:05 | 01:10 | 2026-09-25 02:17 | | |
| C279 | f200b21 | 2026-09-29 | 5c69d86 | before | yes |
| | 2026-09-29 11:38 | 16:14 | 2026-09-29 16:31 | | |
| C280 | b3eac4a | 2026-09-29 | 5c69d86 | before | yes |
| | 2026-09-29 11:45 | 16:14 | 2026-09-29 16:31 | | |
| absences | fa7355e | 2026-09-10 | fa7355e | same | – |
| battery | 2026-09-10 13:22 | 13:22 | 2026-09-10 13:22 | commit | |
| increment | 4148b3b | 2026-08-31 | 4148b3b | same | – |
| battery one | 2026-08-31 09:14 | 09:14 | 2026-08-31 09:14 | commit | |
| increment | 7897041 | 2026-08-31 | 7897041 | same | – |
| battery two | 2026-08-31 09:49 | 09:49 | 2026-08-31 09:49 | commit | |

Table S14: The 29 pre-specified tests bearing on the claims that the paper does not report, each with its outcome and the reason. The other 79 of the 108 are reported in the main text or in this supplement.

| Test | Outcome | Why it is not reported |
|---|---|---|
| C14 | held | The load-management law is not claimed in this version. The timing slope inside the departure and the load-management coefficient come from one regression (Supplementary Section S7). This test belongs to the withdrawn law. The text keeps only the tests that read a league's habit (C275 to C280), and the law's fit as description. |
| C15 | held | The load-management law is not claimed in this version. The timing slope inside the departure and the load-management coefficient come from one regression (Supplementary Section S7). This test belongs to the withdrawn law. The text keeps only the tests that read a league's habit (C275 to C280), and the law's fit as description. |
| C16 | refused | The load-management law is not claimed in this version. The timing slope inside the departure and the load-management coefficient come from one regression (Supplementary Section S7). This test belongs to the withdrawn law. The text keeps only the tests that read a league's habit (C275 to C280), and the law's fit as description. |
| C17 | held | The load-management law is not claimed in this version. The timing slope inside the departure and the load-management coefficient come from one regression (Supplementary Section S7). This test belongs to the withdrawn law. The text keeps only the tests that read a league's habit (C275 to C280), and the law's fit as description. |
| C20 | held | Not reported in this version. It reads the women's leagues' timing against the curve of the load-management law, which this version does not claim. |
| C38 | held | Not reported in this version. It reads rugby union's timing, and rugby union is not one of the paper's sports. |
| C39 | failed | Not in the submitted text. Its outcome reads England's starters' workload gradient, a load measure, which this revision leaves to Paper A. |
| C47 | refused | A refusal of one pair of cells, which the pre-specified fallback replaced; the regime constants it serves (C10 to C12) are in Supplementary Section S2. |
| C48 | held | A check of the interval of the dilution behind the baseline count; the text prints the count (2.84) as description, not this check. |
| C64 | held | Not reported in this version. These clauses read England's climb under three substitutions, the drift of a quantity in Paper A's model of when starters come off, or sports outside this paper, and the supplement keeps only what the paper's points use. |
| C137 | held | The load-management law is not claimed in this version. The timing slope inside the departure and the load-management coefficient come from one regression (Supplementary Section S7). This test belongs to the withdrawn law. The text keeps only the tests that read a league's habit (C275 to C280), and the law's fit as description. |
| C138 | held | The load-management law is not claimed in this version. The timing slope inside the departure and the load-management coefficient come from one regression (Supplementary Section S7). This test belongs to the withdrawn law. The text keeps only the tests that read a league's habit (C275 to C280), and the law's fit as description. |
| C176 | held | The load-management law is not claimed in this version. The timing slope inside the departure and the load-management coefficient come from one regression (Supplementary Section S7). This test belongs to the withdrawn law. The text keeps only the tests that read a league's habit (C275 to C280), and the law's fit as description. |
| C179 | held | Not reported in this version. These clauses read England's climb under three substitutions, the drift of a quantity in Paper A's model of when starters come off, or sports outside this paper, and the supplement keeps only what the paper's points use. |

| Test | Outcome | Why it is not reported |
|---|---|---|
| C250 | held | Not reported in this version. These clauses read England's climb under three substitutions, the drift of a quantity in Paper A's model of when starters come off, or sports outside this paper, and the supplement keeps only what the paper's points use. |
| C251 | held | Not reported in this version. These clauses read England's climb under three substitutions, the drift of a quantity in Paper A's model of when starters come off, or sports outside this paper, and the supplement keeps only what the paper's points use. |
| C252 | held | Not reported in this version. These clauses read England's climb under three substitutions, the drift of a quantity in Paper A's model of when starters come off, or sports outside this paper, and the supplement keeps only what the paper's points use. |
| C253 | refused | Not reported in this version. These clauses read England's climb under three substitutions, the drift of a quantity in Paper A's model of when starters come off, or sports outside this paper, and the supplement keeps only what the paper's points use. |
| C254 | held | Not reported in this version. It reads the drift of the retention coefficient, a quantity of Paper A's, and the supplement keeps only what the paper's points use. |
| C62 | failed | Superseded by C71. The archived register gives the full reason. |
| C63 | failed | Superseded by C72 and C69. The archived register gives the full reason. |
| C135 | failed | Superseded by its amendment; its negative result stands. The archived register gives the full reason. |
| C136 | failed | Superseded by its amendment; its negative result stands. The archived register gives the full reason. |
| C139 | failed | Superseded by its amendment; its negative result stands. The archived register gives the full reason. |
| C69 | held | Superseded by C250; its definition was replaced. The archived register gives the full reason. |
| C71 | held | Superseded by C251; its definition was replaced. The archived register gives the full reason. |
| C73 | held | Superseded by C252; its definition was replaced. The archived register gives the full reason. |
| C103 | failed | Superseded by C253; its definition was replaced. The archived register gives the full reason. |
| C128 | held | Superseded by C254; its definition was replaced. The archived register gives the full reason. |

### S4.4 Predictions for records not yet read

Some quantities were fitted across the leagues of several sports at once. Each such fit is a random-effects model. It gives a combined value, the spread $\tau$ of the leagues around it and a 95% prediction interval for a new league [11]. The fits were made on both halves after the split record's tests, so their values are description. Each prediction for a competition or season not yet read was written down before any of its records was read. None was read more than once. A prediction passes when the new value lies within two standard errors of the combined value. A standard error is the typical size of the chance error in a value. Here it combines the new value's own error with the spread $\tau$. So the pass mark is a 95% prediction interval for a new league. Of the eighteen predictions read, thirteen held. Table S15 gives every one in its pre-specified words. Sections S5 and S6 report them.

Two predictions were written down but not read. Both concern IFAB's limit of ten seconds for a substituted player to leave the field. IFAB is football's rule-maker, and the limit took effect in 2026–27 [12]. The rule does not change how many substitutions a team may make. So the first prediction is that the fourteen European leagues keep their order from 2025–26 into 2026–27. The rule also makes a slow late substitution costly. A substituted player who leaves slowly leaves the side a player short for a minute or more. So the second prediction is that the share of each league's substitutions written at the

ninetieth minute falls in 2026–27, against the league's own seasons before. Both needed the records of the whole season. The study ended before the end of that season, so neither was read. Table S15 gives both in their pre-specified words.

A prediction on baseball's order over eight leagues was read after every regular season of 2026 had ended. It predicted that each league keeps its rank in pitchers used per team and game. The ranks came from each league's fitted level on the seasons read before, which Table S15 calls its standing. The prediction then read each league's next season. That season is 2026 for the five summer leagues and 2025–26 for the three winter leagues. A second reading was written down a day after the first. It counts two leagues again: Japan's NPB on all its clubs, and South Korea's KBO on its finished regular-season games. Both readings found every league at its rank (rank correlation 1.00, exact one-sided $P = 2.5 \times 10^{-5}$, the smallest that eight leagues allow). Every league used more pitchers than its standing, by 0.005 to 0.436 a team and game. So the whole sport drifted up while each league kept its rank.

A prediction on football's second tiers was also read once, and held. It asked whether the order under three predicts the order under five in thirty European second tiers whose substitutions had never been read. It was written down, and reached the repository's host, before their team sheets were collected (Table S13). Each league's limit in each season was first taken from its regulations. Suppose the two orders agreed as closely as the first tiers' do (0.82). Then the prediction would hold in 88 per cent of simulated readings with ten leagues, and in 97 per cent with fourteen. With no link it would hold in 5 per cent. Transfermarkt serves many second tiers without line-ups, or without the seasons before 2019–20. A season whose sheets contradicted its listed limit was also left out. So twelve leagues met the clause's rules on seasons, matches and limits. Over them, the rank correlation was 0.73 (one-sided permutation $P = 0.0041$). That is $-0.47$ standard errors from the first tiers' value of 0.82, so the second tiers sit close to it.

Table S15: The predictions on new records, printed word for word as pre-specified. Each row also gives the statistic, its $P$ and $n$, and the outcome. Each prediction was written down after its constant was fitted and before any record it reads was read. None was read more than once. G17 and G18 were not read, because the study ended before the end of the 2026–27 season.

| Clause | Its words, as pre-specified | Statistic ($P$; $n$) | Outcome |
|---|---|---|---|
| G01 | Registered on 2026-09-27, after stage 283 fitted the constants on every game read so far and before any record this clause reads was read. The 2010 three-point line takes one amount off every competition's share of shots from three. In the competitions that moved the line to 6.75 m in 2010-11 and whose 2009-10 and 2010-11 box scores are read after this registration (the EuroCup, which Euroleague Basketball put under the new line from the 2010-11 qualifying rounds, and Germany's BBL and France's Pro A, both 2010-11 in docs/rule_change_harvest/basketball.md), each competition's change in the team-game mean share of three-point attempts among field-goal attempts, 2010-11 less 2009-10, lies within the fitted constant: \|z\| below 2, where z is the change less -0.0382, over the root of the change's squared standard error plus 0.0022 squared (general_formula_constants.csv, kappa_3). The changes pooled by inverse variance are negative with a one-sided P below 0.05. A competition whose source does not serve both seasons is left out; with none left the clause is unresolvable. | -0.0313 ($4.1 \times 10^{-16}$; 2) | failed |

| Clause | Its words, as pre-specified | Statistic ($P$; $n$) | Outcome |
|---|---|---|---|
| G02 | Registered on 2026-09-27, after stage 283 fitted the constants on every game read so far and before any record this clause reads was read. Three-on-three overtime multiplies a league's odds of settling a tied game in overtime rather than by shootout by one factor. In HockeyAllsvenskan, Sweden's second tier, which the federation's series rules for 2016/2017 (fixed 27 May 2016, data/raw/harvest_documents/ice_hockey/sb-2016-2017.pdf, page 6) put on three skaters and a goalkeeper in overtime, the odds ratio of an overtime decision against a shootout among the regular season's extra-time games, 2016-17 and 2017-18 against 2014-15 and 2015-16, exceeds one with a one-sided P below 0.05 and lies within the fitted constant: \|z\| below 2 on the log scale, where z is the log odds ratio less the log of 1.7056, over the root of its squared standard error plus 0.1033 squared (general_formula_constants.csv, kappa_OT, its interval read as a standard error). The decision is read as stage 279 reads the SHL's. If a rules document or report found before the games are read shows three-on-three in HockeyAllsvenskan's 2015-16 regular season, the clause is refused rather than failed. | 1.58 (0.020; 329) | held |
| G03 | Registered on 2026-09-27, after stage 283 fitted the constants on every game read so far and before any record this clause reads was read. A league keeps its standing a(l) in the habit form h(l, t) = mu(t) + sigma(t) a(l) + e(l, t). The standings in the five busiest players' share of minutes were fitted over 2006-07 to 2018-19, each league's mean distance from the season's mean across the seven leagues (general_formula_standings.csv, measure=top5_share, window ending 'additive': acb -0.0487; lba +0.01536; esake -0.02173; aba -0.009197; winner +0.03388; kbl +0.04435; lkl -0.01396). The Spearman correlation between those standings and each league's mean over 2021-22 to 2024-25 is positive with an exact one-sided permutation P below 0.05, over the leagues of the seven (Spain, Italy, Greece, the ABA League, Israel, Korea and Lithuania, with Spain read from 2023-24 only, since stage 277 read its 2019-20 to 2022-23 seasons) whose sources serve at least one season of that window. The measure is read as stage 277 reads it. With fewer than five leagues the clause is unresolvable. | 0.929 (0.0034; 7) | held |
| G04 | Registered on 2026-09-27, after stage 283 fitted the constants on every game read so far and before any record this clause reads was read. A league keeps its standing a(l) in the habit form h(l, t) = mu(t) + sigma(t) a(l) + e(l, t). The standings in the number of players given ten minutes or more were fitted over 2006-07 to 2018-19, each league's mean distance from the season's mean across the seven leagues (general_formula_standings.csv, measure=used_10, window ending 'additive': acb +0.6618; lba -0.1046; esake +0.1816; aba +0.1497; winner -0.4017; kbl -0.718; lkl +0.2311). The Spearman correlation between those standings and each league's mean over 2021-22 to 2024-25 is positive with an exact one-sided permutation P below 0.05, over the leagues of the seven (Spain, Italy, Greece, the ABA League, Israel, Korea and Lithuania, with Spain read from 2023-24 only, since stage 277 read its 2019-20 to 2022-23 seasons) whose sources serve at least one season of that window. The measure is read as stage 277 reads it. With fewer than five leagues the clause is unresolvable. | 0.679 (0.055; 7) | failed |

| Clause | Its words, as pre-specified | Statistic ($P$; $n$) | Outcome |
|---|---|---|---|
| G05 | Registered on 2026-09-27, after stage 283 fitted the constants on every game read so far and before any record this clause reads was read. A league keeps its standing a(l) in the habit form h(l, t) = mu(t) + sigma(t) a(l) + e(l, t). The standings in the share of shots taken from three were fitted over 2006-07 to 2018-19, each league's mean distance from the season's mean across the seven leagues (general_formula_standings.csv, measure=three_share, window ending 'additive': acb +0.0255; lba +0.02089; esake +0.01374; aba +0.01767; winner -0.02498; kbl -0.0471; lkl -0.005718). The Spearman correlation between those standings and each league's mean over 2021-22 to 2024-25 is positive with an exact one-sided permutation P below 0.05, over the leagues of the seven (Spain, Italy, Greece, the ABA League, Israel, Korea and Lithuania, with Spain read from 2023-24 only, since stage 277 read its 2019-20 to 2022-23 seasons) whose sources serve at least one season of that window. The measure is read as stage 277 reads it. With fewer than five leagues the clause is unresolvable. | 0.893 (0.0062; 7) | held |
| G06 | Registered on 2026-09-27, after stage 283 fitted the constants on every game read so far and before any record this clause reads was read. A league keeps its standing a(l) in the habit form h(l, t) = mu(t) + sigma(t) a(l) + e(l, t). The standings in possessions per 40 minutes were fitted over 2016-17 to 2018-19, each league's mean distance from the season's mean across the seven leagues (general_formula_standings.csv, measure=pace_40, window ending 'additive': acb -1.555; lba +0.01722; esake -1.345; aba +0.4379; winner +1.73; kbl +0.5266; lkl +0.188). The Spearman correlation between those standings and each league's mean over 2021-22 to 2024-25 is positive with an exact one-sided permutation P below 0.05, over the leagues of the seven (Spain, Italy, Greece, the ABA League, Israel, Korea and Lithuania, with Spain read from 2023-24 only, since stage 277 read its 2019-20 to 2022-23 seasons) whose sources serve at least one season of that window. The measure is read as stage 277 reads it. With fewer than five leagues the clause is unresolvable. | 0.321 (0.25; 7) | failed |
| G07 | Registered on 2026-09-27, after stage 283 fitted the constants on every game read so far and before any record this clause reads was read. A coach carries a fixed share of a team's distance from its league. Across the moves that R16 defines, whose season after the move was not read before this registration (Israel from 2019-20, Brazil from 2020-21, Spain from 2023-24), the Spearman correlation of the coach's distance from the league in the five busiest players' share of minutes, before and after the move, is positive with a one-sided P below 0.05 from 10,000 random permutations (seed 20260927), and lies within the fitted constant: \|z\| below 2 on Fisher's scale, where z is atanh of the correlation less atanh 0.3025, over the root of 1/(n - 3) plus 0.0542 squared (general_formula_constants.csv, rho_carry, pooled over football, basketball and baseball). With fewer than ten moves the clause is unresolvable. | 0.23 (0.039; 56) | held |

| Clause | Its words, as pre-specified | Statistic ($P$; $n$) | Outcome |
|---|---|---|---|
| G08 | Registered on 2026-09-28, after stage 290 fitted the forms with added terms on every game read so far and before any record this clause reads was read. A league keeps its standing a(l) in the habit form h(l, t) = mu(t) + sigma(t) a(l, t) + e(l, t), and with six leagues that time the goalie the order test no longer needs the same order twice. The standings in the seconds left in regulation at a trailing team's first late pull (stage 279's measure: a team trailing by one or two goals, its goalie's first exit in the last ten minutes) were fitted over 2015-16 to 2018-19, each league's mean distance from the season's mean of the six league medians (general_forms_hockey_standings.csv: Finland's Liiga +12.42; Sweden's SHL +10.67; Sweden's HockeyAllsvenskan +4.792; Switzerland's National League -6.583; the ICE league -9.083; the Swiss League -12.21). The Spearman correlation between those standings and each league's standing, computed the same way, over 2021-22 to 2024-25 is positive with an exact one-sided permutation P below 0.05 over the six leagues (720 orders), which needs a correlation of 0.829 or more, the squared differences of rank summing to 6 or less. Each league is read by the tool and the phases that served its fitted window; the two second tiers are read as their first tiers are. A league whose four seasons are not all served is left out, and with fewer than five leagues the clause is unresolvable. Under the drift of G09 (phi 0.969 a season, six seasons apart) and the fitted window's sampling error, the clause holds with a probability near 0.39 (20,000 simulated draws), so a failure alone does not refute the form. | 0.771 (0.051; 6) | failed |
| G09 | Registered on 2026-09-28, after stage 290 fitted the forms with added terms on every game read so far and before any record this clause reads was read. A league's standing drifts: a(l, t) = phi a(l, t-1) + noise, so the leagues' standings k seasons apart correlate phi$^k$ once sampling noise is taken out, and phi is one value across sports. Fitted by least squares on every pair of seasons, each correlation divided by the root of the two seasons' reliabilities (the share of the spread between leagues that is not sampling error), phi is 0.9712 in basketball (four measures stacked: the five busiest players' share of minutes, players given ten minutes or more, the share of shots from three and possessions per 40 minutes; seven leagues, 2006-07 to 2024-25), 0.9258 in volleyball (two measures, eight leagues), 0.9981 in ice hockey (the seconds left at the first late pull, the six leagues of G08, 2015-16 to 2018-19) and 0.9406 in baseball (pitchers per team and game, four leagues); pooled by inverse variance 0.9689 (standard error 0.0128), one value (Q 0.79 on 3 degrees of freedom, P 0.85) (general_forms_drift.csv). In each sport with records not read before this registration, phi fitted the same way on the pairs of seasons in which at least one season is new lies within the pooled value: \|z\| below 2, where z is that phi less 0.9689 over the root of its squared standard error (leagues resampled, 500 draws, seed 20260927) plus 0.0128 squared. The new records are ice hockey's 2021-22 to 2024-25 over the six leagues of G08, paired with every season from 2015-16, and basketball's 2025-26 over the seven leagues of G03, paired with every season from 2006-07, its four measures stacked. Spain's and Israel's 2025-26 records were fetched on 2026-09-28 by the harvest for G03 to G06 and none of the four measures was computed on them. The clause holds when every sport read holds. A sport whose new seasons serve fewer than four leagues is left out; with neither sport, the clause is unresolvable. | 0.203 (0.84; 2) | held |

| Clause | Its words, as pre-specified | Statistic ($P$; $n$) | Outcome |
|---|---|---|---|
| G10 | Registered on 2026-09-28, after stage 290 fitted the forms with added terms on every game read so far and before any record this clause reads was read. The 2010 three-point line takes off a competition's share of shots from three a fixed proportion of the share it started from, give or take a spread between competitions: change = kappa_rel x share before + u, the spread of u being tau. Over the eight competitions read (Spain, Italy, Lithuania, Greece, the ABA League, Israel, the EuroCup and the BBL's main rounds), kappa_rel is -0.0991 (standard error 0.00827) and tau is 0.0177, the DerSimonian-Laird spread (general_forms_line.csv: rel_pooled, rel_pooled_se, rel_tau); one proportion without the spread is refused there (Q 17.1 on 7 degrees of freedom, P 0.017). The new competitions, with their seasons before and after the line: the EuroLeague, 2009-10 and 2010-11 (Euroleague Basketball's new rules from the 2010-11 qualifying rounds), every game its box-score service lists; Spain's Liga Femenina, 2009-10 and 2010-11 (the federation's news of 15 October 2010), and Spain's LEB Oro, 2008-09 and 2009-10 (the federation's first competition on the new line, from October 2009), each on the regular season its calendar lists; Poland's PLK, 2009-10 and 2010-11 (the season preview of 8 October 2010), on its regular season; and Portugal's LPB and Liga Feminina, 2009-10 and 2010-11 (the federation's decision of April 2010), on every game their calendars list. In each, the statistic is the team-game share of three-point attempts among field-goal attempts; the change is the after season's mean less the before season's, and the ratio is the change over the before season's mean, its error by the delta method (stage 290, competition()). The clause holds when the ratios lie jointly within the form, the squared distance r'$S^{-1}$ r below the 95th percentile of chi-square with as many degrees of freedom as competitions read, where r is each ratio less -0.0991 and S has each ratio's squared error plus 0.0177 squared on its diagonal and 0.00827 squared in every cell, and the changes pooled by inverse variance are below zero with a one-sided P below 0.05. A game is kept only when each team's points equal its shots (tools/fetch_eurocup_shooting.py –competition E, fetch_feb_shooting.py, fetch_plk_shooting.py, fetch_fpb_shooting.py). A competition that keeps fewer than 100 team-games in either season is left out, and with fewer than two the clause is unresolvable. The competitions considered and left out, each with its reason, are listed in docs/rule_change_harvest/general_formulas.md before any is read. About ten box scores of these seasons, one game each, were opened while the sources were identified, and no share was computed on them. Spain's Liga Femenina pages were fetched on 2026-09-24 by Paper 1's harvest (tools/fetch_lf_endesa.py), which reads that league from 2023-24 only and never its shots. The single amount with its spread (-0.0372, standard error 0.00315, tau 0.00687) is reported beside the clause as a variant. | 8.44 (0.21; 6) | held |

| Clause | Its words, as pre-specified | Statistic ($P$; $n$) | Outcome |
|---|---|---|---|
| G11 | Registered on 2026-09-28, after stage 290 fitted the standings on every game read so far and before any record this clause reads was read. A league keeps its standing in how many pitchers a manager uses, and with eight leagues the order test no longer needs the same order twice. The standings are each league's mean pitchers per team and game over its two seasons before the test (general_forms_baseball_standings.csv): the Dominican LIDOM 6.320 (2023-24, 2024-25); Venezuela's LVBP 5.900 (2023-24, 2024-25); the Mexican League 5.282 (2024, 2025); Mexico's Pacific LMP 5.203 (2023-24, 2024-25); Korea's KBO 4.885 (2024, 2025); Taiwan's CPBL 4.319 (2024, 2025); MLB 4.277 (2024, 2025); Japan's NPB 4.116 (2024, 2025). MLB, the Mexican League, the CPBL and the KBO are read as stage 280 reads them; Japan's NPB is counted as one plus half the game's pitching changes, from the substitution rows of its records, since they keep one pitching line a team; the winter leagues from their box scores (tools/fetch_statsapi_baseball.py). The Spearman correlation between those standings and each league's mean in its next complete season (2026 for MLB, the NPB, the KBO, the CPBL and the Mexican League, read after each regular season has ended; 2025-26 for the three winter leagues) is positive with an exact one-sided permutation P below 0.05 over the eight leagues (40,320 orders), which needs 0.643 or more. Regular-season games only, as each source lists them. A league whose next season keeps fewer than 100 team-games is left out, and with fewer than six leagues the clause is unresolvable. The standings span 4.12 to 6.32 pitchers, far beyond their sampling error; under the drift of G09 (phi 0.969 a season) the clause holds with a probability near 1.00 (20,000 simulated draws). | 1 ($2.5 \times 10^{-5}$; 8) | held |

| Clause | Its words, as pre-specified | Statistic ($P$; $n$) | Outcome |
|---|---|---|---|
| G11a | Registered on 2026-09-29, after stage 300 read the NPB's and the KBO's 2024 and 2025 seasons again and before any record this clause reads was read. G11a is G11 with two of its eight standings counted again. G11 counted Japan's NPB as one plus half the game's pitching changes, on eleven of its twelve clubs, because the harvested records missed every game of DeNA and kept only each side's first pitcher. G11 counted Korea's KBO on every page its source lists, among them games called off before they were official, which the league voids and plays again, a page left in a live state whose pitching table lists the bullpen, and the exhibition and postseason games. G11a counts the NPB from its box scores on all twelve clubs, and the KBO from its finished regular-season games. A league keeps its standing in how many pitchers a manager uses. The standings are each league's mean pitchers per team and game over its two seasons before the test (g11a_standings.csv): the Dominican LIDOM 6.320 (2023-24, 2024-25); Venezuela's LVBP 5.900 (2023-24, 2024-25); the Mexican League 5.282 (2024, 2025); Mexico's Pacific LMP 5.203 (2023-24, 2024-25); Korea's KBO 4.892 (2024, 2025); Taiwan's CPBL 4.319 (2024, 2025); MLB 4.277 (2024, 2025); Japan's NPB 4.117 (2024, 2025). The six standings other than the NPB's and the KBO's are G11's, and stage 300 recomputes each from the same records to the last digit. An NPB game counts when its box score's heading names a Central League game, a Pacific League game or an interleague game, and its state line says the game was played out or called once it was official. A no-game and a cancelled game are not games. A side's pitchers are the pitchers its box score lists. A KBO game counts when its page marks it Final, or Delayed (called once it was official), and when it falls inside the regular season, from the first to the last day the season's summary gives. A Bullpen table is never counted, and each game is counted once, under its own number. Each league's next complete season is read the same way. The NPB's 2026 is read from the records of tools/fetch_npb_games.py, which from 2026-09-29 keep every pitcher and each box score's heading and state line, on all twelve clubs. The KBO's 2026 is read from the records of tools/fetch_kbo_mykbostats.py, which from 2026-09-29 keep each page's state. Its regular season's first and last day are entered from the season's summary after the season ends and before any of its records is read. The other six leagues are read as G11 reads them: 2026 for MLB, the CPBL and the Mexican League, read after each regular season has ended, and 2025-26 for the three winter leagues. The Spearman correlation between the standings and each league's mean in its next complete season is positive with an exact one-sided permutation P below 0.05 over the eight leagues (40,320 orders), which needs 0.643 or more. Stage 300's g11a asks it with stage 291's own function for G11. A league whose next season keeps fewer than 100 team-games is left out, and with fewer than six leagues the clause is unresolvable. The standings span 4.12 to 6.32 pitchers. The sampling error of one season's mean is at most 0.096 pitchers (LIDOM's). Under the drift of G09 (phi 0.969 a season, 1.5 seasons from the fitted window's centre to the test season), the clause holds with a probability near 1.00 (0.9998, 20,000 simulated draws). G11 itself is read as registered: the NPB's 2026 as one plus half the game's pitching changes on the same eleven clubs, and the KBO's 2026 from every page and every row, as stage 280 reads them. Both results are reported. | 1 ($2.5 \times 10^{-5}$; 8) | held |

| Clause | Its words, as pre-specified | Statistic ($P$; $n$) | Outcome |
|---|---|---|---|
| G12 | Registered on 2026-09-28, after stage 290 fitted the drift of game length on every game read so far and before any record this clause reads was read. A first pitch clock takes off a share of the game in step with how tight it is: share = a - kappa x (A - 17.5), where share is one less the mean game length with the clock over the mean without it, and A is the clock's allowance, the mean of the seconds it allows with the bases empty and with runners on (data/manual/pitch_clocks.csv). The first clocks not read before, each from the season before to the season of its clock, on the regular-season games scheduled for nine innings that MLB's statistics service lists as final, with their stated length (tools/fetch_statsapi_baseball.py): Triple-A (A = 16.5), Double-A, High-A and the Single-A leagues other than the California League (A = 16 each), 2021 against 2022, when the clock came to every full-season level but the California League, which had used one in 2021 (MLB, 14 March 2022); the Dominican league, LIDOM (A = 20), 2022-23 against 2023-24; and Venezuela's LVBP (A = 17.5), within 2023-24, games before 1 December 2023 against games from it. Each unit is weighted by one over its share's squared error (delta method) plus (4.53 over its mean before) squared, 4.53 minutes being the drift of a league's mean game length between seasons with the same clock (general_forms_clock.csv, tau_minutes). The clause holds when kappa, fitted by weighted least squares, is positive with a one-sided P below 0.05, and the units lie on the fitted line: Q on m - 2 degrees of freedom with P above 0.05. A unit with fewer than 40 games in either period is left out, and with fewer than five units the clause is unresolvable. The minor leagues' 2022 season also brought larger bases at every level, limits on the shift below Triple-A and the automated ball-strike system in Triple-A and Low-A Southeast; the result is read with them. If the constant were MLB's of 2024 (3.5 minutes a second of allowance, about 0.022 of the game), the clause would hold about eight times in ten (20,000 simulated draws with the units' expected errors). MLB's own first clock (2023, A = 17.5, 13.1 per cent of the game) was read before and is printed beside the result, not in it. | 0.0101 (0.081; 6) | failed |
| G13 | Registered on 2026-09-28, after stage 295 fitted cricket's standings on every innings of the seasons opening in 2022 or earlier, and before any innings of a later season was read. Cricket caps each bowler at a fifth of the innings, four of a T20 match's twenty overs, so a side needs at least five bowlers; how many more it uses is a league's habit under a rule, and a league keeps its standing in it. The standings are each T20 league's mean bowlers used per full-length team-innings (legal balls equal to the 120 scheduled: no side bowled out, no chase won early, no innings shortened by rain) over its seasons opening from 2019 to 2022, from Cricsheet's ball-by-ball records (data/raw/cricsheet/all_json.zip; cricket_bowling_standings.csv): LPL 6.455; CSA T20 6.258; CPL 6.224; WBBL 6.142; WPL 6.136; BPL 6.107; SA20 6.083; Super Smash (women) 6.057; BBL 6.031; ILT20 6.000; Super Smash (men) 5.957; PSL 5.956; IPL 5.797. A season is named by the year it opens (2022/23 is 2022). The Spearman correlation between those standings and each league's mean over its seasons opening in 2023, 2024 and 2025 is positive with a one-sided P below 0.05 from 10,000 random permutations (seed 20260927), over the thirteen T20 leagues. The Hundred and the Rachael Heyhoe Flint Trophy, whose formats differ, are described beside it and not tested. Every match the archive holds for these competitions is read, league games and play-offs alike. A league with fewer than 20 full-length innings in those seasons is left out, and with fewer than ten leagues the clause is unresolvable. Under the drift of G09 (phi 0.969 a season, about 3.5 seasons apart) and the two windows' sampling errors, the clause holds about four times in five (20,000 simulated draws). | 0.626 (0.013; 13) | held |

| Clause | Its words, as pre-specified | Statistic ($P$; $n$) | Outcome |
|---|---|---|---|
| G14 | Registered on 2026-09-28, after stage 295 fitted cricket's standings on every innings of the seasons opening in 2022 or earlier, and before any innings of a later season was read. A league's standing keeps one share of itself a season in cricket as in the four sports of G09. Phi, fitted as in G09 on the pairs of seasons in which at least one season opens in 2023, 2024 or 2025, over the thirteen T20 leagues of G13, with two measures stacked (bowlers used per full-length innings, and the five busiest bowlers' share of its legal balls), lies within the pooled 0.9689 of stage 290: \|z\| below 2, where z is that phi less 0.9689 over the root of its squared standard error (leagues resampled, 500 draws, seed 20260927) plus 0.0128 squared. A season is named by the year it opens, and a season with fewer than four leagues is skipped, as in G09. With fewer than six leagues in the new seasons the clause is unresolvable. | -1.14 (0.26; 13) | held |
| G15 | Registered on 2026-09-29, after stage 290 fitted the drift's persistence and before any season's value of the measure below was computed for any league it names. A league's standing keeps one share of itself a season in how widely its clubs share out their minutes, as it does in the four sports of G09 and in cricket (G14). The measure is the busiest-five share: for every team-game, the time on ice of the side's five busiest skaters over the time of all its skaters, goalies left out, as stage 292 reads it. A league-season's value is the mean over its team-games. The leagues are the time-on-ice leagues that Paper 1's gate reads (stage 259) and whose minutes add up there, that is, which have a minutes era in jsams_cross_sport_eras.csv; the gate's question of professional status plays no part. Each is read by its stage 259 reader as it stands, on the seasons Paper 1 registered it on within that era, from 2017-18, the first season at least four of them serve: the NHL (2017-18 to 2024-25, regular season), the Swiss National League (2017-18 to 2025-26), the Swiss League (2020-21 to 2025-26), Finland's Liiga (2022-23 to 2025-26) and Mestis (2017-18 to 2025-26), the Czech Extraliga (2017-18 to 2025-26), and Germany's DEL and Sweden's SHL (2023-24 to 2025-26). A game whose record breaks the exposure identity is left out whole, both sides, as Paper 1's gate quarantines it. Phi, fitted as in G09 on every pair of seasons that at least four leagues serve (33 pairs if every season named is served), lies within the pooled 0.9689 of stage 290: \|z\| below 2, where z is that phi less 0.9689 over the root of its squared standard error (leagues resampled, 500 draws, seed 20260927) plus 0.0128 squared. With fewer than 20 pairs, fewer than five leagues or no standard error, the clause is unresolvable. Left out before any record was read: DEL2, whose reader keeps one whole season (2025-26) and so serves no pair; and the Elite League and the Czech second tier, whose minutes do not add up in Paper 1's gate (4.4 and 3.4 per cent of their team-games break the identity). The two are read beside the clause as description and change no verdict. What Paper B had read of these records: stage 292 printed each first division's mean busiest-five share over its whole minutes era, and no season's value; nothing of the Swiss League's or Mestis's time on ice had been read. Paper C's stage 293 had printed the NHL's forwards' and defencemen's median share of the game, pooled over every season. Under the drift of G09, with each league-season's sampling error set by Paper 1's count of its team-games, the clause holds 988 times in 1,000 simulated readings. If the standings kept only 0.8 of themselves a season, it would still hold about 81 times in 100, and it fails half the time only near 0.6. So a pass says little, and a failure would say much. | -0.0516 (0.96; 8) | held |

| Clause | Its words, as pre-specified | Statistic ($P$; $n$) | Outcome |
|---|---|---|---|
| G16 | Registered on 2026-09-29, after stage 290 fitted the drift's persistence and before any season's value of the four measures below was computed for any league it names. Basketball's standings keep the share a season that G09 found in the seven leagues of its fit, in leagues outside that fit. The measures are G09's four, per team-game as stage 277 defines them: the five busiest players' share of the side's minutes; players given ten minutes or more; the share of field-goal attempts taken from three; and possessions (field-goal attempts, plus 0.44 free-throw attempts, plus turnovers, less offensive rebounds) per 40 minutes of the side's time. A league-season's value of each is the mean over its team-games. The minutes are each league's stage 259 reader's, as they stand. The shooting counts are summed over the players' lines of the same box scores, so a count credited to a team and to no player is left out in every league alike. A game whose minutes break the exposure identity is left out whole, as Paper 1's gate quarantines it. A season is named by the year it opens (the WNBA's 2016 beside the NBA's 2016-17). The leagues are Paper 1's basketball panels outside phi's fit whose seasons Paper B has not read, on the seasons Paper 1 registered within their minutes era, from 2016, the first season four of them serve: the NBA (2016-17 to 2024-25, regular season), the WNBA (2016 to 2025, regular season), the G League (2016-17 to 2025-26, regular season and Showcase Cup), Japan's B.LEAGUE (its B1 regular season, 2016-17 to 2025-26) and France's Betclic ELITE (every finished game, 2023-24 to 2025-26). Phi, fitted as in G09 with the four measures stacked, on every pair of seasons that at least four leagues serve (38 a measure, 152 in all, if every season named is served), lies within the pooled 0.9689: \|z\| below 2, where z is that phi less 0.9689 over the root of its squared standard error (leagues resampled, 500 draws, seed 20260927) plus 0.0128 squared. With fewer than 20 pairs, fewer than four leagues or no standard error, the clause is unresolvable. Left out before any record was read: Turkiye's BSL, whose seasons on disk (2006-07 to 2018-19) Paper B read for R03, R12 and R14; the seven leagues of the fit; and Germany's BBL, whose harvest had not ended. What Paper B had read of these records: stage 292 printed the NBA's, the WNBA's, the G League's and the B.LEAGUE's means over their whole minutes eras (the busiest-five share, players given ten minutes, players used), and no season's value; Paper B had read nothing of the Betclic ELITE's box scores. Paper A's stage 294 had printed each league's share of appearances that were starts, pooled over its seasons, and the former Paper 2's stage 240 had counted the NBA's substitutions a game in three seasons; neither is one of the four measures by season. Under the drift of G09, with each league-season's sampling error set by Paper 1's count of its team-games, the clause holds 998 times in 1,000 simulated readings. If the standings kept only 0.8 of themselves a season, it would still hold about 79 times in 100, and it fails half the time only near 0.65. So a pass says little, and a failure would say much. | 0.0933 (0.93; 5) | held |

| Clause | Its words, as pre-specified | Statistic ($P$; $n$) | Outcome |
|---|---|---|---|
| G17 | Registered on 2026-10-15, after stage 343 read the fourteen leagues' seasons to 2025-26 for its base rates and before any record of 2026-27 was read. IFAB's Laws of the Game 2026/27, in force from 1 July 2026, give a substituted player ten seconds to leave the field; a player who takes longer still leaves, but the substitute may enter only at the first stoppage after one minute of play. The rule applies in all fourteen leagues from their first match of 2026-27 (data/manual/paper_b_ifab_2026_rule.csv gives each league's source). It does not change how many substitutions a team may make, so the fourteen European leagues of Paper B's fan (Belgium, Denmark, England, France, Germany, Greece, Italy, the Netherlands, Portugal, Russia, Scotland, Spain, Turkey and Ukraine) keep their order in substitutions per team-match from 2025-26 into 2026-27: Spearman's rho between the two seasons' values is at least 0.70. The measure is stage 298's: players fielded per club-match less eleven, in the domestic league matches of each competition's own season, over the club-matches that record at least eleven players fielded and every starter their line-up lists; a league-season enters with at least 100 such matches and only when stage 62's signature certifies it five. Both seasons are read from the first transfermarkt-datasets snapshot retrieved on or after 1 July 2027. With fewer than twelve leagues read in both seasons the clause is unresolvable. The bar is fixed from the persistence measured so far: with the limit unchanged, the fourteen leagues' order held from one season to the next at 0.74 to 0.90 (Spearman, six pairs of seasons), and the drift fitted to their seasons under five (phi 0.919, standard error 0.054, fitted as stage 290 fits it) gives rho of 0.70 or more 93 times in 100 simulated readings, each league-season's sampling error set by its 2025-26 team-matches (99 in 100 given 2025-26's known levels). If the standings kept only 0.8 of themselves that season, it would still hold 63 times in 100; it holds half the time near 0.75. So a pass says little: the rule left the leagues where they were. A failure would say much: the rule reordered the leagues. | – | not read |

| Clause | Its words, as pre-specified | Statistic ($P$; $n$) | Outcome |
|---|---|---|---|
| G18 | Registered on 2026-10-15, with G17 and on its terms: the rule, the fourteen leagues, the snapshot and stage 298's completeness rules. The ten-second limit makes a late substitution that wastes time costly, because a player who leaves slowly leaves their side a player short for a minute or more. So the share of a league's substitutions recorded at minute 90 falls in 2026-27, each league by its own amount, judged as Paper B advises against each league's own seasons before the rule. The source records every substitution made in stoppage time at minute 90 and no later minute. A league-season's late share is its substitution events at minute 90 over all its substitution events, over the club-matches that pass stage 298's completeness rules and whose events count as many substitutions as their appearances (stage 338's check), in a league-season of at least 100 such matches certified five by stage 62's signature. Each league's 2026-27 share is set against its own mean share of 2022-23, 2023-24, 2024-25 and 2025-26, the four seasons under five that all fourteen leagues played in full; all five seasons are read from the first transfermarkt-datasets snapshot retrieved on or after 1 July 2027. The clause holds if the share falls in all but at most two of the leagues read, with at least twelve read in all five seasons; otherwise it fails, or it is unresolvable with fewer than twelve leagues read. Beside it, changing no verdict: each league's change and its error, the changes' heterogeneity (Cochran's Q and tau), and the order of the late share (Spearman between each league's four-season mean and its 2026-27 share). The base rates, from the seasons to 2025-26: under five the leagues' late shares ran from 0.034 (Italy) to 0.084 (Turkey); from season to season they moved by a spread common to all leagues of 0.0025 and a league-season residual of 0.0083; between consecutive seasons under five they fell in 4, 8 and 8 of the fourteen leagues; and 2025-26, set against each league's own mean of 2022-23 to 2024-25, fell in ten. Without any effect of the rule the clause would pass 3 times in 100 (9 in 100 if the common spread were the 0.0047 measured across both eras); if the rule removed a tenth of each league's late substitutions it would pass 31 to 36 times in 100, a fifth 75 to 83 times, and three tenths 97 to 99 times; it passes half the time if the rule removes 13.5 per cent. So a pass would say that the rule cut late substitutions in nearly every league; a failure would say that it cut them by less than about a seventh in most leagues, or that something else lifted them. | – | not read |

| Clause | Its words, as pre-specified | Statistic ($P$; $n$) | Outcome |
|---|---|---|---|
| G19 | Registered on 2026-10-15, before the second tiers named below were harvested and before any of their substitution counts was computed. Their limits are compiled from their regulations into data/manual/paper_b_second_tier_limits.csv before the clause is read. A league's order under three substitutions foretells its order under five in football leagues whose substitutions Paper B has never read. The leagues are the 30 second tiers of the harvest's own index named in stage 344: England's Championship, Germany's 2. Bundesliga, Spain's LaLiga 2, Italy's Serie B, France's Ligue 2, the Netherlands' Keuken Kampioen Divisie, Portugal's Liga Portugal 2, Turkey's 1. Lig, Belgium's Challenger Pro League, Scotland's Championship, Greece's Super League 2, Russia's First League, Ukraine's Persha Liga and Denmark's 1st Division, and the second tiers of Austria, Switzerland, Poland, Czechia, Romania, Serbia, Croatia, Bulgaria, Hungary, Israel, Cyprus, Armenia, Sweden, Norway, Finland and Kazakhstan, read from the project's Transfermarkt sheets in data/raw/second_tiers. A league is included with at least three seasons under three among the four before 2019-20 (2015-16 to 2018-19; 2016 to 2019 for a league played in calendar years) and at least one full season under five from 2020-21 to 2025-26 (calendar 2020 to 2025), each by its regulations, each with at least 100 matches whose sheets are complete (eleven different starters a side, every substitution a new player for one on the field) and certified for its limit by stage 62's signature read in sheet form (substitutes entering before minute 90 per complete team-match: three at most 3.05, five at least 3.60). A season is named by its match dates, and a league's mean under each limit weights its seasons by their complete team-matches. Spearman's rho between the included leagues' means under three and under five is positive, with a one-sided P below 0.05 from 100,000 random permutations (seed 20261015). With fewer than ten leagues included the clause is unresolvable. Beside it, changing no verdict: the new leagues' rho against the first tiers' 0.82 (stage 336), by Fisher's z with variance 1.06/(n - 3) for each. What Paper B had read of these records: nothing; no deposit names a second tier beside a substitution column, and no sheet of one was on disk on 15 October 2026 (paper_b_order_new_leagues_coverage.csv). If the new leagues' means under the two limits correlate as the first tiers' do (0.82), the clause holds 88 times in 100 with ten leagues included, 97 with fourteen and more than 99 with twenty; with no relation it holds 5 times in 100; with fourteen leagues it holds half the time near a true correlation of 0.49. So a pass would say that the order under three foretold the order under five beyond the leagues it was found in; a failure would say that the first tiers' kept order does not travel to the tier below. | 0.734 (0.0041; 12) | held |

# S5 Direct effects and the coach's carry

## S5.1 A rule's direct effect

A rule's direct effect is a change that is the same in every league. Each league keeps its own value of the measure, and the rule moves each value by one common amount or factor. For example, three-on-three overtime in ice hockey multiplied every league's odds of settling a tied game before a shootout by one factor, $\kappa_{\mathrm{OT}}$:

$$\mathrm{odds}_{\mathrm{after}} \;=\; \kappa_{\mathrm{OT}}\,\mathrm{odds}_{\mathrm{before}}. \tag{S1}$$

Over both halves, $\kappa_{\mathrm{OT}} = 1.71$ (95% interval 1.39 to 2.09). The five leagues' own factors are consistent with that one value ($P = 0.58$). In Sweden's second tier of ice hockey, read later as a prediction,

three-on-three overtime multiplied the odds by 1.58 ($P = 0.020$). That is $-0.3$ standard errors from the combined value, so the new league fits it.

Section S4 gives a larger factor, because the pre-specified test of overtime read half B alone. There the factor was 2.21 (1.65 to 2.97), over 756 games that went beyond regulation time. Half A gave 1.33 (1.01 to 1.77). Both halves together give 1.71 over 1,538 games, the combined value above. The method, the leagues and the seasons are the same in all three. The halves differ by a ratio of 1.66 (1.11 to 2.49, two-sided $P = 0.014$). A gap this large is rare. But the games went to the halves by a rule that works like a coin toss, so the gap can only be chance. Half B's value is high by chance, and the combined value uses every game.

A second example is the 2010 line, which moved basketball's three-point line further out. It took one amount off every league's share of shots from three: 3.8 percentage points (3.4 to 4.3). The six leagues' own amounts are consistent with that one value ($P = 0.15$). Studies of single leagues also found a cut [13, 14, 15, 16]. But the amount did not carry to every new competition. The line took 4.5 points off the share of the EuroCup, Europe's second-tier club competition. That lies $-1.0$ standard errors from the combined value, inside the pass mark of two. It took 2.3 points off the share of Germany's top basketball league. That lies $+2.8$ standard errors away, outside the pass mark. Over the eight competitions read, the cut was not one proportion of each starting share either. The test of one proportion gave $P = 0.017$, and the test of one amount $P = 0.013$. The eight started from shares between 0.35 and 0.39. Those shares were too close together to tell an amount from a proportion.

So the competitions spread around any one common value. Write the cut as a proportion of the share $s_c$ that competition $c$ started from:

$$\frac{\Delta s_c}{s_c} \;=\; \kappa_{\text{rel}} + u_c. \tag{S2}$$

Then the line took 9.9 per cent of the share. Each competition's own part, $u_c$, has a spread of 1.8 per cent. With that spread, the model held on six competitions not read when it was written down. Two of them were women's leagues. They started from shares between 0.27 and 0.38 and lost between 2.0 and 4.4 points. Together they lay within the model ($P = 0.21$). One amount with its own spread fitted them as well ($P = 0.30$).

### S5.2 The pitch clock

The pitch clock took no common amount of minutes. The leagues' clocks differ in the seconds they allow, in the season they began and in how they were enforced. So the leagues' spread in game length after the clock partly reflects different rules, and not different use of one rule. Each league's clock was therefore read from its rules, and we looked for a common value per second of clock. A clock's allowance is the mean of the seconds it allows with the bases empty and with runners on. Only a change that moves the allowance and nothing else can give that value. In the four leagues only one change did so. In 2024 MLB took two seconds off its clock with runners on, so its allowance fell by one second. Its games were 3.5 minutes shorter. But a league's mean game length moves by 4.5 minutes from season to season under the same clock (25 pairs of seasons). So the common value, 3.5 minutes for each second of allowance, carries an error of 4.6 minutes a second. For example, the top leagues of Taiwan and South Korea, the CPBL and the KBO, enforced the same clock. Over both halves, their games were 15.9 and 7.9 minutes shorter. A prediction for the next cut of two seconds in the allowance would be 7.0 minutes, give or take 20 minutes. No result could prove a prediction that wide wrong, so none was written down.

The pre-specified test read half B alone. On half B, MLB's games were 24.6 minutes shorter, the Mexican League's 16.3, the CPBL's 15.3 and the KBO's 9.8. The values in this section read both halves: 24.4, 15.1, 15.9 and 7.9 minutes. The seasons compared are the same.

Within MLB the clock also made game lengths more uniform. Their spread fell from 23.4 minutes in 2022 to 19.7 in 2023, a ratio of 0.84 (0.81 to 0.89). Over 2023 to 2025, against 2021 and 2022, the ratio was 0.76. The ratios in the Mexican League, the CPBL and the KBO were 0.77, 0.79 and 0.88. The records do not separate extra innings. MLB's changes of 2023 also limited the defensive shift and enlarged the bases. So the clock was not the only change behind this.

A league's first clock is a larger change, so it gives a stronger test. We wrote down a model: a first clock takes one share of the game for each second of its allowance. We then read six first clocks. They were in MLB's minor leagues and in two winter leagues, the Dominican Republic's and Venezuela's. On them the model failed. The share of the game fell by only 0.010 for each second of allowance, which is not beyond chance ($P = 0.081$). And the leagues did not lie on one line ($P = 0.025$). The same clock took 10 per cent of one league's game and 17 per cent of another's.

### S5.3 The coach's carry

A team's distance from its league is its substitutions per team-match minus its league's mean that season. It is averaged over each coach's spell. Basketball's distance uses the five busiest players' share of minutes, and baseball's the players used beyond the starters, each minus its league's mean. The coach's carry is the rank correlation between the old team's distance and the new team's, across the coaches who moved. The old team's distance is read over the coach's spell there, and the new team's over the spell that follows the move. Table S16 gives every reading of the carry. It also gives the slope of a straight line fitted to the distance after a move against the distance before.

Suppose a team's distance is the sum of three parts: the coach's, the club's and the season's. Suppose too that the parts are unrelated to one another, and that the distance has the same spread before and after a move. Then the correlation across a move is the coach's share of the variance in distance, that is, of the differences between teams. We call these assumptions the additive model. Read as a share, the carry assumes it. Under it, a combined carry of 0.30 means that about 30 per cent of a team's distance travels with its coach. For example, suppose a coach leaves a team 0.4 substitutions a match above its league. The coach then takes about 0.12 of them to the new team, about 4 a season. The slope needs no such assumption. It was 0.29 in football (0.13 to 0.45), 0.26 in basketball and 0.26 combined. In football the spreads before and after a move were close, 0.42 and 0.40, and so were basketball's. So in both sports the correlation and the slope agree. In MLB the spreads were not close: 0.66 before a move and 0.44 after.

Football's managers who changed club within one league carried 0.32 over 192 moves, in a pre-specified test (two-sided $P = 6.5 \times 10^{-6}$). Those who changed league carried 0.27 over 99 moves, in another (two-sided $P = 0.0063$). Basketball's coaches carried 0.29 over 133 moves on every game. On half B, where the split record's clause read them, they carried 0.27 over 134 moves ($P = 0.0015$). In basketball, a coach counts as moving when fewer than fifteen per cent of the new team's players played for the coach's team the season before. The rule finds a move from each reading's own box scores, so the two readings differ. They share 131 moves. Half B alone finds three, and every game alone finds two. Two coaches stayed with a team whose roster changed by more than the rule's bar on half B's box scores, but not on every game's. So half B counts them as moves. One coach led two teams in a season, and the two readings date that move a season apart. One coach's only game at a new team fell in half A. Basketball's moves also include ten coaches who stayed with a team whose players almost all changed. Without them the carry is 0.28. No basketball move crosses leagues. In 56 moves made since 2019 and read later as a prediction, the carry was 0.23 ($P = 0.039$). That is $-0.5$ standard errors from the combined value, so the new moves fit it.

Baseball's pre-specified test of the carry read the 24 moves of MLB's managers. It failed: 0.26 ($-0.16$ to 0.60), with a two-sided $P$ of 0.22. A second pre-specified test asked what share of the differences between teams the managers explain. It failed too: knowing the manager explained only a

further 0.0014 of them, its added $R^2$ ($P = 1.00$). A third reading measured each spell against its own team's season instead of the league's. On that reading the moves gave −0.29. But a spell that fills a whole season is its team's season, so its distance on that reading is zero. In twelve of the 24 pairs that distance is zero at both clubs. No other measure of the distance for MLB was fixed in advance. So the carry is shown in two sports, and baseball is inconclusive.

Combined over the three sports, the carry is 0.30 (0.20 to 0.40). Without MLB it is 0.31 (0.20 to 0.40). In the combined value MLB carries 6.2 per cent of the weight, football 56 and basketball 38. The three sports are consistent with one value ($P = 0.93$). With only three sports, the prediction interval for a further sport is wide, from −0.36 to 0.76 on one degree of freedom. Its width reflects the number of sports, not their disagreement.

Football's moves within a league enter the combined value because basketball's and baseball's moves are all within one league. So the combined value compares one kind of move across the sports. Football's moves between leagues are the stricter test, because the coach meets a new league with its own habit. They carried 0.27 (0.08 to 0.45). With them in place of the moves within a league, the combined value is 0.28 (0.16 to 0.39).

Table S16: Every reading of the coach's carry. Each row gives the moves and the rank correlation between the old team's distance from its league before the move and the new team's distance after it, with its 95% interval. It also gives the slope of the straight line fitted to the distance after against the distance before, with its 95% interval. The combined rows pool football's moves within a league, basketball's moves on every game and MLB's moves.

| Reading | Moves | $\rho$ (95% interval) | Slope (95% interval) |
|---|---|---|---|
| *Football* | | | |
| Managers who changed club within one league | 192 | 0.32 (0.19 to 0.44) | 0.29 (0.13 to 0.45) |
| under the rule on players who follow | 189 | 0.31 (0.18 to 0.43) | 0.28 (0.12 to 0.44) |
| allowing for the new club's season before | 159 | 0.26 (0.11 to 0.40) | 0.25 (0.03 to 0.46) |
| Managers who changed league | 99 | 0.27 (0.08 to 0.45) | 0.23 (0.07 to 0.39) |
| allowing for the new club's season before | 78 | 0.23 (0.01 to 0.43) | 0.20 (0.02 to 0.39) |
| Every mover | 291 | 0.30 (0.19 to 0.40) | 0.27 (0.15 to 0.39) |
| *Basketball* | | | |
| Coaches who moved, every game | 133 | 0.29 (0.12 to 0.43) | 0.26 (0.05 to 0.47) |
| half B, the pre-specified test | 134 | 0.27 (0.10 to 0.42) | 0.27 (0.09 to 0.46) |
| to a new team only | 123 | 0.28 (0.11 to 0.43) | 0.25 (0.02 to 0.48) |
| allowing for the new club's season before | 107 | 0.31 (0.13 to 0.47) | 0.27 (0.03 to 0.51) |
| Moves since 2019, read later | 56 | 0.23 (−0.03 to 0.47) | 0.31 (0.03 to 0.59) |
| *Baseball* | | | |
| MLB's managers, the pre-specified test | 24 | 0.26 (−0.16 to 0.60) | 0.19 (−0.11 to 0.48) |
| residual within each team's season | 24 | −0.29 (−0.62 to +0.13) | – |
| allowing for the new club's season before | 24 | 0.34 (−0.08 to 0.66) | 0.28 (0.00 to 0.55) |
| *Combined* | | | |
| Three sports | 349 | 0.30 (0.20 to 0.40) | 0.26 (0.15 to 0.38) |
| Without MLB | 325 | 0.31 (0.20 to 0.40) | 0.28 (0.15 to 0.40) |
| Football's moves between leagues in place | 256 | 0.28 (0.16 to 0.39) | 0.23 (0.12 to 0.35) |
| Given the destination's season before | 290 | 0.29 (0.17 to 0.39) | 0.26 (0.13 to 0.40) |

### S5.4 The new club's own distance

Coaches who make many changes could be hired by clubs that make many changes. We call this sorting. With sorting, the correlation would reflect the clubs, not the coach. So each move was set against the new club's own distance in the season before the coach arrived, over its team-matches under other managers. Clubs did not hire coaches like themselves. A manager's distance at the old club correlated 0.08 with the new club's distance the season before (−0.08 to 0.23, over 159 moves). In basketball and MLB the same correlations lay between −0.06 and 0.11. With the new club's own distance held fixed, the carry was 0.26 in football (0.11 to 0.40) and 0.31 in basketball (0.13 to 0.47). Combined, it was 0.29 (0.17 to 0.39).

The new club's distance in the season before the move needs that season to be under five. So moves into a league's first season under five, and moves to promoted clubs, drop out. A distance read over one season moves by chance, so holding it fixed adjusts only in part for any sorting. With sorting near zero this cannot matter much.

An event study of movers reads the same question another way [17]. It sets the new club's change in distance against the gap between the coach's old distance and the club's distance the season before. In football the slope was 0.50. But regression to the mean inflates it. A club far from its league one season tends to move back the next, whatever the coach does. The club's distance the season before sits in both quantities, so the slope picks up that move back. To measure it, the coaches' old distances were shuffled across moves, and the slope was then 0.36. So the coach's own part is the excess, 0.14 (0.05 to 0.22).

Managers rarely brought their players with them. The upper quartile of the share of players who followed was 3.8 per cent. So in three moves of every four, at most 3.8 per cent of the new team's players had played for the manager before. Basketball's moves already require that fewer than fifteen per cent of the new team's players played for the coach before. Applied to football, that rule removes three of the 192 moves within a league, and none between leagues. Without those three moves the carry was 0.31.

### S5.5 Managers and clubs in the variance

The data here are 46,392 team-matches of the fourteen leagues under five, from 2020–21 to 2025–26. The variance in substitutions measures how much these team-matches differ from one another. League-season means explained 0.156 of it. Means by league, season and manager explained a further 0.177, over 1,158 managers and 342 clubs in 82 league-seasons. That share does not separate the managers from their clubs.

A placebo repeats an analysis with the labels shuffled, to show what it gives when the labels mean nothing. The first placebo shuffled the manager labels across team-matches within each league-season. That breaks every spell, and the shuffled labels follow no team. So it explains little (0.043) whenever teams differ at all. It is the wrong comparison for a manager's own part beyond the club. The right one keeps each spell whole [18]. Each club's spells were put in a random order, each keeping its manager and its length. On means by league, season, club and manager, the real labels explained 0.181, and the reordered spells 0.181 ($P = 0.44$). So the share credited to managers belongs to the clubs.

A model with terms for league-season, club and manager separates the two where managers move between clubs [19, 20]. It was fitted on the largest set of clubs that moving managers connect, 284 clubs and 378 movers over 41,561 team-matches. Of the movers, 120 moved between leagues, and 290 had ten matches or more at each of two clubs. The median club had three movers, and 53 clubs had only one. Beyond league-season and club, manager terms added 0.075 of the variance. Reordered spells added 0.058 ($P = 1.0 \times 10^{-3}$). The excess is the managers' own part. As a spread between managers, it is about 0.13 substitutions a match, or 4 a season.

The variances of the manager and club terms themselves are inflated where few managers link the

clubs [21]. Read directly, the manager terms carry 0.248 of the variance and the club terms 0.184. Their covariance measures how the two sets of terms move together, and twice it is −0.254. That strongly negative value is the known sign of this inflation. The correction of Andrews and colleagues gives 0.166 to the managers, and the leave-one-out correction 0.175 [22]. Both corrections treat team-matches as independent, so chance swings shared within a spell remain. So the corrected shares are upper bounds. The share of the variance beyond the reordered spells is the figure that holds.

In playing terms, clubs and coaches inside a league-season differed about as much as leagues did. Their spread was 0.36 substitutions a match against 0.39 for league-seasons, about 12 and 13 a season. In substitute minutes it was 11.2 against 8.7 minutes a match. For example, in the Premier League of 2023–24 the clubs' mean share of the limit used ran from 57 to 93 per cent.

The design that follows movers has two published criticisms, and each applies here. First, what follows a departure forced on a firm can reflect its circumstances rather than a style [23]. Mid-season changes of manager usually follow poor results [24], so we do not read them as a style. Second, manager effects can be a statistical artefact, because spells reassigned at random explain as much [25]. Here the reordered spells explained as much as the real managers. What survives both critiques is the movers' carry, which follows the coach and not the club, and a small share beyond the club.

Two studies separate managers from their setting in other ways. In MLB, the match between a club's general manager and its field manager explains part of what the usual models credit to managers [26]. In the Bundesliga, coaches' age, experience and tenure with the team shaped their substitutions across the change from three to five [27].

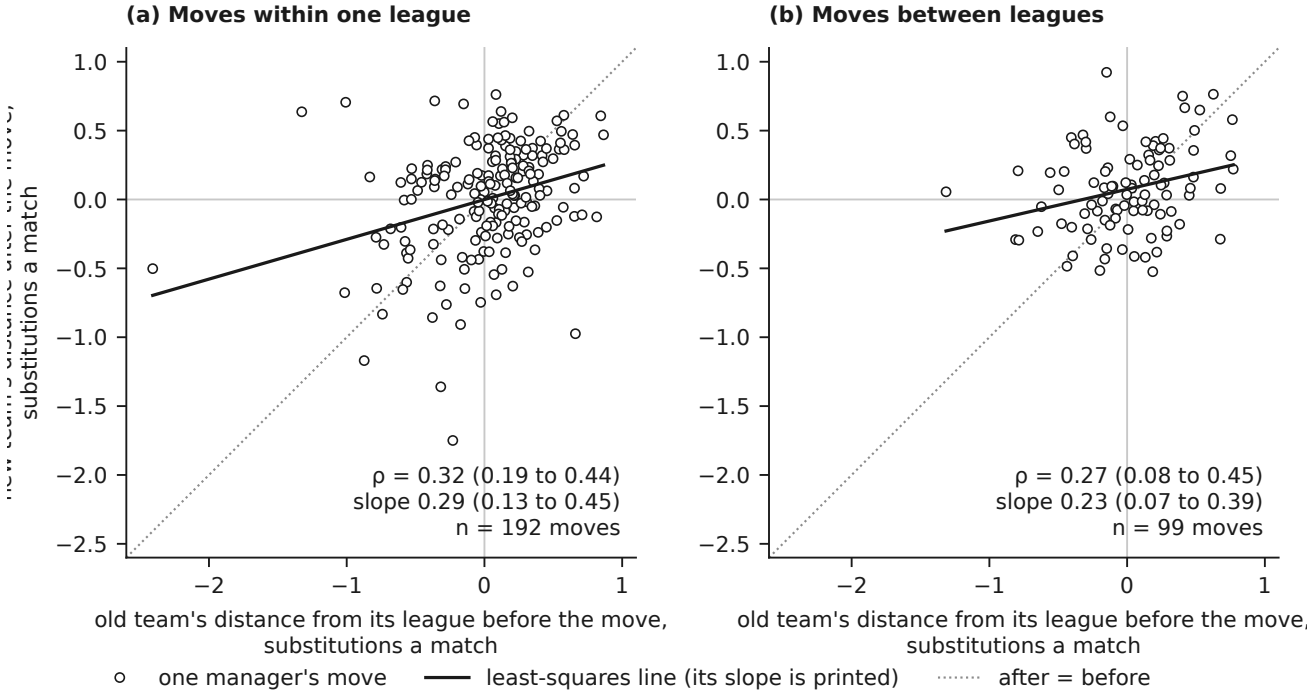



Figure S4: Football managers who moved, (a) within one league and (b) between leagues. Each point is one move. It sets the old team's distance from its league under the manager before the move against the new team's distance after it. The solid line is the straight line fitted to the points, and each panel prints its slope beside the rank correlation $\rho$ and the number of moves. The dotted line marks a distance carried whole.

### S5.6 Mid-season changes of manager

A mid-season change was read against clubs that kept their manager, as in a matched design for coaching changes [24]. Each club's first change in a season needed ten matches on each side. It was matched to three club-seasons under one manager, at the same match of their season. These three controls were the club-seasons nearest on points per match and on the distance, each over the ten and the five matches before. Matching brought the gap in points before the change from −0.87 to −0.10 standard deviations, and the gap in distance to −0.02. Over 323 changes and 969 controls, the changed teams' distance moved by 0.42 substitutions a match on average, in either direction. The controls' moved by 0.34. So after a change of manager, a team's distance moved 0.09 substitutions a match more than its controls' did (95% interval 0.05 to 0.13; $P = 5.0 \times 10^{-4}$).

The signed shift was −0.05 (−0.11 to 0.00), so changed teams moved towards fewer substitutions, if anything. With one manager over the ten matches after, the excess was 0.13 (0.06 to 0.18; 166 changes). With controls from the same league-season it was 0.10 (0.05 to 0.14), but points stayed apart there (−0.60 standard deviations). The matching uses actual points, because the records hold no expected points. Such changes usually follow poor results, so we do not read the shift as a style.

## S6 Persistence

### S6.1 A league's place

On each measure, a league's value in a season has three parts. The first is its sport's mean that season. The second is the league's own place, times the spread between the leagues. The third is sampling noise, the chance swings of single games in that season's mean:

$$h_{l,t} \;=\; \mu_t + \sigma_t\, a_{l,t} + e_{l,t}. \tag{S3}$$

The mean $\mu_t$ and the spread $\sigma_t$ belong to the sport's rules in that season. The place $a_{l,t}$ belongs to the league, and $e_{l,t}$ is the sampling noise. So a league's place is how far it stands from its sport's mean, in units of the spread. In the field's terms it is a standardised league effect. Suppose each league kept one place over all the seasons. On each measure read, that one place accounts for 59 to 92 per cent of the differences between the leagues.

A rule that shifts every league moves $\mu_t$ and keeps the order. The 2010 move of basketball's three-point line is an example. A rule that tightens shrinks $\sigma_t$. Basketball's rules of 2014 are an example: they shortened the shot clock's reset after an offensive rebound. The spread of the basketball leagues' pace was 1.9 to 2.2 possessions in the three seasons before those rules. In the four seasons from 2015–16 it was 1.0 to 1.5. As the leagues drew together, their order in pace was lost. A rule that loosens widens $\sigma_t$. The larger match sheets in volleyball are an example. Football fits the same form. Under three substitutions the leagues sat close to the limit, so $\sigma_t$ was small. Under five it grew, and each league's place showed in its increment.

The place drifts. A league keeps a share $\phi$ of its place from one season to the next (main text, Methods). We call that share the persistence. It is the autocorrelation of the standardised league effect from season to season.

### S6.2 How the persistence is estimated

In symbols, a league's place in a season is $\phi$ times its place in the season before, plus a part that is new:

$$a_{l,t} \;=\; \phi\, a_{l,t'} + \eta_{l,t}. \tag{S4}$$

Here $a_{l,t}$ is league $l$'s place in season $t$, $t'$ is the season before, and $\eta_{l,t}$ is the part of its place that is new in season $t$. So the places of two seasons $k$ apart correlate by $\phi^k$, once sampling noise is taken out.

A season's reliability is the share of the spread between its leagues that is not sampling noise. For each pair of seasons, we correlate the places of the leagues read in both. We divide each correlation by the square root of the two seasons' reliabilities. The corrected correlation estimates what the correlation would be without sampling noise. Each reliability is floored at 0.05: a lower one is raised to it. The corrected correlations are clipped to the range from minus one to 1.5: a value outside it is set to the nearer end. The model $\phi^k$ is then fitted by least squares to the corrected correlations of every pair of seasons $k$ apart, each pair weighted equally. The fit keeps $\phi$ at one or below, and its standard error comes from resampling leagues.

The estimator's choices do not move the result in the four sports of the fit. Their lowest reliability is 0.58, so the floor never binds. Every corrected correlation lies between −0.98 and 1.14, so the clipping never binds either. Corrected correlations above one are a known sign of low reliability in small samples [28]. We varied the floor to 0.10, 0.25 and 0.50 and removed it. We dropped pairs with a reliability below 0.50, a rule stated here and not pre-specified. We lifted the clipping or clipped to one, and we lifted the bound. We took each pair's reliabilities over the leagues the pair shares. We weighted the pairs, and we fitted the raw correlations. Each sport's persistence moved by 0.003 or less, except ice hockey's when clipped to one. That lowered it by 0.02. The four sports' pooled value stayed between 0.959 and 0.974. In the resamples, though, some corrected correlations do pass the clip. So without the clipping the resamples are much noisier: basketball's error is 0.035 instead of 0.013.

The fit assumes that each sport has one persistence, the same in every season. We also fitted the earlier seasons and the later seasons apart. In both parts the persistence was lower: 0.939 and 0.959 in basketball, and 0.911 and 0.917 in volleyball. Each part holds only pairs of seasons that are close together. So the lower values fit a decay that slows as the seasons grow further apart (Section S6.5) better than a persistence that changed over time.

Resampling only four to eight leagues is unreliable. Tests on few clusters, here few leagues, tend to understate the error [29]. So Table S17 also gives each sport's error from leaving out one league at a time. These errors are smaller than the resampled ones in every sport. So the resampled errors are the more cautious, and the main text uses them. Baseball has no such error. Leaving out one of its four leagues leaves three, below the four leagues that a season needs to enter the fit. With the errors from leaving out one league, the four sports' pooled value is 0.970 (standard error 0.007). Over the eight measures of Table S17, with the resampled errors, the pooled value's standard error is 0.023.

### S6.3 The persistence of each sport and measure

Table S17 gives each sport's and each measure's persistence. Beside it are its interval, its half-life, its weight, and the leagues and pairs of seasons behind it. The four sports are consistent with one value, 0.969 (0.94 to 0.99; $Q = 0.79$, $P = 0.85$). This is the pooled value, which combines the four sports. Basketball counts for 92 per cent of its weight. The test of one value has little power, because three of the sports are imprecise. At the pooled value, a league loses half of its place in 22 seasons (12 to 113). This is the half-life. In basketball, most of a league's place is permanent (Section S6.5), so a half-life is the wrong measure there.

The data rule out only large differences between sports. Each sport was compared with the pooled value of the other three. The data rule out a gap larger than 0.10 for basketball, 0.14 for volleyball, 0.19 for ice hockey and 0.23 for baseball. For example, volleyball's persistence could lie 0.14 below the others'. That is a half-life of 3.8 seasons, rather than 24. With the errors from leaving out one league, these gaps shrink to 0.05, 0.11, 0.08 and 0.23. So the data could have shown another sport to differ only if it stood well apart from the rest. In volleyball, ice hockey and baseball, a test against the other three sports catches a gap four times in five only when the gap is 0.16 to 0.33 or more. A test against the pooled value of all four catches a gap half the time only when the gap is at least 0.04 in basketball, 0.11 in volleyball, 0.19 in ice hockey and 0.24 in baseball. The spread between sports, $\tau$, is zero, with

an upper bound of 0.092. The prediction interval for a new sport runs from 0.914 to 1.024. A model that compares sports in one fit found that the persistence of team strength differs between sports [30]. So one value across sports describes these data, and is not a law.

Within a sport, the measures' values differ. Basketball's pace keeps 0.930 of its place a season, and the five busiest players' share keeps 0.987. Volleyball's substitutions per set keep 0.815, and its players used keep 0.984. That gap in volleyball comes largely from a counting error in Japan's sets (Section S4.2). Counted as played, substitutions per set keep 0.98. But basketball's four measures are consistent with one value ($Q = 0.34$, $P = 0.95$), and so are volleyball's two ($Q = 2.08$, $P = 0.15$). These tests find one value too easily, because the measures share their leagues.

Table S17: The persistence of each sport and measure, fitted on the pairs of seasons read before the later tests (Section S6.7). A sport's first row combines its measures. Leagues and Pairs count the leagues and the pairs of seasons behind each value. Lag is the largest number of seasons between the two seasons of a pair. The persistence $\phi$ has its 95% interval from resampling leagues. The next column gives its standard error from leaving out one league at a time. The half-life is the number of seasons until a league has lost half of its place. The weight is a sport's share of the four sports' pooled value, or a measure's share of the eight measures' pooled value. The last column is the state-space model's persistence (Section S6.4).

| | Leagues | Pairs | Lag | $\phi$ (95% interval) | One league out | Half-life | Weight (%) | State space |
|---|---|---|---|---|---|---|---|---|
| Basketball | 7 | 544 | 18 | 0.971 (0.95 to 1.00) | 0.008 | 24 (12 to 253) | 92 | 0.945 (0.90 to 0.99) |
| busiest five | 7 | 136 | 18 | 0.987 (0.91 to 1.06) | 0.010 | 51 (7 to $\infty$) | 33 | 0.969 (0.93 to 0.99) |
| ten minutes | 7 | 136 | 18 | 0.985 (0.88 to 1.09) | 0.013 | 45 (6 to $\infty$) | 19 | 0.960 (0.91 to 0.99) |
| shots from three | 7 | 136 | 18 | 0.970 (0.86 to 1.08) | 0.018 | 23 (5 to $\infty$) | 17 | 0.921 (0.85 to 0.97) |
| pace | 7 | 136 | 18 | 0.930 (0.74 to 1.12) | 0.048 | 9 (2 to $\infty$) | 5 | 0.874 (0.78 to 0.95) |
| Volleyball | 8 | 306 | 17 | 0.926 (0.82 to 1.03) | 0.035 | 9 (3 to $\infty$) | 5 | 0.926 (0.85 to 1.00) |
| players used | 8 | 153 | 17 | 0.984 (0.85 to 1.12) | 0.008 | 42 (4 to $\infty$) | 11 | 0.947 (0.89 to 0.98) |
| substitutions a set | 8 | 153 | 17 | 0.815 (0.63 to 1.00) | 0.109 | 3 (2 to 636) | 6 | 0.880 (0.77 to 0.96) |
| Ice hockey, goalie pulls | 6 | 6 | 3 | 0.998 (0.81 to 1.18) | 0.033 | 367 (3 to $\infty$) | 2 | 1.000 (0.79 to 1.00) |
| Baseball, pitchers | 4 | 15 | 6 | 0.941 (0.71 to 1.17) | – | 11 (2 to $\infty$) | 1 | 0.942 (0.79 to 0.99) |

## S6.4 A model that includes the sampling noise

The estimator above corrects each pair of seasons for sampling noise. A state-space model instead puts the noise into the model itself [31]. In it, each league-season's mean is the season's mean, plus the season's spread times the league's true place, plus sampling noise. The true place is not seen directly. The variance of the noise is known: it is the squared standard error of the league-season's mean. From one calendar season to the next, the true place follows the persistence equation of Section S6.2, with a variance of one. This is a first-order autoregression. The model is fitted by restricted maximum likelihood. A sport's measures share one persistence, so the fit over several measures is a composite likelihood. Intervals come from the profile likelihood for one measure, and from leaving out one league for several.

The model gives 0.945 in basketball, 0.926 in volleyball, 1.000 in ice hockey, at its bound, and 0.942 in baseball (Table S17). Pooled, it gives 0.94 (0.90 to 0.98). In basketball it is lower than the pairs' value, because one rate of decay does not fit basketball's pairs (Section S6.5). So in basketball a single value of the autoregression is a compromise, and not the quantity to report.

## S6.5 A permanent part and a part that decays

A league's place may have two parts. A permanent part holds a share $w$ of its variance. The other part decays, keeping a share $r$ of itself each season. Places $k$ seasons apart then correlate $w + v\,r^k$, where $v$ is the share that decays. This is the trait-state-error model [32], a random-intercept autoregression [33].

It was compared with the single rate $\phi^k$ in three ways. Each way allows for the overlap of the pairs, since every season enters many pairs. The first left out each league in turn, and scored how well each model predicted that league's whole series. The second was a parametric bootstrap of the likelihood ratio, in panels simulated under the single rate. The third was the same bootstrap of the drop in the residual sum of squares on the pairs.

In basketball the permanent part held 78 per cent of a league's place (64 to 93), over nineteen seasons. The rest decayed, keeping 0.72 of itself each season. The likelihood ratio was 17.4 ($P = 0.0050$). Leaving out one league at a time, the model with a permanent part predicted the left-out league better. Summed over the leagues, it gained 17.0 in log-likelihood (standard error 5.4). It did better in six of the seven leagues. On the pairs themselves the model gave a permanent part of 0.75 and a share kept each season of 0.63 ($P = 0.037$). Basketball's corrected correlations average 0.91 one season apart. From 13 seasons apart and beyond they average 0.76. So the correlations level off. The single rate implies 0.65 there, and the permanent part plus decay 0.78. Seventeen seasons apart, the model with a permanent part implies a correlation of 0.78, the state-space autoregression 0.38 and the pairs' single rate 0.61.

The same held in volleyball, with a permanent part of 0.67 ($P = 0.025$). It did not show clearly in ice hockey or in baseball. In ice hockey the permanent part was 0.84 ($P = 0.12$), over all its seasons, those read after the fit included. In baseball $P$ was 0.35. Both records are too short to separate the two parts. When a sport's measures are fitted together, part of the permanent part's advantage comes from forcing one rate on measures that decay differently. Measure by measure, leaving out one league clearly favours the permanent part only twice. The two are basketball's pace, by 3.4 (standard error 0.9), and volleyball's players used, by 6.1 (standard error 2.0). Permanent here means lasting through the seasons read. A very slow decay would look the same. Figure S5 draws both models in the four other sports.

### S6.6 Quantities unrelated to rule use

League quantities unrelated to rule use kept almost as much of their place. By the same estimator, goals per match kept 0.93 of their place a season in football's fourteen leagues. Points per forty minutes kept 0.93 in basketball, goals in regulation time 0.96 in ice hockey and runs per full innings 0.94 in cricket. Volleyball's sets per match kept 0.95. The state-space model gives 0.88, 0.81, 0.92 and 0.92 for the first four. No baseball benchmark was computed, because the baseball records read keep no score in a usable form. Home advantage and the share of draws were read too, but with almost no reliability. The median reliability is 0.02 for football's home goal difference and 0.15 for its draws. So their persistence means nothing.

Some measures are built from events a scorer records. A league's scorekeeping practice can itself persist [34]. So such a measure could persist because the scorekeeping does. Basketball's turnovers, offensive rebounds and shots, as the scorer records them, kept 0.93 to 0.95. That is no more than other league quantities, and less than the measures read from players' minutes. These records cannot separate scorekeeping from play. They show only that counts of events persist no more than other quantities.

### S6.7 Later seasons and new leagues

The basketball places were fitted on the seasons up to 2018–19. Before any later season was read, predictions were written down for the seasons from 2021–22 to 2024–25 (Section S4.4). The leagues kept their order in the five busiest players' share of minutes (rank correlation +0.93, $P = 0.0034$). They also kept it in the share of shots from three (+0.89, $P = 0.0062$). The order was lost in players given ten minutes (+0.68, $P = 0.055$), and in pace (+0.32, $P = 0.25$).

Drift also limits what an order test can show. With four leagues, only a perfect order is rare enough to pass. With the six ice hockey leagues whose records time their goalie pulls, an order no longer has to

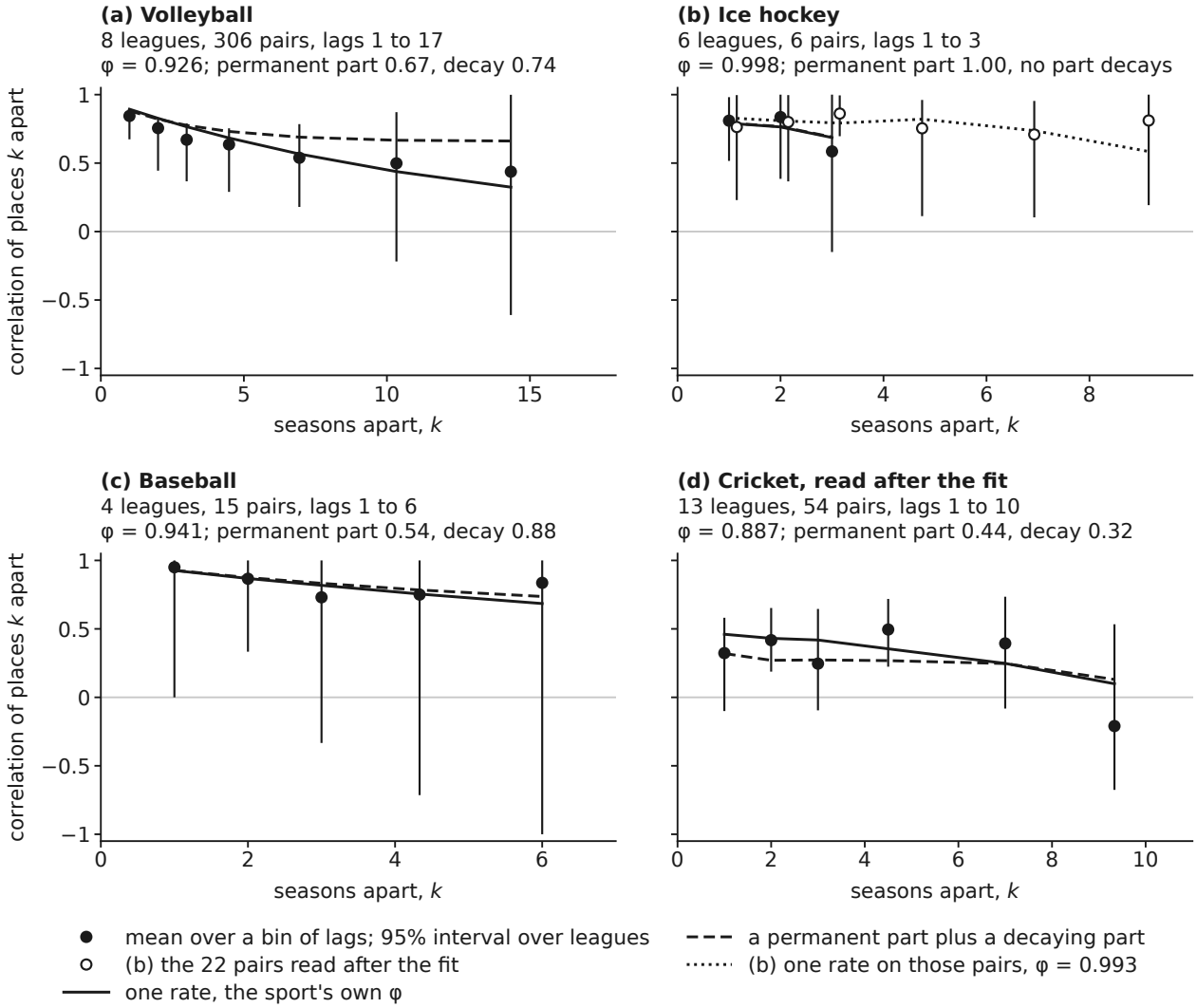



Figure S5: Persistence in the four other sports, drawn as basketball's is drawn in the main text, on a wider scale. Each point is the mean raw correlation of the leagues' places $k$ seasons apart, over a group of nearby $k$. Its 95% interval is over leagues, cut at plus and minus one. Each line is the mean raw correlation that a model predicts for the same pairs, once their sampling noise is allowed for. So a line need not fall steadily. The solid line has one rate, the sport's own $\phi$, and the dashed line a permanent part plus a part that decays. Each panel prints both fits, on the scale corrected for sampling noise. In ice hockey, on the seasons read before the later tests, the whole place is permanent. So no part decays there, and the two lines coincide. Section S6.5 gives ice hockey's fit over all its seasons. Ice hockey's open points are the pairs read after the fit, and the dotted line is one rate fitted on them. Cricket's pairs were all read after the fit, and are drawn as the other panels' fitted pairs.

be perfect to pass. But six seasons of drift still move some of them. Their places of 2015–16 to 2018–19 and of 2021–22 to 2024–25 correlated +0.77 ($P = 0.051$). That was short of the pass mark that the pre-specified test set. We had written down that a failure was more likely than not.

Three later tests read the persistence on records it was not fitted on. Each gives $z$, the gap between its value and the pooled value, in standard errors. A prediction passes when its value lies within two standard errors of the pooled value (Section S4.4). The first test used a measure that had never been read season by season. It is the five busiest skaters' share of a side's time on ice, with goalies left out. It was read in the eight ice hockey leagues whose time on ice adds up, game by game, to the length of the game. A game that does not add up was left out whole. The share kept 0.963 of its place a season, over 33 pairs of seasons (standard error 0.11, $z = -0.1$). In two more leagues too many games do not add up. They are the Elite League, the top league in the United Kingdom, and the Czech second tier. They were read beside the test as description only. With them the value is 0.901 (36 pairs of seasons, ten leagues).

The second test read basketball leagues outside the fit. They were North America's top men's and women's leagues, the NBA and the WNBA, and the NBA's development league, the G League. They were also the top men's leagues of Japan and France, the B.LEAGUE and the Betclic ELITE. On the four basketball measures above, these leagues kept 0.979 of their place a season (standard error 0.11, $z = +0.1$). That is over 152 pairs of seasons. The third test read later seasons of the leagues the persistence was fitted on. It used the pairs that include a season read after the fit. On them, a league kept 0.993 of its place a season in ice hockey and 0.965 in basketball. They lie at $z = +0.2$ and $z = -0.1$.

These passes say little. Each later test would fail four times in five only at a much weaker persistence, about 0.64 to 0.84 or below. Each would fail half the time at a persistence whose half-life is 2.2 to 5.2 seasons. The predictions' own simulations say the same. The prediction on ice hockey's busiest skaters would still hold 81 per cent of the time at a persistence of 0.8. It would fail half the time only near 0.60. The prediction on basketball leagues outside the fit would hold 79 per cent of the time at that persistence. It would fail half the time near 0.65. So the later tests were consistent with the persistence fitted, but they did not confirm it.

### S6.8 Cricket

A bowler may bowl at most a fifth of an innings, four of a T20 match's twenty overs. So a side needs at least five bowlers. How many more it uses is a league's habit under a rule. Over thirteen T20 leagues, a prediction asked whether the order in bowlers used per full innings held. It held from the seasons opening 2019 to 2022 to those opening 2023 to 2025 (rank correlation +0.63, $P = 0.013$). An earlier analysis had used the same archive, but pooled across leagues and seasons. So this order had not been examined when the test was written down. A second prediction read two measures: bowlers used per full innings, and the five busiest bowlers' share of the overs. On the two together, cricket's leagues kept 0.887 of their place a season ($z = -1.1$ against the pooled value). The IPL, the Indian Premier League, rose the most. It went from 5.80 to 6.01 bowlers per full innings. It is the one league with an Impact Player rule. From 2023 that rule lets a side bring in a player during the match.

Cricket's second measure, the five busiest bowlers' share of the overs, sits near the bound that the rule on overs sets. Its reliabilities are low, with a median of 0.69. Those of bowlers used are low too, with a median of 0.56. Each measure has one season whose reliability was raised to the floor of Section S6.2. So the busiest bowlers' share is floored no more often than bowlers used. Of the 54 corrected correlations of the prediction, nineteen lie outside the range from minus one to one. The prediction read both measures as it was written down, so both stay in it. Without the busiest bowlers' share, cricket's leagues kept 0.91 of their place a season on the same pairs. That share alone kept 0.86. Dropping the pairs of low reliability gives 0.92, and fitting the raw correlations gives 0.92. The

state-space model, which includes the sampling noise, puts cricket's persistence near 0.71 (0.51 to 0.90). The pairs' estimator floors and clips cricket's pairs, and gives a higher value.

# S7 Timing in football

## S7.1 What is measured

The timing measures are read on starters, the players who begin a match. The withdrawal rate $s_w$ is the share of starters taken off during a match. Under three it ran from 0.24 to 0.27 across the 17 timing leagues, and under five from 0.34 to 0.42. A starter's recent workload $x$ is the minutes the starter played in the seven days before the match, divided by ninety. So a full match in that week gives $x$ of one.

The measured timing slope $b$ is read on the starters who were taken off. It is the least-squares slope of the logarithm of their minutes played on $x$. So a positive $b$ means that starters with more recent minutes come off later. The fit allows for the day of the week through the sine and cosine of a weekly and a half-weekly cycle. We call these the calendar terms. The fit's errors are clustered by player.

The retention coefficient $\theta$ describes who comes off. The match is cut into three parts: up to the sixtieth minute, from the sixty-first to the seventy-fifth, and from the seventy-sixth to the eighty-ninth. For a starter still on the field, the log-odds of being taken off in part $k$ is $\alpha_k + \theta x + \gamma' z$, where $z$ holds the calendar terms. A loaded starter is one with a heavy workload in the previous seven days. A negative $\theta$ means that a loaded starter is less likely to be taken off, in every part of the match. The coefficient is negative wherever it has been measured. The companion paper [35] reports it as $-0.35$ across 15 leagues under three. The coefficient first appeared in a preprint on nine sports [36].

That lower chance alone makes loaded starters come off later. In each part, a smaller share of them is taken off, so more of them last into the later parts. The slope that $\theta$ implies, $b_\theta$, is the least-squares slope on $x$ of the expected log minute under the fitted model:

$$b_\theta = \text{slope}_x \sum_k \pi_k(x)\, \bar{\ell}_k, \qquad \pi_k(x) = \frac{h_k(x) \prod_{j<k} \bar{h}_j(x)}{\sum_m h_m(x) \prod_{j<m} \bar{h}_j(x)}. \tag{S5}$$

Here $h_k(x) = \Lambda(\alpha_k + \theta x + \gamma' z)$ is the chance of being taken off in part $k$, and $\Lambda$ is the logistic function. $\bar{h}_j(x)$ is the chance of staying on through part $j$, and $\bar{\ell}_k$ is the mean log minute of the starters taken off in part $k$. So $\pi_k(x)$ is the share of withdrawn starters who come off in part $k$. The fitted model is held fixed, so $b_\theta$ has no error of its own. It uses the same calendar terms as $b$.

A league's timing departure is $g = b - b_\theta$. A positive departure means that loaded starters come off later than their lower chance of being taken off alone would place them. Its unit is the change in log minutes per full match of prior play. The companion paper [35] derives the model and reports its one attenuation constant, 0.81. The constant is the observed fall in a starter's chance of being taken off with workload, as a share of the fall the model predicts.

The product $C = g\, s_w$ is the departure times the withdrawal rate. A league that takes off more starters spreads the same few late decisions about loaded starters over more withdrawals. So its departure per withdrawal shrinks. The product counts the departure per starter who began the match, not per starter taken off. So leagues that take off different shares of their starters can be compared.

Load management is the field's term for matching how much a player plays to how much the player has played recently [37]. A league's load-management coefficient $M$ is its measured timing slope $b$ with the starter's market value added to the fit. Holding market value fixed separates workload from how much the coach values a starter.

## S7.2 Timing under three substitutions and under five

The spread between the leagues is $\tau$, the between-league standard deviation of a random-effects fit. Its 95% interval comes from the Q-profile, and Table S18 also gives a 95% prediction interval for a new league. Cochran's $Q$ alone mixes the spread with the precision of each league's estimate. Departures under three are less precise, because fewer starters are taken off. Their typical error was 0.014, against 0.010 under five. So the same spread would give a smaller $Q$ under three, and the limits are compared by $\tau$.

Under three, the products' spread was small but not shown to be zero. Its $\tau$ was 0.0009, with a 95% upper bound of 0.0037 ($Q(16) = 16.5$, $P = 0.42$). Under five it was 0.0031 (0.0014 to 0.0069; $Q(16) = 35.8$, $P = 0.0030$). The departures' $\tau$ was 0.0042 under three (0.0000 to 0.0148) and 0.0081 under five (0.0037 to 0.0176). So the leagues' timing spread more under five. The two limits' intervals overlap, though, and the spread under three could have been as large as 0.0037. The test under three had power 0.72 to detect a spread as large as the one under five.

Each league's step is its product under five less its product under three. The steps averaged +0.0021. With the law's load-management term subtracted (Section S7.4), they averaged +0.0024. They showed no spread: $\tau$ was zero, with 95% upper bounds of 0.0044 and 0.0024. The step test is weak, though. Its power was 0.41 at a spread equal to the products' $\tau$ under five. It was 0.45 if the two limits' league effects were independent, and 0.20 at the least spread that the two limits' $\tau$ allow. It needs a spread of 0.0048 for power 0.80. So the steps' lack of spread is weak evidence. The adjusted products under five are alike because of how the adjustment is built (Section S7.4). So their $\tau$ of zero is evidence of nothing.

Table S18: The timing departure and the timing product under each limit, and their steps from three to five, across the 17 timing leagues. Each row gives the random-effects mean with its 95% interval, the between-league standard deviation $\tau$ with its 95% interval from the Q-profile, the 95% prediction interval for a new league and Cochran's $Q$ on 16 degrees of freedom with its $P$. The adjusted product subtracts the load-management term.

| Quantity | Mean (95% interval) | $\tau$ (95% interval) | Prediction interval | $Q$ ($P$) |
|---|---|---|---|---|
| *Departure g* | | | | |
| under three | −0.0111 (−0.0168 to −0.0055) | 0.0042 (0.0000 to 0.0148) | −0.0220 to −0.0003 | 17.2 (0.37) |
| under five | −0.0002 (−0.0056 to +0.0051) | 0.0081 (0.0037 to 0.0176) | −0.0184 to +0.0179 | 36.0 (0.0029) |
| step | +0.0091 (+0.0027 to +0.0154) | 0.0000 (0.0000 to 0.0103) | +0.0022 to +0.0160 | 10.6 (0.83) |
| *Product C* | | | | |
| under three | −0.0029 (−0.0043 to −0.0015) | 0.0009 (0.0000 to 0.0037) | −0.0054 to −0.0003 | 16.5 (0.42) |
| under five | −0.0001 (−0.0022 to +0.0019) | 0.0031 (0.0014 to 0.0069) | −0.0070 to +0.0068 | 35.8 (0.0030) |
| under five, adjusted | −0.0006 (−0.0018 to +0.0007) | 0.0000 (0.0000 to 0.0000) | −0.0019 to +0.0008 | 6.7 (0.98) |
| step | +0.0021 (+0.0002 to +0.0040) | 0.0000 (0.0000 to 0.0044) | +0.0001 to +0.0042 | 13.9 (0.61) |
| step, adjusted | +0.0024 (+0.0005 to +0.0043) | 0.0000 (0.0000 to 0.0024) | +0.0004 to +0.0045 | 9.7 (0.88) |

## S7.3 England as the only comparison league

The rule change of 2020 came with a crowded calendar, so the timing may have moved with the calendar rather than with the rule. Seven of the eight European leagues measured under both limits moved to five in 2020. England kept three for 2020–21 and 2021–22. So for two seasons the calendar was the same for every league, and the rule was not. It is the simplest case of staggered adoption, with two groups and two switch dates [38]. Rules adopted at different dates have been read across many countries before, for example the award of three points for a win [39]. Two studies used England's late adoption as a comparison for the counts of substitutions [40, 41]. Here it is applied to timing, at both switches.

Each league's measured timing slope $b$ was read in three windows: before 2020, between the two switches and after 2022. In 2020 the seven leagues' slope rose by +0.0125 and England's moved by

−0.0024, a difference of +0.0149. In 2022 England's rose by +0.0217 and the seven's moved by −0.0001, a difference of +0.0218. So at both switches the slope moved with the rule rather than with the calendar, before England's own variation is allowed for.

With one comparison league, the main uncertainty is England's own movement from season to season [42]. With few leagues, tests that treat them as many give too many false alarms [29]. So Table S19 reads each contrast six ways. The first, a $t$-test across the seven, takes England's change as known. The second allows for England's own sampling error. The third and fourth put each league in turn in England's place, with England left out or kept in. With eight leagues, the smallest $P$ either can give is 0.125. The last two ask how large one league's change against the others' mean is, in consecutive seasons in which no league changed its limit. These changes form a placebo distribution. The raw placebo takes the changes as they are, and the studentised one divides each by its own standard error.

The placebo's changes from season to season vary 1.22 times as much as their standard errors alone would make them. So England's slope wanders a little beyond sampling noise. The placebo's windows are single seasons, shorter than the real windows of two or three seasons. So the raw placebo overstates England's noise, and the studentised one may understate it. The right $P$ lies between theirs. With England's own error allowed for, the 2020 contrast has $P = 0.24$ and the 2022 contrast $P = 0.045$. So the contrast at 2022 is the stronger. With one comparison league, this is suggestive, not proof.

The calendar windows put England's restart, played under five from 17 June 2020 to 26 July 2020, into its windows under three. It holds 464 of the 4,363 withdrawn starters between the switches, and 186 of the 5,698 before them. Those matches were played under five, like the seven's. So they pull England toward the seven, and shrink the contrast. Read on windows of whole seasons with the restart left out, the contrasts are +0.0200 for 2020 and +0.0251 for 2022. With England's error allowed for, they have $P = 0.12$ and $P = 0.028$. Some league-seasons lost more than a tenth of their team-matches to the record check. Without these six league-seasons, the contrasts are +0.0183 and +0.0252, with $P = 0.17$ and $P = 0.032$.

Table S19: England as the only comparison league: the two-sided $P$ of each contrast in the measured timing slope, by six methods. The contrast at 2020 is the seven leagues' change less England's, and at 2022 England's change less the seven's. Both switches together sum the two. The windows are calendar windows, as the pre-specified test set them.

| Method | 2020 (+0.0149) | 2022 (+0.0218) | Both (+0.0367) |
|---|---|---|---|
| $t$-test, England's change taken as known | 0.023 | $1.6 \times 10^{-4}$ | 0.0019 |
| $t$-test allowing for England's sampling error | 0.24 | 0.045 | 0.076 |
| Each other league in England's place | 0.500 | 0.125 | 0.125 |
| Each of the eight in England's place | 0.500 | 0.125 | 0.125 |
| Placebo distribution, raw | 0.71 | 0.49 | 0.59 |
| Placebo distribution, studentised | 0.41 | 0.098 | 0.18 |

England's rules also changed in its two seasons as the comparison league. Its bench grew from seven named substitutes to nine from 19 December 2020. Concussion substitutes, beyond the three, came from 6 February 2021. The team sheets show both. Nine were named in 93 per cent of team-matches from that December, and in 96 per cent in 2021–22. Only four team-matches in 2020–21 and fifteen in 2021–22 recorded a fourth substitution. England's 2020–21 was also played behind closed doors in 92 per cent of team-matches. Most of the seven leagues enlarged their benches when they moved to five. About eleven were named in Spain, and about nine in Germany, France and Portugal. So the bench changed in both groups, and the limit on substitutions in one.

Under the old limit and the crowded calendar of 2020–21, England's teams did not use more of their three. They made 2.66 substitutions a match in 2020–21, against 2.75 to 2.78 in 2015–16 to 2018–19. The match events agree (2.68). That season 27 per cent of team-matches came after three days of rest

or fewer, against 17 to 19 per cent before. Reweighted to the earlier seasons' mix of rest, the count is still 2.66. Reweighted to their mix of months, it is 2.66. The fall had begun before the pandemic, at 2.70 in 2019–20 before play stopped. The 62 team-matches played with a crowd averaged 2.74, and the 698 without one 2.67. Too few had a crowd to separate the crowd from the season. In 2021–22 England also stayed under three. Its count that season from the appearances, 2.72, falls short of its match events, 2.80. The appearances miss some incoming substitutes that season (Section S1.4).

### S7.4 The load-management law, and why it is not claimed

The preprint stated a load-management law [36]. It ties a league's departure to its withdrawal rate and its load-management coefficient:

$$g \;=\; \frac{C_5}{s_w} \;+\; \lambda\,\big(M - \bar{M}\big). \tag{S6}$$

Here $C_5$ is the dilution constant. It is the product $C$ of Section S7.1, which the law took as one value fitted across both limits. $\bar{M}$ is the mean coefficient across leagues, and $\lambda$ is one slope shared by the leagues. Fitted on 12 leagues, the law gave $\lambda = 0.87$ (95% interval 0.54 to 1.21). With the load-management term subtracted, the leagues' departures were consistent with one value ($Q(10) = 3.9$, $P = 0.95$). Without it, they differed between leagues ($Q(11) = 30.9$, $P = 0.0011$).

That agreement is built in. The measured slope inside $g$ and the coefficient $M$ are one regression on the same withdrawn starters. They share the log minutes played, the workload $x$, the calendar terms and the errors. $M$ adds one more term, market value. In the top men's leagues of Brazil, Argentina, Japan and Mexico, the two were fitted on the same starters. In Europe the starters behind $g$ are the five-substitution part of those behind $M$. Across the 12 leagues the slope and the coefficient correlate 0.97. The other part of $g$, the slope $b_\theta$ that $\theta$ implies, varies little between leagues. It accounts for about 1 per cent of the spread of $g$. So $g$ is close to $M$ minus a constant. A fitted $\lambda$ near one then follows from the definitions. If the measured slope itself is put in place of $M$, the law's $\lambda$ is 0.94. The spread left between the leagues then falls from $Q = 30.3$ on 11 degrees of freedom to 0.34. Both values are computed after the law's first term, $C_5/s_w$, is subtracted from each departure. The fitted law therefore shows how the two measures were built, not how leagues behave. This paper does not claim it.

The same reading applies to what the preprint offered as the law's support. Each placement was the law's prediction of one league's departure. Its $z$ is the gap between the measured and the predicted departure, in standard errors. A league lands inside the bar when $|z| < 2$. Five placements were written down before their leagues were measured. Brazil and Argentina landed at $z = 0.25$ and $z = 0.17$, and Mexico at $z = 0.26$. The NWSL, the top women's league in the United States, landed at $z = 0.07$. The top French women's league was predicted at $-0.037$ and measured at $+0.007$ ($z = 2.45$). A flat model sets every league's departure at the mean of the leagues the law was fitted on. Across all 7 placed leagues, these five among them, the flat model places 7 inside the bar, and the law places 6. Each placement drew its $M$ from records that also gave the slope inside $g$. The French league, the one miss, is where the two shared the fewest starters.

### S7.5 The test on separate matches

We wrote down four pre-specified tests on separate matches. They asked whether a league's load-management coefficient, measured on its three-substitution matches, predicted its departure under five. For the European leagues, the three-substitution matches are those before play was suspended in March 2020. For England they also include its 2020–21 and 2021–22 seasons. For Brazil, Argentina, Japan and Mexico they are each league's last season under three. Market values come from that season's squad pages. The departures under five are unchanged. On separate matches the coefficient and the measured slope correlate 0.70, against 0.97 on shared ones. So a pass of these tests would not be built in.

Across the 12 leagues, the coefficients measured before the change predicted the departures after it. The slope was 0.51 (standard error 0.24, one-sided $P = 0.019$). That result rests on Japan. Without Japan the slope is 0.23 ($P = 0.25$). Leaving out any other single league keeps the slope between 0.47 and 0.61. Japan's leverage, a measure of how far its coefficient lies from the others', is 0.51 against a conventional threshold of 0.50. Japan's coefficient was already 0.036 (standard error 0.016) under three substitutions. So in Japan the load-management habit, which the coefficient measures, was there before the rule change.

In Europe the habit could not be read before the change. Before it, every European coefficient sat between $-0.020$ and $+0.0004$. After it they ran from $-0.003$ to $+0.024$, and seven of the eight rose. Across the 8 European leagues, the coefficients measured before the change did not predict the departures (slope 0.21, standard error 0.46, one-sided $P = 0.33$). They also ranked the leagues differently from the coefficients measured after the change (rank correlation $+0.05$, exact one-sided $P = 0.47$). Put into the law, the coefficients measured before the change also left the 12 leagues' departures different from one another ($Q(10) = 21.2$, $P = 0.020$).

We pre-specified two further tests, and each was read once. The first fitted the coefficient on one half of each league's five-substitution matches, and the departure on the other half, in both directions. This split of football's matches is separate from the split record of the other sports. Before any half was fitted, each match's half was fixed by a hash of its identifier, a number computed from the identifier alone. The split removes the noise the two measures share, but not the regression they share. So the test's pre-specified text said it could hardly fail. The two directions gave slopes of 0.74 and 0.41. The two halves' coefficients agreed with a reliability of 0.63. Divided by that reliability, the mean of the two slopes was 0.91. Its 95% interval, by Fieller's method for a ratio, came from 2,000 resamples of players within league. It ran from 0.70 to 1.04. The interval lies above zero and contains one, so the test held. The leagues' spread about each fit was within sampling error ($Q = 12.8$, $P = 0.23$, and $Q = 12.6$, $P = 0.24$, on 10 degrees of freedom). The pass says only that noise shared by the two measures did not make the agreement. It says nothing about whether leagues manage their players' load.

The second test asked whether leagues that rest heavily loaded starters between matches take them off earlier within matches. Its measure of resting is computed by the companion paper [35] and is not a timing slope. In each league it takes starters who had played a full match or more in the previous seven days, and starters who had played none. It asks how much more often the first group is left on the bench for the club's next league match. The two groups are compared within the same player's season, and with the same number of days to that match. The test asked for a negative rank correlation between the leagues' resting measures and their departures, with an exact one-sided $P$ below five per cent. Across the eight European leagues the correlation was $-0.02$ ($P = 0.49$). The test failed. With so few leagues it had little power, and its pre-specified text said so.

## S8 How each competition shares out its minutes

The leagues of one sport share out their minutes differently under the same rules. Table S20 shows how, in eighteen competitions of five sports. These are the other sports whose records give each player's minutes or, in baseball, each pitcher's pitches. Volleyball and handball record neither, so they are absent here. Australian football and rugby league record minutes, so they are here. No test uses them. They show the same measure in sports that field many more players at once.

The busiest-$n$ share is the share of a team's minutes played by its $n$ busiest players. Here $n$ is the number the rules put on the field: five in basketball, five skaters in ice hockey, eighteen in Australian football and thirteen in rugby league. Goalies are left out in ice hockey. A share of one would mean that nobody came off. Baseball is read one pitcher at a time. Its row gives the starter's share of the team's pitches. Its last column gives the pitchers a team used. In basketball and ice hockey, "Ten minutes" is the number of players given ten minutes or more. "Players" is the number who played. Each value

is the mean over team-games. A team-game is one team's side of one game, as a team-match is in football. In brackets is the standard deviation across team-games: roughly, a typical team-game's gap from the mean. Each competition is read over the seasons in which its minutes are recorded alike. Its records are the league's own box scores and time-on-ice records.

In basketball the five busiest players took between 0.64 and 0.74 of a team's minutes across nine leagues. Treating the leagues as a sample, the spread between them was 0.034 (95% interval 0.023 to 0.065). In ice hockey the five busiest skaters took between 0.34 and 0.37 of the skaters' time across six leagues, with a spread of 0.010 (0.006 to 0.025). Each competition's mean is far more precise than these gaps. Treating team-games as independent, its 95% interval reaches at most 0.002 either side. The busiest players' share is one of the measures whose order and persistence the main text reports (main text, Results).

Table S20: How a team shares out its minutes, or in baseball its pitches, by competition. The table is description.

| Competition | Seasons | Team-games | Busiest-$n$ share | Given ten minutes | Players |
|---|---|---|---|---|---|
| *Basketball* ($n = 5$) | | | | | |
| Spain, ACB | 2006-07–2022-23 | 10,876 | 0.641 (0.060) | 9.0 | 10.6 |
| Italy, LBA | 2006-07–2018-19 | 7,298 | 0.712 (0.061) | 8.1 | 9.5 |
| South Korea, KBL | 2006-07–2018-19 | 7,562 | 0.741 (0.065) | 7.5 | 10.1 |
| Turkey, BSL | 2006-07–2018-19 | 6,746 | 0.718 (0.075) | 8.0 | 9.9 |
| Lithuania, LKL | 2006-07–2025-26 | 7,088 | 0.673 (0.077) | 8.6 | 9.9 |
| NBA | 2015–2024 | 23,958 | 0.658 (0.061) | 9.1 | 10.6 |
| WNBA | 1997–2025 | 12,328 | 0.729 (0.069) | 7.8 | 9.7 |
| G League | 2013–2025 | 15,398 | 0.681 (0.060) | 8.7 | 9.7 |
| Japan, B.LEAGUE | 2016-17–2025-26 | 12,492 | 0.681 (0.063) | 8.3 | 10.4 |
| *Ice hockey* ($n = 5$) | | | | | |
| NHL | 2008-09–2024-25 | 40,592 | 0.372 (0.020) | 16.3 | 18.0 |
| Switzerland, National League | 2017-18–2025-26 | 6,774 | 0.349 (0.022) | 16.6 | 19.5 |
| Finland, Liiga | 2022-23–2025-26 | 4,130 | 0.343 (0.021) | 17.2 | 18.8 |
| Czechia, Extraliga | 2014-15–2025-26 | 10,045 | 0.357 (0.023) | 16.3 | 18.7 |
| Germany, DEL | 2023-24–2025-26 | 2,428 | 0.358 (0.023) | 16.1 | 18.5 |
| Sweden, SHL | 2023-24–2025-26 | 2,442 | 0.346 (0.018) | 17.0 | 19.2 |
| *Australian football* ($n = 18$) | | | | | |
| AFL | 2017–2025 | 3,690 | 0.846 (0.011) | | 22.5 |
| *Rugby league* ($n = 13$) | | | | | |
| NRL | 2022–2025 | 1,680 | 0.895 (0.016) | | 17.0 |
| *Baseball* (pitchers, one at a time) | | | | | |
| MLB (pitchers) | 2008–2025 | 84,410 | 0.633 (0.155) | | 4.1 |

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